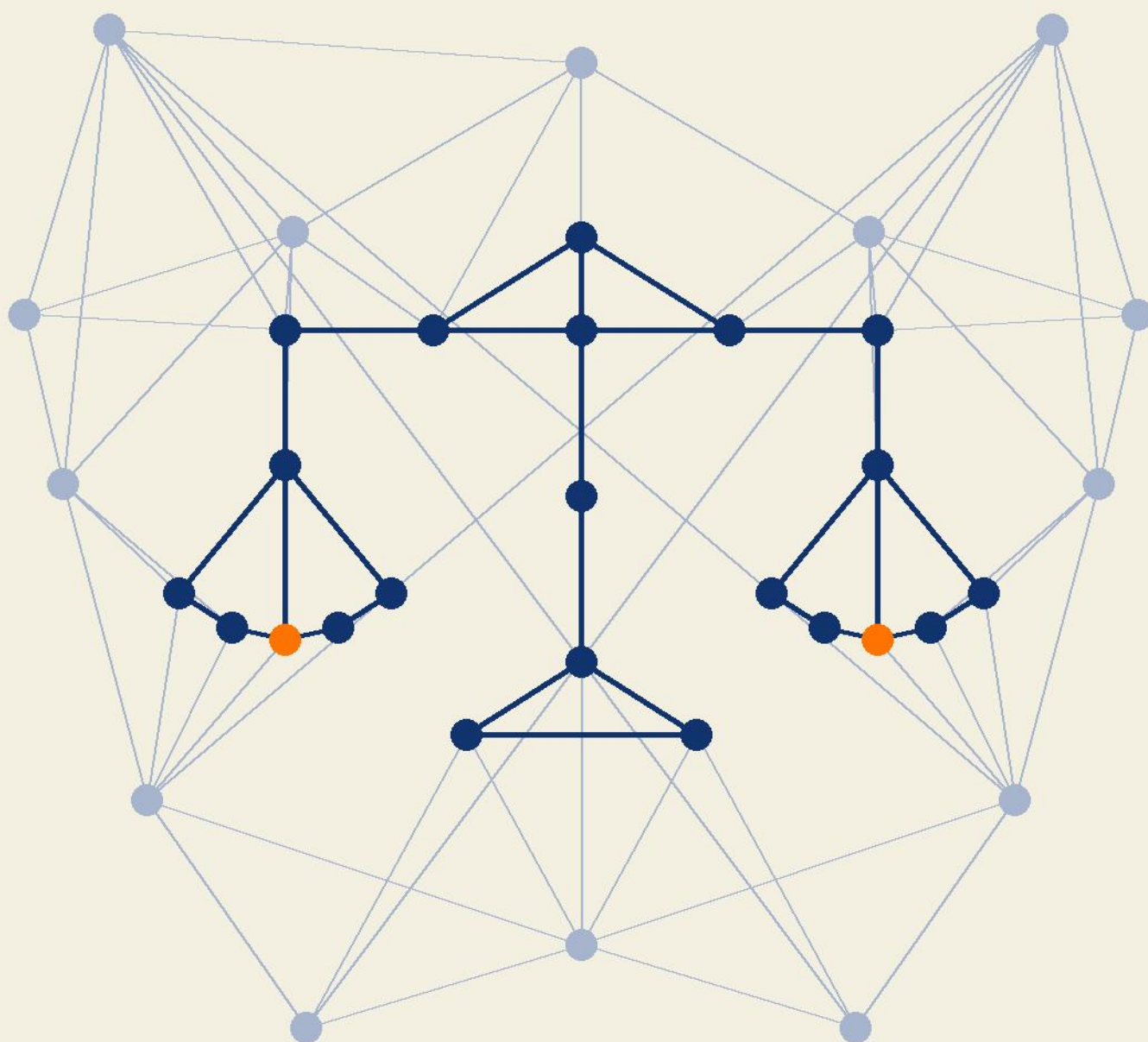


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Faculty of Education

Perceived Justice and Its Measurement

Among Czech Pre-Service Teachers



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Faculty of Education

Perceived Justice and Its Measurement

Among Czech Pre-Service Teachers

Habilitation thesis

(a monograph)

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Jan Nehyba

Table of contents

Declaration	1
Abstract	3
Acknowledgements	5
1. Introduction	7
Aim of the chapter	7
1.1 Six Villages: An Opening Scene	7
1.2 The Problem: Justice as an Everyday Act of Teaching	8
1.3 Aim of the Book and Research Questions	9
1.4 One Research Programme, Two Streams, Seven Cohort-Years (2019–2026) . . .	10
1.5 Structure of the Book	11
1.6 Note on Prior Publications, Data and Code Availability	12
Chapter summary	13
Foundations	15
2. Theoretical Foundations: Justice from Philosophy to the Classroom	17
Aim of the chapter	17
2.1 Justice and Fairness: A Contested Concept	17
2.2 Organizational Justice and Its Rules	20
2.3 Principles of Distributive Justice: The Wide and the Narrow Taxonomy	25
2.4 Classroom Justice: The Teacher as Allocator	29
2.5 The Four-Principle Framework of This Book	32
2.6 What This Book Excludes and Why	34
2.7 An Exploratory Conceptual-Similarity Map	36
2.8 Pre-Service Teachers as the Population of Interest	38
Chapter summary	40
3. Justice in Czech Educational Research: The Macro Tradition and the Micro Gap	43
Aim of the chapter	43
3.1 Equity of Access and Educational Inequality: The Dominant Czech Paradigm .	43
3.2 Taking Stock: The ČAPV 2021 Volume	46
3.3 Micro-Level Threads Already Present	48

Table of contents

3.4 The Gap and the Claim: Perceived Justice at the Level of Teacher–Pupil Interaction	51
Chapter summary	55
4. Research Design and Methodology	57
Aim of the chapter	57
4.1 Overall Design: Two Streams, Seven Cohorts, Four Research Questions	57
4.2 Samples and Data Collection 2019–2026	59
4.3 Instruments and Their Evolution	60
4.4 The Three-Tier Comparability Framework	61
4.5 Analytic Strategy and Software	62
4.6 Transparency and Reproducibility	64
4.7 Ethics and Anonymization	64
Chapter summary	65
Evidence	67
5. Study 1: Which Distributions Do Future Teachers Call Just? The Six Villages	69
Aim of the chapter	69
5.1 Focused Background: Vignettes in Justice Research	69
5.2 Materials: Six Villages, Six Principles	71
5.3 Sample and Procedure	72
5.3.1 The pedagogical setting	72
5.3.2 Sample	73
5.3.3 Analysis plan	74
5.4 Results: A Hierarchy of Principles	74
5.4.1 Descriptive hierarchy	74
5.4.2 Omnibus and pairwise tests	76
5.4.3 Robustness on the latent scale	78
5.5 Respondent Profiles	78
5.6 Interim Discussion: The Equality Preference, the Equal-Opportunity Paradox, and the Road to Classrooms	81
5.6.1 The equality preference	81
5.6.2 The equal-opportunity paradox	81
5.6.3 From villages to classrooms, and the limits of that road	82
5.7 From Six Principles to Four: Rationale for the Instrument’s Narrower Taxonomy	85
5.8 Chapter Summary	86
6. Study 2: Building an Instrument for Self-Perceived Justice of Teaching	89
Aim of the chapter	89
6.1 Focused Background: How Justice Has Been Measured	89
6.2 Version 1 (2019): Third-Person Descriptive Items, a Documentary Reconstruction	92
6.3 Version 2 (2020): Third-Person Normative Items	93

6.4 Version 3 (2021): The 36-Item First-Person Pool and the Published 26-Item EFA	94
6.5 Version 4 (2022): Consolidation on the Agreement Scale	97
6.6 Version 5 (2023): Rewording and the Scale Decision	98
6.7 The Final Instrument	100
6.8 Chapter Summary: Open Questions for Validation	103
7. Study 3: The Structure of Perceived Justice: Factor and Network Views	105
Aim of the chapter	105
7.1 Focused Background: Two Ways to Think About a Construct	105
7.2 Sample and Data	107
7.3 The Factor View	108
7.3.1 Exploration: the 2023 wave	108
7.3.2 Confirmation: competing models on the 2024 wave	109
7.3.3 Reliability at the latent level	111
7.3.4 Holdout: the 2026 wave, reported without cosmetics	111
7.3.5 Pooled estimates and sensitivity	112
7.4 The Network View	113
7.4.1 Estimation and stability	113
7.4.2 Centrality and predictability	114
7.4.3 Communities: the network's answer to the dimensionality question	114
7.4.4 The same network in all three cohorts	115
7.5 Convergence and Divergence of the Two Views	117
7.6 Is the Structure Contaminated by Social Desirability?	118
7.7 Chapter Summary	119
8. Study 4: Stability Across Cohorts and Instrument Versions	121
Aim of the chapter	121
8.1 What Can and Cannot Be Compared: The Three-Tier Framework Applied	121
8.2 The Identical-Instrument Tier: Invariance and Latent Means 2023–2024–2026	123
8.3 The Same-Items Tier: Structural Replication 2021 and 2022	125
8.4 The Different-Frame Tier: The 2020 Normative Cohort	126
8.5 Reliability Across Cohorts	128
8.6 Chapter Summary: A Structure That Travels, Scores That Do Not	129
Integration	131
9. Discussion	133
Aim of the chapter	133
9.1 The Arc of Findings: From a Hierarchy of Principles to a Stable Measured Structure	133
9.2 A Micro Lens on a Macro Debate: Confronting the Czech Equity Tradition	135
9.3 Theoretical Contribution	137
9.4 Limitations	138

Table of contents

9.5 Implications for Teacher Education	139
9.6 Future Research	139
Chapter summary	140
10. Conclusion	141
Aim of the chapter	141
Chapter summary	144
References	145
List of Abbreviations	155
Appendices	157
Appendix A: The Perceived-Justice Instrument (Final 26-Item Version)	157
A.1 Instructions and Response Scale	157
A.2 Items by Principle	158
A.2.1 Procedural justice (8 items)	158
A.2.2 Equality (6 items)	160
A.2.3 Needs (6 items)	161
A.2.4 Merit (6 items)	162
A.3 Version Crosswalk	164
Appendix B: The Six Villages Vignettes	165
B.1 Instruction and Response Scale	165
B.2 The Six Vignettes	166
B.2.1 Blue settlement: <i>equality</i> (v_rovnost)	166
B.2.2 Green village: <i>need</i> (v_potreby)	166
B.2.3 Orange town: <i>merit / desert</i> (v_zasluchy)	167
B.2.4 White settlement: <i>chance / lottery</i> (v_nahoda)	168
B.2.5 Red village: <i>equal opportunity</i> (v_rovne_prilezitosti)	168
B.2.6 Yellow town: <i>caste / social groups</i> (v_kasty)	169
B.3 Ranking Task	170
Appendix C: Supplementary Tables	171
C.1 Full Exploratory Factor Loadings (2023)	171
C.2 Pair-Level Redundancy Diagnostics	173
Appendix D: Ethics, Informed Consent, and Anonymization	175
D.1 Information and Consent Statement	175
D.1.1 As administered (Czech)	175
D.1.2 English translation	176
D.2 Ethics Review	177
D.3 Anonymization Protocol	178

Appendix E: A Conceptual-Similarity Map of Justice Concepts: Method	179
E.1 Design and concept set	179
E.2 Theory-defined feature representation	180
E.3 What the feature tree shows	182
E.4 Wording-based representation	182
E.5 Reproducibility and limits	184

Declaration

I declare that this thesis has been composed solely by myself using only the cited literature, information and resources and that it has not been submitted in any previous application for a degree.

In Brno on

.....

author's handwritten signature

Abstract

Educational research knows a great deal about justice as a property of school *systems* and much less about justice as a property of teachers' everyday *acts*. This monograph turns to the allocator's side of the classroom dyad and asks what those about to become teachers consider just, and whether their self-perceived justice can be measured well enough to study. It draws on two never-pooled data streams gathered from Czech pre-service teachers between 2019 and 2026. In the first (Study 1, $n = 315$), respondents judged six "villages," each distributing a windfall by a different principle: equality, need, merit, chance, equal opportunity, and caste; equality was judged most just and caste least, and the profile of judgments maps a recognizable moral hierarchy. The second stream documents the construction and validation of a first-person self-report instrument for perceived justice toward pupils, organized around four principles: procedural justice, equality, needs, and merit. Chapter 6 reconstructs the instrument's five-version lineage (2019–2023), including the shift from third-person to first-person items and from agreement to fairness response anchors. Chapter 7 tests its structure on the pooled identical-instrument cohorts ($n = 744$) with two convergent lenses (ordinal confirmatory and exploratory factor models and Gaussian graphical / exploratory graph analysis) and shows the four-principle organization to be recoverable under both and independent of socially desirable responding (BIDR). Chapter 8 examines stability across cohorts within an explicit three-tier comparability framework that assigns each cross-cohort claim only the strength its measurement affords: full invariance (under a continuous approximation) within the identical-instrument, identical-scale cohorts; structural congruence across the near-identical earlier cohorts; and configural comparison only where the item wording differs. A supplementary exploratory conceptual-similarity map (Appendix E) situates the

Abstract

four measured principles within a broader, literature-bounded concept set without testing a pre-given taxonomy. Throughout, the analysis is ordinal-first, fully scripted, and reproducible; sum scores are not recommended for individual diagnostics, and no claim is made that self-perceived justice equals enacted justice. The contribution is threefold: it extends organizational-justice theory into the teacher's allocative role, delivers a documented and reusable instrument with its full measurement lineage, and models a disciplined way to compare cohorts across an evolving instrument.

Keywords: perceived justice, distributive justice, pre-service teachers, self-report instrument, measurement invariance, network psychometrics, social desirability, vignettes.

Acknowledgements

This book rests on the trust of the many cohorts of pre-service teachers who answered its questionnaires and judged its six villages; without their willingness to reflect on their own sense of fairness, before practice had shaped it, there would be nothing to study here. I thank my colleagues at the Department of Education of the Faculty of Education, Masaryk University, for the intellectual home in which this work took shape, and the colleague who served as a second coder in the small conceptual-typology plausibility check (Appendix E) for their careful work. Responsibility for all remaining errors is mine alone.

1. Introduction

Aim of the chapter

This chapter opens the monograph with the Six Villages scene and motivates perceived justice as an everyday, micro-level act of teaching; states the aim of the book and its four research questions; introduces the research programme (two separate data streams collected across the cohort-years 2019–2026 from Czech pre-service teachers) and maps the structure of the book.

1.1 Six Villages: An Opening Scene

Imagine six villages, each with wealth to divide and the same question: how to divide it justly.

In the Blue Village, every citizen receives the same share: the elderly and the young, the industrious and the idle, those who need much and those who need nothing. Nobody asks who you are; the portion is the portion. In the White Village, the shares are unequal, but nobody chose them: the village draws lots, and fortune decides who prospers. Everyone had the same chance; the outcomes could hardly differ more. The Green Village asks one question only: *what do you need?* The sick receive more than the healthy, the family with five children more than the couple with none; those who need little get little. The Orange Village asks a different question: *what have you done?* Those who worked, contributed, served long: they receive more, and the village considers this only fair. In the Red Village, everyone starts

1. Introduction

at the same line, and then, everyone grabs what they can. The start was equal; the scramble is not, and nobody corrects its results. And in the Yellow Village, your share depends on your caste: the village has three castes, and within each caste all are equal; between castes, the shares differ.

Which of these villages is just? Before reading further, choose, and notice how quickly you chose, and how certain you feel. In the spring of 2019, three hundred and fifteen Czech students preparing to become teachers rated each of these six villages on a scale from *completely unjust* to *completely just*. Their answers (remarkably patterned, in one respect startling) open the empirical heart of this book (Chapter 5). (The villages are retold here freely; the exact texts the respondents rated are reproduced in Chapter 5 and Appendix B.) But the villages are not really about villages. Every teacher runs a small polity: there is never enough time, attention, praise, patience, or leniency for everyone, and each lesson quietly distributes them all. The village question (*which distribution is just?*) walks into the classroom every morning, wearing the teacher's clothes.

1.2 The Problem: Justice as an Everyday Act of Teaching

Educational research knows a great deal about justice as a property of *systems*: who gains access, who is tracked where, whose origins predict whose outcomes. It knows much less about justice as a property of *acts*: the everyday allocations a teacher performs, mostly without noticing that they are allocations at all. Who gets called on, and how often; who gets three extra minutes on a test, and on what grounds; whose disruption is punished and whose is overlooked; who is praised for effort and who for results; classroom life is dense with distributive and procedural decisions (Jackson, 1968), and the teacher performs them under time pressure, in public, with an audience of experts, because nobody monitors fairness as attentively as children. Pupils' reactions to these micro-allocations are well documented: perceived unfairness corrodes motivation, cooperation, and the relationship with the teacher (Chory-Assad & Paulsel, 2004; Resh & Sabbagh, 2016). The allocator's side of the dyad is

far less charted. What do those who are *about to become* teachers consider just? Which principles (equal shares, need, merit, fair procedure) organize their thinking before practice has shaped it? And can their perception of their own (anticipated) justice be measured well enough to study it?

These questions have a particular urgency in the Czech context, where a strong research tradition has documented the system-level inequities of education (Chapter 3) and a national strategy demands their reduction (MŠMT, 2020), while the classroom-level counterpart, the justice of the teacher as an everyday allocator (Lipsky, 1980), has remained an under-measured blind spot. Policies about equal opportunities will ultimately be enacted, or quietly re-interpreted, by people whose own intuitions about just distribution predate the policy. This book is about those intuitions, in the population that matters most and is easiest to reach before practice begins: pre-service teachers.

1.3 Aim of the Book and Research Questions

The aim of the book is twofold: to describe which distributive principles Czech pre-service teachers consider just, and to build and validate an instrument measuring the self-perceived justice of their (anticipated) teaching. The aim unfolds into four research questions, each answered by one empirical study:

- RQ1 (Study 1, Chapter 5): Which principles of distribution do pre-service teachers judge just, and in what hierarchy? Answered with the six-villages vignette task (2019, $n = 315$).
- RQ2 (Study 2, Chapter 6): Can self-perceived justice toward pupils be operationalized as a four-principle self-report instrument? Answered by the documented development of the questionnaire through five versions (2019–2023).
- RQ3 (Study 3, Chapter 7): Do the factor-analytic and network views of the final instrument converge on the same four-principle structure (procedural justice, equality,

1. Introduction

needs, merit) and is that structure independent of social desirability? Answered on the pooled identical-instrument cohorts (2023, 2024, 2026; $n = 744$).

- RQ4 (Study 4, Chapter 8): Is the structure stable across cohorts and instrument versions? Answered under a three-tier comparability framework that assigns every cross-cohort comparison exactly the strength its shared measurement affords.

1.4 One Research Programme, Two Streams, Seven Cohort-Years (2019–2026)

The evidence comes from one research programme with two strictly separated streams. Stream 2, the vignettes, was collected once, in 2019 (315 respondents). Stream 1, the questionnaire, was collected in six waves between 2020 and 2026, totaling 1,815 responses, while its instrument matured through the five versions that Chapter 6 documents. Table 1.1 gives the programme at a glance. Two design commitments, argued in full in Chapter 4, are worth registering from the start: the streams are never pooled (no individual-level link exists between them), and the cohorts are repeated independent cross-sections, not a panel; this book never claims to measure change within persons, only the recurrence of structure across samples.

Table 1.1. The research programme at a glance (details and collection windows in Chapter 4)

Cohort-year	Stream	n	Instrument	Response scale
2019	vignettes	315	six villages	fairness 1–7
2019	questionnaire	(wording only)	v1: 21 items, 3rd person descriptive	

Cohort-year	Stream	n	Instrument	Response scale
2020	questionnaire	375	v2: 23 items, 3rd person normative	agreement 1–7
2021	questionnaire	433 (346 published)	v3: 36-item pool → 26 items, 1st person	agreement 1–7
2022	questionnaire	263	v4: 26 items, 1st person	agreement 1–7
2023	questionnaire	247	v5 (final): 26 items, 1st person	fairness 1–7
2024	questionnaire	333	v5 (final)	fairness 1–7
2026	questionnaire	164	v5 (final)	fairness 1–7

Note. Source: data/processed/DATA_README.md. There was no 2025 collection. The bold entry marks the agreement→fairness scale change, the programme’s deepest comparability boundary (Chapters 4.4, 8.1).

1.5 Structure of the Book

The book proceeds in three movements. Foundations (Chapters 2–4): Chapter 2 builds the theoretical frame, from philosophical taxonomies of justice through organizational justice to the classroom, and narrows the wide space of justice types to the four principles this book measures, including an explicit map of what it deliberately does not measure, and an empirical check on that map (Section 2.7, with the method in Appendix E). Chapter 3 reads

1. Introduction

the Czech research tradition and locates the micro-level gap this programme fills. Chapter 4 fixes the methodological rules (two streams, the three-tier comparability framework, the ordinal-first analytic strategy, and the transparency infrastructure) before any result is shown. Evidence (Chapters 5–8): four studies answer the four research questions in sequence: the villages hierarchy (5), the instrument’s construction (6), its structure under two convergent lenses and a social-desirability audit (7), and its tiered stability across cohorts (8). Integration (Chapters 9–10): the discussion synthesizes the arc, confronts the Czech equity tradition, states limitations candidly, and derives implications for teacher education; the conclusion distills the book’s claims. Appendices provide the final instrument in both languages with its full version crosswalk, the vignette texts, supplementary tables, and the ethics and anonymization documentation.

1.6 Note on Prior Publications, Data and Code Availability

Only two early outputs of this research programme have been published, and both concern its 2021 stage: the exploratory factor analysis of the questionnaire’s item pool, released openly as data and analysis code (Nehyba, 2021a), and a short report to the Czech educational research community in the ČAPV conference volume on justice in education (Nehyba, 2021b). These are the author’s only publications on perceived justice; the remainder of the author’s scholarly output, some ninety works listed in the accompanying curriculum vitae, addresses unrelated topics and is not drawn upon here. This monograph does not reuse even the two 2021 outputs as they stand: the 2021 analysis is independently replicated here from the archived data within a reproducible pipeline (Chapter 6.4), and everything beyond it (the vignette study, the final instrument’s validation, the network analyses, and the cross-cohort stability programme) rests on a body of data assembled specifically for this book and is published here for the first time. The monograph is therefore an original scholarly work written for this habilitation; it is not identical with, and does not reproduce, any dissertation or other qualification thesis previously submitted by the author for an academic degree. All analyses are scripted; every number in the text traces to a versioned dataset and a frozen number man-

ifest (Chapter 4.6). The anonymized datasets, preparation scripts, analysis notebooks, and rendered reports are openly available in the [companion repository](#)[↗] (Nehyba, 2026b).

Chapter summary

Six villages, one question (*which distribution is just?*), and a population, future teachers, who will answer it in practice every working day of their lives. The book asks what they answer now, before practice: in judgment (RQ1), in measurable self-perception (RQ2–RQ3), and in the recurrence of that structure across cohorts and instrument versions, as far as measurement permits comparison (RQ4). Chapter 2 begins where every justice question must: with what, exactly, justice means.

Foundations

2. Theoretical Foundations: Justice from Philosophy to the Classroom

Aim of the chapter

This chapter builds the conceptual scaffolding of the book: it traces justice from its contested philosophical roots through organizational and classroom justice research to the two taxonomies that anchor the empirical streams, the wide six-principle taxonomy behind the vignette study and the four-principle framework (procedural, equality, needs, merit) that the self-report instrument operationalizes, and it defines pre-service teachers as the population whose perceived justice the following chapters measure.

2.1 Justice and Fairness: A Contested Concept

Few concepts carry as much normative weight, and as little definitional consensus, as justice. Western philosophy has argued about it continuously since Plato asked in the *Republic* whether justice is anything more than the advantage of the stronger, and the argument shows no sign of resolution. Gallie (1956) offered a useful diagnosis of why: justice belongs to the class of *essentially contested concepts*: concepts whose proper application is, by their very nature, the object of endless and rationally irresolvable dispute. On this reading, the disagreement about justice is not a temporary embarrassment awaiting a better theory; it is constitutive of the concept itself. Different users of the word attend to different features of

2. Theoretical Foundations: Justice from Philosophy to the Classroom

the same practice (the shares people receive, the process that produced them, the standing of the people involved) and each usage can be defended with reasons that the other side cannot simply dismiss. For an empirical research programme, this diagnosis has a practical consequence. A study of justice cannot begin from a settled definition handed down by philosophy, because no such definition exists. It must instead map how the concept fragments, choose a deliberate slice of that fragmented terrain, and defend the choice. That is the work of this chapter.

The fragmentation has structure, and philosophers have repeatedly tried to chart it. The oldest chart is Aristotle's: in Book V of the *Nicomachean Ethics* he separates justice as the whole of virtue in relation to others from justice as a particular virtue, and within the latter distinguishes distributive justice, the proportional division of honors and goods among members of a community, from corrective (rectificatory) justice, which restores a balance disturbed by a wrongful transaction (Aristotle, trans. 2009). A modern analytic descendant of this charting is found in Ernst Tugendhat's lectures on ethics (Tugendhat, 1993). Tugendhat divides the question of justice in two. The first question is the justice of norm *application*: whatever rules are in force, are they applied impartially, treating like cases alike, without favoritism or arbitrary exception? The second question is the justice of the *norms themselves*: can the rules by which goods and burdens are distributed be justified equally to everyone subject to them? A grading scheme, to take an example this book will dwell on, can be administered with scrupulous impartiality and still rest on a distributive rule that is itself contestable, and conversely, a defensible rule can be applied capriciously. Tugendhat's division is philosophical, but the reader will recognize in it the exact silhouette of the distinction that empirical justice research would draw independently: the distinction between procedural and distributive justice, developed in Section 2.2.

Even within the second, distributive question, no single principle commands the field, and no one has made this point more vividly than Deborah Stone. In *Policy Paradox and Political Reason*, Stone (1988) stages the problem with a deliberately homely example: a chocolate cake to be divided among the students of a public policy seminar. Equal slices for every student is the obvious answer, until someone proposes larger shares for those who did the best

work, or division by rank, or parity between groups rather than individuals, or slices proportional to hunger, or a lottery, or simply setting the cake in the middle of the room and letting the quickest forks win. Stone's point is not that one of these answers is right and the others confused. Her point is the paradox that gives her book its title: *every one of these distributions can be described as equality*, equality of slices, of reward per unit of merit, of treatment within rank, of shares between groups, of satisfaction relative to need, of chances, of opportunity to compete. The same distribution is simultaneously just and unjust depending on which dimension (recipients, the good itself, or the process of allocation) one chooses to equalize. Distributions therefore cannot be scored as just or unjust in the abstract; the judgment is always relative to a principle, and the principles are plural, context-bound, and in genuine competition with one another. Walzer (1983) generalizes the lesson: different social goods inhabit different "spheres" with different distributive logics, and much perceived injustice consists in a principle native to one sphere colonizing another. Stone's cake will return in this book twice: once as the direct inspiration for the six-villages vignettes of Chapter 5, in which the same scarce good is distributed under six different principles, and once as the reason the classroom, where one person continually divides scarce goods among thirty claimants, is such a natural site for justice research.

If philosophy cannot settle which principle is correct, empirical social science can ask a different and answerable question: which principles do *people* treat as just, in which contexts, and with what consequences? This descriptive turn, from justice as a normative ideal to justice as a perception, is the founding move of the research tradition this book works in. The turn requires terminological care, however, because everyday English (and everyday research practice) uses *justice* and *fairness* almost interchangeably. Following Colquitt and Rodell (2015), this book distinguishes them. *Justice*, in their usage, refers to perceived adherence to normative rules: the judgment that an authority did or did not follow specific, nameable standards of decision-making, distribution, and treatment. *Fairness*, by contrast, is a global evaluative judgment: an overall, holistic appraisal of the appropriateness of one's treatment, formed downstream of (and not reducible to) the specific rule-adherence perceptions. The distinction is not pedantry; it has an empirical basis. Ambrose and Schminke

2. Theoretical Foundations: Justice from Philosophy to the Classroom

(2009) showed that such overall fairness judgments mediate the effects of specific justice perceptions on outcomes, and Rodell and Colquitt (2009) demonstrated that global, even anticipatory, fairness judgments behave as a construct in their own right. The working terminology of this book follows directly. *Perceived justice*, the book's central construct, means perceptions of adherence to identifiable distributive and procedural principles; when the text speaks of a *fairness* judgment, it means the global evaluation. One bilingual remark is in order, since this monograph is maintained in English and Czech in parallel: Czech collapses both terms into the single word *spravedlnost*, and the book's glossary deliberately maps *justice* and *fairness* onto that one Czech term. The conceptual distinction must therefore be carried by context in the Czech mirror, a small but instructive illustration of how language itself partitions the contested concept differently. Fittingly, the response scale of the final instrument asks respondents to judge concrete teaching behaviors on a scale anchored from "completely unjust" to "completely just": it asks for justice judgments about specific rule-following behaviors, not for a single global fairness verdict.

2.2 Organizational Justice and Its Rules

The descriptive study of justice perceptions found its most productive home in an unglamorous setting: the workplace. Greenberg (1987) gave the field its name and its first map, coining the term *organizational justice* and organizing the existing theories into a taxonomy along two dimensions: whether a theory is reactive or proactive, and whether it concerns the content of outcomes or the process that generates them. A decade later, Cropanzano and Greenberg (1997) could already describe researchers as "tunneling through the maze" of a large and rapidly professionalizing literature. The maze had grown in identifiable waves, and each wave contributed a family of what the field came to call *justice rules*: the criteria people demonstrably use when they judge whether an allocation, a procedure, or an interaction was just (Colquitt & Rodell, 2015).

The first wave was distributive. Adams (1965), building on the social-exchange tradition,

proposed equity theory: people judge outcomes by comparing their own ratio of outcomes to inputs with the ratio of a referent other, and experience distress when the ratios diverge, in either direction. Equity, in this technical sense, is the merit principle in psychological dress: outcomes should be proportional to contributions. Subsequent work established that equity is one allocation rule among several rather than the whole of distributive judgment; Leventhal (1976) documented equality and need as rules that allocators demonstrably apply alongside contribution, a pluralism to which Section 2.3 returns in detail.

The second wave was procedural, and it began in the courtroom rather than the workplace. Thibaut and Walker (1975), studying dispute-resolution procedures, found that disputants' satisfaction depended not only on verdicts but on the structure of the process, above all on *process control*, the opportunity to voice one's case and have it heard before a decision is reached. What entered the literature as "voice" turned out to be one of the most robust findings in justice research: people accept unfavorable outcomes far more readily when the procedure gave them a hearing. Leventhal (1980) then carried procedural justice out of the courtroom and into allocation generally, proposing six rules by which procedures are judged: *consistency* across persons and time, *bias suppression*, *accuracy* of the information used, *correctability* of wrong decisions, *representativeness* of the interests of those affected, and *ethicality*: compatibility with prevailing moral standards. Leventhal's list has proved remarkably durable; the reader will meet several of its rules again, translated into teaching behaviors, in Section 2.5.

The third wave concerned the human encounter in which procedures are enacted. Bies and Moag (1986) argued that people evaluate not only the formal procedure but the quality of interpersonal treatment they receive during its enactment: they judge whether they were treated with *respect* and *propriety*, and whether the communication they received was *truthful* and whether decisions were adequately *justified*. Greenberg (1993) proposed splitting this interactional family in two: an *interpersonal* facet (respect, propriety) and an *informational* facet (truthfulness, justification), a split that Colquitt's (2001) construct-validation study supported empirically, yielding the four-factor structure (distributive, procedural, interpersonal, informational) that has since served as the field's default measurement model.

2. Theoretical Foundations: Justice from Philosophy to the Classroom

Colquitt and Rodell (2015) codify the whole apparatus as a table of justice rules; Table 2.1 reproduces its logic.

Table 2.1. Justice rules by family (adapted from Colquitt & Rodell, 2015).

Family	Rule	The question the rule answers	Primary sources
Procedural	Process control / voice	Could those affected express their views before the decision?	Thibaut & Walker (1975)
Procedural	Consistency	Are procedures applied uniformly across persons and time?	Leventhal (1980)
Procedural	Bias suppression	Is the decision-maker neutral, without self-interest or prejudice?	Leventhal (1980)
Procedural	Accuracy	Is the decision based on accurate, complete information?	Leventhal (1980)
Procedural	Correctability	Can flawed decisions be appealed and corrected?	Leventhal (1980)
Procedural	Representativeness	Are the concerns of all affected parties reflected?	Leventhal (1980)

2.2 Organizational Justice and Its Rules

Family	Rule	The question the rule answers	Primary sources
Procedural	Ethicality	Does the procedure conform to prevailing moral standards?	Leventhal (1980)
Distributive	Equity (merit)	Are outcomes proportional to contributions?	Adams (1965)
Distributive	Equality	Do all recipients receive the same?	Leventhal (1976)
Distributive	Need	Do outcomes track recipients' needs?	Deutsch (1975); Leventhal (1976)
Interpersonal	Respect	Was the person treated with dignity and courtesy?	Bies & Moag (1986); Greenberg (1993)
Interpersonal	Propriety	Were improper remarks and prejudicial statements avoided?	Bies & Moag (1986); Greenberg (1993)
Informational	Truthfulness	Was communication candid and honest?	Bies & Moag (1986); Tyler & Bies (1990)
Informational	Justification	Were decisions explained adequately and in time?	Bies & Moag (1986); Tyler & Bies (1990)

The dimensionality of this apparatus, however, has never been fully settled, and one seam in particular matters for this book: the boundary between the procedural and interactional

2. Theoretical Foundations: Justice from Philosophy to the Classroom

families. Bies and Moag (1986) introduced interactional justice precisely as the fairness of treatment *during the enactment of procedures*, and an influential line of argument has held that it is best understood as the interpersonal context of procedural justice rather than a separate construct: Tyler and Bies (1990) titled their statement of this view “beyond formal procedures” and treated respectful, well-justified enactment as procedural justice’s informal face. Outside organizational psychology proper, the same consolidation appears; McCollough et al. (2000), for example, operationalize interactional treatment within the procedural family in their study of service recovery. This book adopts that consolidating position, and the decision is documented here as a deliberate design choice rather than an oversight. Three considerations motivate it. First, and conceptually decisive: the agent-based rationale for separating interactional from procedural justice, that formal procedures belong to the organization while their enactment belongs to a particular manager, loses its force in the classroom, where the teacher is simultaneously the maker of the rules, their applier, and their communicator. When one person designs the grading procedure, applies it, and explains it face-to-face, the seam between “the procedure” and “how the procedure was enacted toward me” has no organizational counterpart to attach to. Second, parsimony: a self-report instrument for pre-service teachers, answered by respondents who have not yet run classrooms of their own, cannot sustain fine-grained distinctions between the respect facet and the justification facet of their imagined conduct without multiplying items beyond what the format bears. Third, the consolidated reading has the precedent just cited. The consequence, visible in the instrument of Chapter 6, is that the book’s *procedural* principle is a broad family: it includes transparency of criteria, advance announcement of rules, voice and co-creation, and the invitation to contest results, items that in Bies and Moag’s (1986) vocabulary would be partly informational (justification, truthfulness) but that function here as facets of fair process in the teacher’s hands.

Why build a book about teachers on organizational justice at all? Because schools are organizations, classrooms are allocation systems embedded in them, and the justice rules of Table 2.1 were formulated at exactly the level of abstraction that travels: they attach to any authority who distributes valued goods among dependents through repeated, visible pro-

cedures. Organizational justice supplies this book with its framework: the vocabulary of principles and rules, the distinction between distribution and procedure, and a half-century of evidence that perceptions organized by these rules predict trust, legitimacy, and cooperation. Section 2.4 documents that the transfer of this framework to schools is not this book's invention; before that, Section 2.3 examines the distributive rules more closely, because they carry the heaviest load in the chapters to come.

2.3 Principles of Distributive Justice: The Wide and the Narrow Taxonomy

Within the distributive family, the canonical taxonomy is a triad. Deutsch (1975) argued that equity, equality, and need are the three fundamental values of distributive justice and, crucially, that which one governs a given allocation depends on the goal of the social relation: where economic productivity is primary, equity (proportionality to contribution) dominates; where the maintenance of enjoyable social relations is primary, equality dominates; where personal welfare and development are primary, need dominates. Leventhal (1976) reached a compatible conclusion from the allocator's side, showing that people distributing rewards demonstrably mix the contribution rule with equality and need rules depending on context and anticipated consequences. Schwinger (1980) systematized the triad as three allocation norms (the contribution (merit) norm, the equality norm, and the need norm) and linked each to a type of relationship in which it is the socially expected default. The triad is the load-bearing core of this book's framework: equality, needs, and merit are three of its four principles, and Deutsch's functional argument will echo loudly when the classroom enters the picture, because the classroom is precisely a setting in which productivity goals, relational goals, and welfare goals coexist and compete, often within a single lesson.

Canonical, however, does not mean exhaustive. Stone's (1988) cake seminar already put lotteries, ranks, and open competition on the table alongside the triad, and her analysis of the *process* dimension of equity (allocation by lottery, by competition, by election) identifies

2. Theoretical Foundations: Justice from Philosophy to the Classroom

distributive principles that the workplace triad ignores. Elster (1992) documented the same plurality in institutional practice: his study of *local justice* (how real institutions actually allocate scarce goods and necessary burdens, from kidneys to military service to kindergarten places) catalogs lotteries, queues, need criteria, merit points, and status categories operating side by side, largely untroubled by philosophical tidiness. Political philosophy contributes a further principle with enormous public resonance: equal opportunity, the rule that positions and rewards should be open to all under fair conditions of competition, canonized in Rawls's (1971) principle of fair equality of opportunity. And history contributes its dark counterpart: allocation by caste or status group, equality within strata combined with categorical inequality between them, a principle almost universally rejected in modern normative theory yet perfectly recognizable as a description of how many societies have in fact distributed goods (Walzer, 1983, discusses caste as the limiting case of a society in which all spheres collapse into one dominant rank order).

This wider space of principles is exactly what the book's first empirical stream explores. The vignette study of Chapter 5 presents respondents with six villages, each distributing the same scarce good under a different principle, and asks them to judge each distribution on a justice scale. The six principles form the *wide taxonomy*: equality (the Blue Village: an identical share for every citizen, regardless of anything), need (the Green Village: shares proportioned to each citizen's needs), merit (the Orange Village: more for those who worked, contributed, or served longest), chance (the White Village: a lottery in which every citizen has an equal ticket and luck decides), equal opportunity (the Red Village: an identical starting line, after which each grabs what they can), and caste (the Yellow Village: goods divided between three hereditary strata, with equality only within each stratum). The two non-triad, non-caste principles deserve a word of theoretical placement. Chance is not a frivolous entry: lottery allocation has a serious pedigree in the local-justice literature as the purest implementation of equal treatment when goods are indivisible (Elster, 1992), and in Stone's (1988) terms it equalizes the *process*, everyone's chances, while guaranteeing nothing about outcomes. Equal opportunity, likewise, is a process principle wearing distributive clothes: it equalizes starting conditions and then lets competition allocate, which is why Stone can present it

2.3 Principles of Distributive Justice: The Wide and the Narrow Taxonomy

as one more meaning of equality and why its empirical reception among Czech pre-service teachers, one of the more striking results of Chapter 5, repays theoretical attention. Table 2.2 aligns the wide taxonomy with the narrow one.

Table 2.2. The wide taxonomy (vignettes) and the narrow taxonomy (questionnaire).

Principle	Village (Chapter 5)	In the questionnaire?	Reason for inclusion/exclusion in the narrow set
Equality	Blue	Yes	Everyday teaching referents abound (time, attention, criteria)
Need	Green	Yes	Everyday teaching referents abound (illness, hardship, difficulty)
Merit	Orange	Yes	Everyday teaching referents abound (effort, performance, contribution)
Chance / lottery	White	No	No legitimate first-person referent for allocating grades or care by lot
Equal opportunity	Red	No	In the classroom, decomposes into procedural transparency plus equal treatment

2. Theoretical Foundations: Justice from Philosophy to the Classroom

Principle	Village (Chapter 5)	In the questionnaire?	Reason for inclusion/exclusion in the narrow set
Caste / status groups	Yellow	No	Normatively proscribed; a self-report item would measure only social desirability
Procedural	(not a vignette)	Yes	The vignette task is purely distributive; procedure enters with the teacher's role

The book's second stream, the self-report instrument developed in Chapter 6, operationalizes a *narrow taxonomy*: procedural justice plus the distributive triad of equality, needs, and merit. The narrowing is deliberate, and its rationale is worth stating carefully because it explains why the two streams are kept methodologically separate throughout the book. Vignettes and self-report items do different epistemic work. A vignette asks the respondent to judge a third-party social arrangement from the outside; nothing in the task requires the respondent to imagine enacting the principle, and so even morally marginal or proscribed principles can be presented and rated. One can meaningfully ask whether the Yellow Village's caste allocation is just; the answer (a resounding no, as Chapter 5 shows) is informative precisely because the principle could be put before respondents at all. A self-report instrument, by contrast, asks respondents how just specific behaviors of their own (prospective) teaching are, and this format imposes three constraints that shrink the taxonomy. First, every item needs a plausible first-person behavioral referent in the teacher's role. There is no defensible everyday teaching behavior that instantiates caste allocation, and grading by lottery has no legitimate referent either; the nearest classroom use of chance (randomly se-

lecting who is called upon) is, on inspection, a *procedural* device for equalizing chances and suppressing bias, not a distributive lottery of outcomes. Second, psychometrics: a measurable factor requires multiple, non-redundant items with genuine variance. Items describing proscribed principles would floor out under social desirability pressure, contributing nothing but restriction of range, a concern the book takes seriously enough to test directly, with a social desirability measure, in Chapter 7. Third, the streams answer different questions: the vignettes map the *preference space* of lay distributive intuitions (RQ1), for which width is a virtue; the instrument measures *self-perceived justice of teaching* (RQ2–RQ4), for which validity within the teacher’s role is the criterion, and width beyond that role is noise. Equal opportunity illustrates the collapse: translated into teaching behaviors, “an equal starting line for every pupil” decomposes into transparent, uniformly applied procedures and equal treatment, that is, into the procedural and equality principles the instrument already contains. The bridge from six principles to four is walked in full, with the vignette results in hand, in Section 5.7.

2.4 Classroom Justice: The Teacher as Allocator

The transfer of justice research into education did not have to be argued from scratch; by the early 2000s it had a name and a small but coherent literature. In the instructional-communication tradition, Chory-Assad and Paulsel (2004) defined *classroom justice* as “perceptions of fairness regarding outcomes or processes that occur in the instructional context” (p. 254) and imported the organizational taxonomy wholesale: distributive justice of grades and other instructional outcomes, procedural justice of the rules by which they are produced, and interactional justice of the instructor’s treatment of students. Their empirical results gave the construct immediate practical bite: perceived classroom injustice predicted student aggression and resistance toward the instructor, and related work in the same programme linked justice perceptions to motivation and learning (Chory-Assad, 2002). Whatever else a classroom is, it is a place where justice judgments are being made continuously, and where their consequences are behavioral, not merely attitudinal.

2. Theoretical Foundations: Justice from Philosophy to the Classroom

A second, European line reached compatible conclusions from social-psychological and developmental starting points. Berti et al. (2010), studying Italian secondary classrooms, mapped students' and teachers' representations of classroom justice and connected them to students' psychological engagement with school, finding the organizational families (distributive, procedural, interactional) recognizable in how both parties spontaneously reason about classroom life. The justice-motive tradition assembled in Dalbert and Sallay (2004) approached the classroom through the belief in a just world and its developmental consequences, and is of particular methodological relevance here: work in that tradition examined the distributive justice of grading directly, including vignette-based pilot studies in which respondents judged the justice of grading decisions under systematically varied circumstances, a direct methodological ancestor of this book's vignette stream, and one to which Chapter 5 returns. From the administration side, Hoy and Tarter (2004) carried organizational justice into schools as workplaces, distilling the literature into ten principles of organizational justice for school administration and binding the construct to its nearest neighbor with a phrase this book happily borrows: there is *no justice without trust*. Justice perceptions, in schools as in firms, are not an ornament on top of otherwise functioning relationships; they are among the conditions of those relationships (see also Resh & Sabbagh, 2016, for a synthesis of justice in teaching as the distribution of grades, attention, and praise).

What makes the classroom such a dense site for justice, however, is best seen not through any single study but through a reframing that this book adopts as its organizing image: *the teacher as allocator*. Jackson (1968), in his classic observational account of classroom life, noted that pupils live in conditions of crowds, praise, and power: thirty claimants, one adult, and a chronic scarcity of everything that matters. Under scarcity, every teacher action is an allocation. The teacher allocates grades, the classroom's most formalized good; but also time: who gets called on, how long an oral examination lasts, who receives an extension; attention: whose raised hand is seen, whose problem is pursued; help: who gets the additional explanation, the adapted homework, the second chance; and opportunity: who represents the class, who runs the activity, whose effort is publicly recognized. In Elster's (1992) terms, the classroom is a system of local justice in miniature: an institution allocating scarce goods

and necessary burdens under principles that are only partly codified, applied by a single gatekeeper, in full view of all claimants. Three features of this allocation system deserve emphasis. It is *dense*: allocation decisions occur many times per lesson, not annually as in salary reviews. It is *public*: nearly every allocation is witnessed by the whole class, so consistency, Leventhal's (1980) first procedural rule, is continuously auditable by the recipients themselves, who are famously vigilant comparators. And it is *discretionary*: however detailed the school rules, the teacher retains wide latitude in distributing time, attention, and recognition, which is precisely the space in which distributive principles, rather than regulations, govern conduct.

Why do pupils' justice experiences matter? The classroom-justice literature supports a conservative three-part answer. Perceived justice is associated with engagement: students who perceive their classroom as just report higher psychological engagement with school (Berti et al., 2010). It underwrites legitimacy: the procedural-justice tradition has shown, most systematically in Tyler's (1990) work on legal authorities, that people who experience fair procedures are more likely to view authority as legitimate and to comply voluntarily rather than under surveillance, a mechanism with obvious classroom analogues in the findings on resistance and aggression toward instructors perceived as unjust (Chory-Assad & Paulsel, 2004). And it touches well-being: the justice-motive tradition links experiences of justice in school to adolescents' adjustment and their developing belief in a just world (Dalbert & Sallay, 2004). This book does not test these downstream consequences and claims nothing about them beyond what the cited literature reports. But they explain the stakes, and they motivate the book's own angle of attack. If the teacher is the allocator, then the justice of classrooms runs through the teacher's own principles: which distributions the teacher considers just, and how just the teacher perceives their own allocative conduct to be. It is that upstream variable, the (future) teacher's perceived justice, rather than the pupils' perceptions, that the remainder of this book sets out to conceptualize and measure.

2.5 The Four-Principle Framework of This Book

The preceding sections assembled the parts; this section states the framework. The book conceptualizes the perceived justice of teaching along four principles, one procedural and three distributive, each defined in the register of the teacher's role and each anchored in the traditions reviewed above. Together they constitute the a priori structure that the instrument of Chapter 6 operationalizes and that the factor-analytic and network analyses of Chapters 7 and 8 put to the test.

Procedural justice is the fairness of the processes by which the teacher decides, evaluates, and manages the classroom: transparency of criteria, advance announcement of rules, consistency of application, voice, and correctability. In the teacher's register, procedural justice looks like clearly establishing the rules of classroom life at the start of the year, or announcing in advance the criteria by which written work will be graded; it also looks like telling pupils they may question and discuss the marking of a test at any time, and giving them room to co-create the classroom's rules of behavior. The lineage is direct: voice and process control from Thibaut and Walker (1975); consistency, accuracy, and correctability from Leventhal (1980); and, per the design decision documented in Section 2.2, the justificatory and communicative facets that Bies and Moag (1986) called interactional, folded here into the procedural family because in the classroom the maker, applier, and explainer of procedures are one person.

Equality is the distributive principle of identical treatment: every pupil receives the same share of the teacher's divisible goods, regardless of performance, circumstance, or status. Enacted, it looks like devoting the same amount of time to each pupil during oral examinations, giving all pupils (and all pupils' problems) the same degree of attention, applying identical criteria when judging every pupil's behavior, and calling on all pupils with the same regularity. It is Leventhal's (1976) equality rule and the Blue Village's principle, transposed to goods a teacher actually holds: time, attention, evaluative criteria, chances to speak.

Needs is the distributive principle of responsiveness to individual circumstance: shares track

what each pupil requires, not what an identical division would give. In the teacher's register, it looks like taking a difficult life situation (their parents' divorce, say) into account when grading; giving a pupil who was ill extra time to catch up on the material; and assigning different homework to a pupil with a long-term difficulty in understanding the subject. It is Deutsch's (1975) welfare-oriented rule and the Green Village's principle, and it stands in visible, deliberate tension with equality: the teacher who gives more care to the pupil who needs it most is, by that very act, not giving every pupil the same.

Merit is the distributive principle of proportionality to contribution: recognition and reward track effort and performance. Enacted, it looks like giving due recognition to a pupil whose effort goes beyond the assigned task, praising a pupil who successfully represents the school in a knowledge competition, and rewarding pupils who finish their classwork ahead of time. It is Adams's (1965) equity rule, Schwinger's (1980) contribution norm, and the Orange Village's principle, narrowed, in the classroom, mostly to symbolic goods: praise, recognition, small privileges.

Several features of the framework should be made explicit. First, the example behaviors above are not decorative: each paraphrases an item of the final instrument, whose full wording in both languages appears in Appendix A. The instrument comprises 26 items (eight procedural and six each for equality, needs, and merit) spanning the recurring contexts of the teacher's allocative work: grading, oral examination, classroom behavior and rules, homework, and the recognition of pupil performance. Respondents judge each behavior on a seven-point scale from "completely unjust" to "completely just" . Second, the framework is *pluralist by construction*. Following Deutsch (1975) and Stone (1988), it does not treat the principles as mutually exclusive commitments of which each person holds exactly one; a teacher may sincerely endorse equal attention for all and extra care for the neediest, and the interesting question is how these endorsements are structured: whether they cohere into the four hypothesized dimensions, how the dimensions relate, and which items bridge them. That question is precisely what Chapter 7 answers twice, once with factor-analytic and once with network models, and what Chapter 8 examines for stability across cohorts. Third, the framework is *a priori* in the strict sense: it was fixed on theoretical grounds, the

organizational-justice rules of Section 2.2 and the distributive triad of Section 2.3, before the structural analyses were run, so that those analyses constitute a genuine test rather than a redescription.

2.6 What This Book Excludes and Why

A construct is defined as much by what it excludes as by what it contains, and three neighboring justice families are deliberately outside this book's framework. Naming them now, with reasons, guards the construct against a foreseeable misreading: that an instrument for the perceived justice of teaching ought to have covered everything a teacher does that could be called just or unjust. Figure 2.1 is a didactic simplification of that argument: it marks the book's measured perimeter and its nearest excluded areas, not a complete taxonomy of justice.

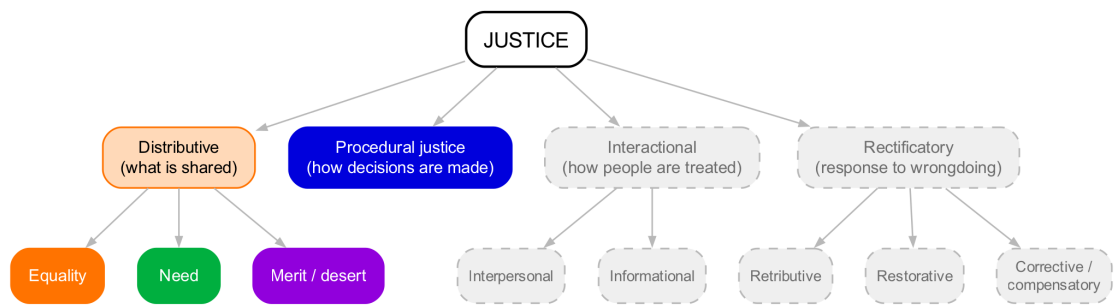


Figure 2.1. A didactic simplification and delineation of the book's measured perimeter. The diagram is neither a complete taxonomy nor a hypothesis tested by the dendrogram in Figure 2.2. Source: [02_typologie.dot diagram](#)[↗]. The arrangement of the rectificatory branch is one defensible ordering among several; the boundary between corrective, retributive, and restorative justice is itself contested.

The first exclusion is *retributive justice*: the justice of punishment, centrally the proportionality of sanctions to wrongdoing. Classrooms unquestionably contain a retributive dimension (discipline, sanctions, detentions), and the justice of punishment has its own venerable literature. It is excluded here because the book's object is the everyday allocation of goods and

the fairness of routine procedure, not the response to transgression. The instrument's items describe grading, attention, time, recognition, and rules; where discipline appears at all, it appears procedurally, as in informing pupils in advance how conflicts in the class will be handled, which concerns the transparency of process, not the justice of any sanction. Measuring retributive judgments properly would require a different item logic (transgression scenarios, sanction magnitudes) and a different theoretical apparatus.

The second exclusion is *restorative justice*: the paradigm that responds to harm not with proportional punishment but with the repair of relationships, typically through structured encounters between those who caused and those who suffered harm. Its school-based variant is well developed; Karp and Breslin (2001) describe restorative practices in school communities as an alternative to zero-tolerance discipline, and it is arguably the most influential justice vocabulary in contemporary educational practice. It is excluded for a categorical reason: restorative justice is primarily a *practice paradigm*, a family of interventions and processes, rather than a perceptual dimension along which everyday allocative behavior varies. A self-report instrument of the present kind has no natural place for it; studying it well means studying programs and encounters, not rating routine behaviors.

The third exclusion is *corrective justice* in the classical, Aristotelian sense: rectificatory justice, which restores a balance disturbed by a wrongful transaction between parties (Aristotle, trans. 2009). Corrective justice was discussed during the framework's development and excluded on the grounds that it is remedial by definition (it presupposes a wrong to be undone), whereas the book's construct is the justice of *ordinary, non-remedial* allocation and procedure: the daily division of time, attention, grades, and recognition when nothing has yet gone wrong. Folding remedial judgments into the same self-report format would blur exactly the boundary the construct definition works to keep sharp.

These exclusions are documented author decisions, made when the construct was specified, and they have an honest price: the instrument developed in Chapter 6 does not measure the justice of discipline, punishment, or repair, and no finding of this book should be read as speaking to them. The corresponding limitation is acknowledged in Chapter 9, where the

2. Theoretical Foundations: Justice from Philosophy to the Classroom

retributive and restorative families reappear on the agenda for future research. Within its declared perimeter, however, the construct is the cleaner for the exclusions: perceived justice of teaching, in this book, means the perceived adherence of everyday teaching conduct to the four principles of Section 2.5, no less, and deliberately no more.

2.7 An Exploratory Conceptual-Similarity Map

The didactic diagram in Figure 2.1 is not translated here into a tested hypothesis. This section offers an exploratory similarity map of 24 concepts with no pre-assigned concept families. The map is not theory-free: the concept set, working definitions, and 16 coded features are authorial, theory-conditioned choices. The same concepts are represented in two complementary ways: first by theory-defined features scored by model coders, and then by *text embeddings*, numeric representations of the concept names and definitions designed so that similarly worded passages lie closer together. Because both routes share the same concepts and working definitions, their agreement is only a sensitivity check on representation choice, not an independent replication or proof of a uniquely correct taxonomy. Appendix E documents the full method and its limits.

Three separately run model coders from different providers scored the concepts on 16 features. Agreement was uneven rather than uniformly high: nine features met the reporting threshold and seven did not, so the analysis includes reliable-only and reliability-weighted sensitivity variants. A small five-concept human pilot provided a plausibility check, not a ground-truth validation. Grouping the consensus feature scores yields the dendrogram in Figure 2.2. Concepts joined farther to the right are more similar; the vertical label order itself has no substantive meaning.

2.7 An Exploratory Conceptual-Similarity Map

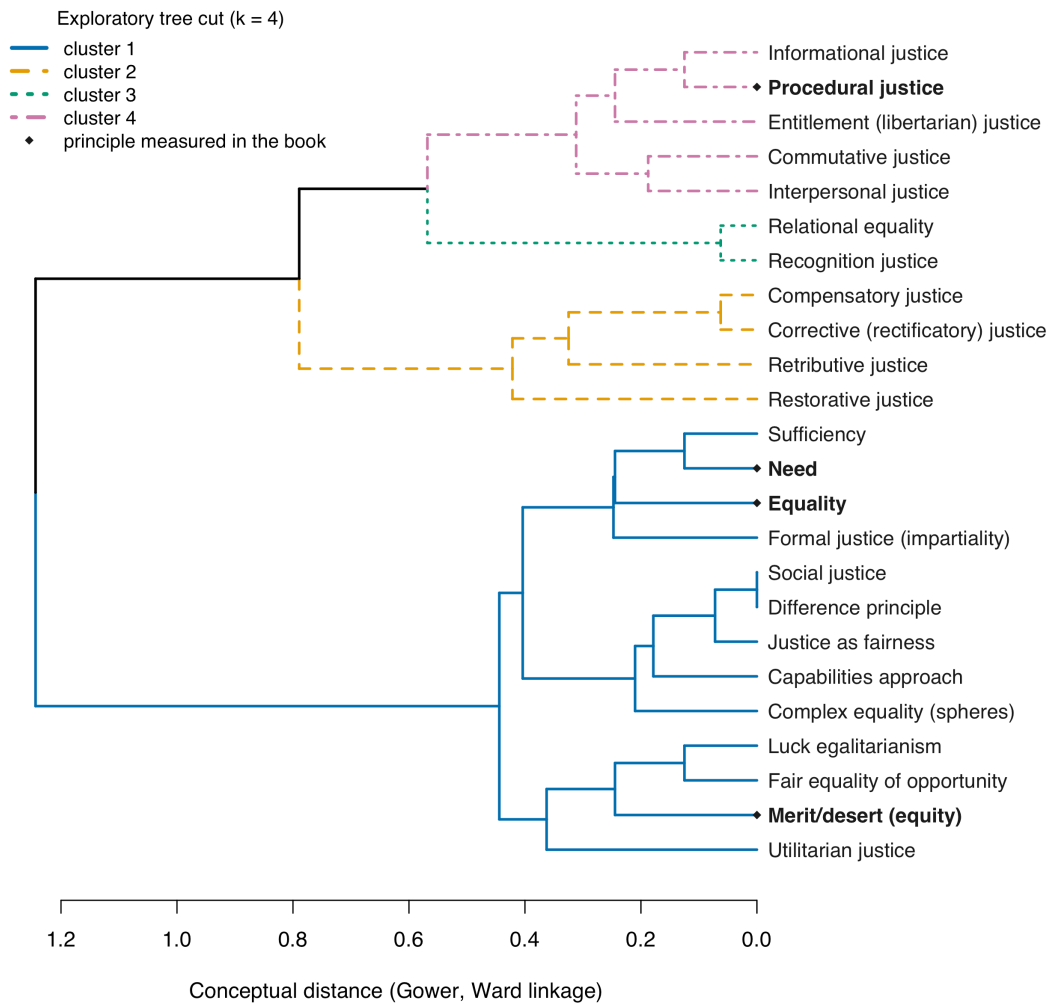


Figure 2.2. Exploratory conceptual-similarity dendrogram of 24 justice concepts based on consensus scores for 16 features. Branch colors identify four groups obtained by cutting the computed tree; black diamonds and bold labels mark the four principles measured in the book. A simple average-silhouette diagnostic slightly preferred four groups, but with only 24 concepts this is not a validated typology. The result is conditional on the selected concepts, definitions, features, and Ward procedure. Method and limitations in Appendix E. Source: [fig_E5_trees.R script](#); [E5_cluster_diagnostics.csv diagnostics](#).

For this book, it is descriptively useful to note where its four measured principles land. Procedural justice lies close to informational and interpersonal justice. Equality and need are

2. Theoretical Foundations: Justice from Philosophy to the Classroom

immediate neighbours beside sufficiency; merit lies in the same broad coloured branch but near fair equality of opportunity and luck egalitarianism. These are locations in this particular map, not a test of the instrument's four-principle structure. That test belongs to the item data in Chapters 7 and 8.

Average silhouette reached .395 for four groups and .381 for three; the .014 difference is small and, with $n = 24$, does not identify a “true” number of types. The complementary wording-based map partly agrees on neighbourhoods but differs substantially in detailed branching (Appendix E). The appropriate conclusion is therefore procedural: the dendrogram transparently displays similarity conditional on the selected concepts, definitions, features, and method. It confirms neither Figure 2.1 nor any pre-given typology.

2.8 Pre-Service Teachers as the Population of Interest

The final element of the scaffolding is the choice of population. Every empirical chapter of this book studies *pre-service teachers*, university students preparing for the teaching profession, rather than practicing teachers or pupils. The choice is substantive, not merely convenient, and it rests on four arguments.

The first argument is formability. Beliefs about teaching are not blank at entry into teacher education; Lortie (1975) famously observed that prospective teachers arrive having already served a years-long “apprenticeship of observation” as pupils, from which they carry vivid, first-personal templates of what teachers do, including, one may add, thousands of hours of watching teachers allocate attention, grades, and praise, and of feeling those allocations from the receiving end. Such early-formed beliefs are robust and act as filters on later professional learning (Pajares, 1992). Teacher education is the last structured, institutional opportunity to surface and examine these dispositions before they harden into practice. If one wants to know what justice conceptions future teachers will bring to their classrooms, and whether teacher education could work with them, the time to measure is *before* practice begins. A

baseline taken during preparation is also the precondition for any later study of how practice reshapes those conceptions.

The second argument concerns what self-reports of this population can and cannot mean, and the book prefers to state it plainly rather than discover it in the limitations section. Argyris and Schön (1974) distinguished *espoused theories*, the beliefs people profess when asked, from *theories-in-use*, the beliefs actually governing their conduct, and the two are notoriously imperfectly aligned. Pre-service teachers, most of whom do not yet have classrooms of their own, can only espouse: the instrument's first-person items are anticipatory self-characterizations of a professional self under construction. The book measures espoused justice and says so. This is a limitation for predicting classroom behavior (no finding here licenses an inference about what these respondents will actually do as teachers, a boundary Chapter 9 restates), but it is arguably an advantage for the book's central task of instrument construction and structural analysis: espoused principles are exactly where justice conceptions live before institutional constraints, workload, and school cultures begin to compress them, and their structure can be studied in a population whose members share a common situation rather than thirty different school contexts.

The third argument is contextual and practical. In Czechia, teaching is a Master's-level profession by law (Act No. 563/2004 Coll.), and teacher preparation is correspondingly long, organized as five-year integrated Master's programs or as a Bachelor's degree followed by a Master's, delivered largely at faculties of education. This long runway makes pre-service teachers an unusually stable and reachable population for a multi-cohort research programme: successive year-groups pass through the same institution, in the same phase of professional formation, and can be sampled under near-identical conditions year after year. That is precisely the design of this book's questionnaire stream: seven questionnaire cohort-years across 2019 to 2026, of which six (2020–2026) yielded questionnaire data, the 2019 instrument surviving only as documented wording. Each data-bearing wave is an independent cohort of pre-service teachers at a single faculty of education: repeated cross-sections, emphatically not a panel, a distinction with analytic consequences that Chapter 4 spells out and Chapter 8 respects throughout. The concentration on one faculty

2. Theoretical Foundations: Justice from Philosophy to the Classroom

buys comparability across cohorts at the price of generalizability across institutions; the price is acknowledged and paid openly in Chapters 4 and 9.

The fourth argument closes the circle with Section 2.4. If the teacher is an allocator, then today's pre-service teachers are tomorrow's allocators, and their justice dispositions are pedagogically consequential in a way that few other student attitudes are: they will be enacted daily, publicly, and upon minors, across careers measured in decades. Understanding which distributive principles this population finds just (RQ1), whether their self-perceived justice can be measured reliably along the four principles (RQ2), what structure those perceptions form (RQ3), and whether that structure holds across cohorts and instrument versions (RQ4) is therefore not only a psychometric exercise; it is groundwork for teacher education, whose implications Chapter 9 draws. One more piece of scaffolding is needed before the empirical work can begin, however. Justice in Czech educational research has overwhelmingly meant something else: equity of access and structural inequality at the level of the system. Chapter 3 maps that macro tradition and locates, within it, the micro-level gap at the scale of teacher–pupil interaction that this book is built to fill.

Chapter summary

This chapter assembled the book's conceptual framework in four moves. It began with justice as an essentially contested concept, adopting Tugendhat's division between the justice of norms and of their application and Stone's demonstration that the same distribution is defensible under multiple principles, and it fixed the book's terminology after Colquitt and Rodell: justice as perceived adherence to normative rules, fairness as the global evaluative judgment. It then took organizational justice as the working framework: distributive, procedural, and (by documented design decision) interactional-as-procedural justice rules, and unfolded the distributive principles into a wide taxonomy of six, suited to vignettes, and a narrow taxonomy of four, suited to a self-report instrument. Classroom justice research licensed the transfer of this apparatus to schools and yielded the book's organizing image of

the teacher as allocator of grades, time, attention, and recognition. The result is the four-principle framework (procedural, equality, needs, merit) defined in the teacher's register, delimited against retributive, restorative, and corrective justice, and aimed at pre-service teachers as the population whose espoused justice conceptions are still forming. Chapter 3 now situates this micro-level lens within the macro-oriented Czech tradition of educational justice research.

3. Justice in Czech Educational Research: The Macro Tradition and the Micro Gap

Aim of the chapter

This chapter situates the book within Czech educational research on justice, showing that the strong domestic tradition of studying equity at the macro level of access and inequality has largely left unexamined the micro level of justice as perceived in teacher–pupil interaction. It thereby defines the gap that the book’s four empirical studies address, as a complement to, not a rejection of, the macro tradition.

3.1 Equity of Access and Educational Inequality: The Dominant Czech Paradigm

When Czech educational research has spoken of justice over the past two decades, it has overwhelmingly meant equity: the fair distribution of educational opportunities and outcomes across social categories. This is not a marginal preference of a few research groups but the constitutive paradigm of the domestic field. It took shape after 1989, when Czech sociology of education reconnected with international stratification research and began to ask, with newly available data, how family background translates into educational careers in a system that had long described itself as egalitarian. A consolidating moment came with the volume *(Ne)rovné šance na vzdělání* [(Un)equal chances in education], which brought

3. Justice in Czech Educational Research: The Macro Tradition and the Micro Gap

together the sociological evidence on educational inequality in the Czech Republic and set the agenda for a generation of researchers (Matějů & Straková, 2006). From that point on, the question “Is Czech education just?” was, in domestic research practice, effectively synonymous with the question “How unequal are educational opportunities, and along which social lines?”

Within pedagogy proper, the conceptual work of translating the international equity discourse into Czech terms is associated above all with David Greger. His theoretical reflection on educational inequalities introduced to a Czech readership the successive conceptions of equality of educational opportunity developed in the international literature (from formal equality of access, through equality of treatment, to equality of outcomes) and argued that a defensible notion of *spravedlivost* (equity, fairness) in education must be anchored in these distinctions rather than in intuition (Greger, 2006). Two features of this framing deserve emphasis, because they define what the paradigm can and cannot see. First, justice is predicated of the *system*: it is the education system, its structures and its policies that are just or unjust, not the individual encounter between a teacher and a pupil. Second, justice is *measured structurally*: through indicators of selectivity, tracking, socioeconomic gradients of achievement and attainment, and between-school variance. Both features are analytic strengths (they make claims about justice empirically decidable at scale) and both, as this chapter will argue, come at a price.

The empirical backbone of the paradigm was supplied to a large extent by Jana Straková’s analyses of international large-scale assessment data. Drawing on studies in the tradition of PISA and its predecessors, this line of work argued that the architecture of the Czech education system, its early selection of pupils into tracks of differing prestige and its comparatively differentiated school network, contributes to the development of educational inequalities, with pupils’ results and trajectories tied to family background to a degree that the research community judged troubling (Straková, 2007). The argumentative pattern of these studies is characteristic of the whole tradition: an internationally comparative measurement instrument, a structural feature of the system as the explanatory variable, and social background as the axis along which injustice is diagnosed.

3.1 Equity of Access and Educational Inequality: The Dominant Czech Paradigm

No institution has focused this debate more sharply than the *víceletá gymnázia*, the multi-year academic gymnasia that admit pupils out of basic school as early as after the fifth grade. For critics, this early branching is the paradigm case of unjust selectivity: a mechanism that sorts children at an age when measured aptitude is still heavily entangled with family resources, drains motivated pupils and engaged parents from mainstream basic schools, and converts social advantage into institutional advantage. Defenders have countered with arguments about excellence and parental choice. Czech researchers entered this debate empirically, asking among other things whether the gymnasia add educational value over and above the pupils they select, with analyses suggesting that a substantial part of their apparent advantage reflects intake rather than instruction (Straková, 2010). The details of that exchange matter less for present purposes than its form: the *víceletá gymnázia* controversy is a controversy about justice conducted entirely in the currency of access, selection, and aggregate outcomes. The tradition subsequently extended the same lens downward to the earliest stages of schooling, examining unequal starting conditions in preschool education and at the transition to basic school (Greger et al., 2015).

The macro paradigm has not remained an academic affair; it has been written into the language of the state. The *Strategy for the Education Policy of the Czech Republic up to 2030+* names the reduction of inequalities in access to quality education, and the development of the full potential of every pupil, as one of its two strategic goals (MŠMT, 2020). This is a remarkable success of the research tradition: concepts forged in equity research (inequality of opportunity, selectivity, dependence of results on family background, regional disparity) became the official vocabulary in which Czech education policy describes its own ambitions. It also means that any new research programme on justice in Czech education must position itself with respect to this tradition, because the tradition now defines not only the scholarly but also the political meaning of the word.

It is worth stating precisely what this paradigm accomplishes and where its purview ends. Its unit of analysis is the system, the school type, or the social category; its instruments are administrative datasets, international assessments, and sociological surveys; its operationalization of justice runs through structural indicators; and its normative benchmark is

3. Justice in Czech Educational Research: The Macro Tradition and the Micro Gap

the equalization of educational opportunity across groups defined by circumstances of birth. Within these bounds it has been cumulative, internationally comparable, and consequential for policy, a genuinely successful research programme. But by construction it does not observe the site at which schooling is actually experienced: the classroom interaction in which a teacher distributes grades, attention, help, praise, and sanctions among particular pupils. In the macro paradigm the teacher appears, if at all, as an aggregate variable, and the pupil as a member of a category. Whether the daily conduct of teaching is fair, and whether it is *perceived* as fair by those subject to it, are questions the paradigm neither asks nor is equipped to answer. That such questions were nonetheless beginning to stir in the field is documented by the stocktaking volume to which we now turn.

3.2 Taking Stock: The ČAPV 2021 Volume

In September 2021, the Czech Educational Research Association (ČAPV) devoted its twenty-ninth annual conference, hosted in Brno by the Department of Educational Sciences of the Faculty of Arts, Masaryk University, to the theme of justice in education. The resulting proceedings, *Na cestě ke spravedlnosti ve vzdělávání: pedagogický výzkum pro lepší praxi a politiku* [On the road to justice in education: Educational research for better practice and policy], edited by Roman Švaříček and Hana Voňková, offer something rare: a census of what an entire national research community counts as research on justice at a given moment (Švaříček & Voňková, 2021). For the argument of this book, the volume is a natural reference point, and this section reads it as such, as the Czech field taking stock of itself.

The editorial framing is instructive in itself. The title's journey metaphor, *on the road to justice*, presents justice as a destination that research helps the system approach, and the subtitle binds that journey explicitly to practice and policy. The animating concern is the reduction of unjust inequality: not every difference between pupils or schools is treated as an injustice, but those inequalities that flow from circumstances rather than from pupils' own efforts are marked as the proper target of research and reform. The distinction has

a recognizably Rawlsian tone; it echoes the idea that inequalities require justification and are tolerable only under conditions that a fair system could endorse (Rawls, 1971), and it aligns the volume with the policy horizon of Strategy 2030+ discussed above (MŠMT, 2020). In its official self-presentation, then, the Czech field in 2021 understood justice research primarily as equity research in the service of better policy: a faithful continuation of the macro tradition.

The body of the volume largely confirms this self-understanding. Contributions include work on monitoring educational inequalities through administrative and geographic data on pupil mobility between basic schools, and on the predictive validity of the state *maturita* (school-leaving) examination and university admission procedures, justice as a property of data infrastructures and selection instruments (Švaříček & Voňková, 2021). Most tellingly, an entire section pairs the COVID-19 pandemic with educational inequality and distance schooling, reading the pandemic through the lens the tradition had prepared: as an amplifier of unequal access, to technology, to parental support, to instruction itself. Even a historical contribution on the politics of egalitarianism in the everyday life of secondary vocational schools in socialist Czechoslovakia during the normalization era extends the distributive frame backward in time, showing how a state ideology of equality was translated, and distorted, in school practice (Švaříček & Voňková, 2021).

Yet the same volume also documents a field beginning to probe beneath the system level, and these openings matter for the present book. The most explicit is the symposium “Lze hovořit o spravedlnosti? Neúspěch u maturitní zkoušky očima maturantů” [Can we speak of justice? Failure in the *maturita* examination through the eyes of the examinees], whose papers examine how repeatedly unsuccessful examinees perceive their failure, to whom they attribute it, what barriers they see, and how just the Czech secondary school system appears *from their vantage point* (Švaříček & Voňková, 2021). Here justice is no longer an attribute of structures read off aggregate statistics; it is a perception held by particular persons about an institutional procedure that decided their fate, precisely the epistemic shift this book advocates. A second opening is a contribution on teachers’ productive handling of pupils’ errors as an expression of a fair approach to pupils (“Produktivní práce učitele s chybou jako

3. Justice in Czech Educational Research: The Macro Tradition and the Micro Gap

projev spravedlivého přístupu k žákům”), which locates fairness in a concrete, recurrent micro-practice of instruction (Švaříček & Voňková, 2021). A third is a theoretical study of teacher authority and its legitimacy, a topic that, as Section 3.3 argues, stands conceptually adjacent to procedural justice. And the volume’s symposium on the ethnography of diversity in pre-service teacher education attends to how future teachers learn to deal fairly with heterogeneous pupils, bringing the justice question into teacher preparation itself (Švaříček & Voňková, 2021).

For the sake of transparency it should be noted that the present author’s own contribution appears in the volume’s section on pre-service teachers: an early report on the development of the self-report instrument for perceived justice whose full documentary reconstruction and validation occupy Chapters 6 through 8 of this book (Nehyba, 2021b).

Read as a whole, the ČAPV 2021 volume thus yields a double finding. On the one hand, it confirms the dominance of the macro-equity paradigm: the field’s center of gravity lies in inequality of access, selection instruments, and policy. On the other hand, it reveals unmistakable openings toward perception and interaction: the examinee’s experience of an unjust exam, the teacher’s fair handling of an error, the legitimacy of authority, fairness in the preparation of future teachers. What the volume does not yet contain is a framework that would bind these openings together: they share no common conceptual vocabulary (the organizational-justice tradition introduced in Chapter 2 is absent as an integrating frame), no measurement programme, and no defined population of study. They are openings, not yet a research line. The next section argues that the raw materials for such a line were nonetheless already present in Czech research.

3.3 Micro-Level Threads Already Present

The claim of this book is not that Czech educational research has ignored everything below the system level. It is that several existing threads touch the micro level of justice without naming it as such, and that they remain unconsolidated. Four threads deserve explicit

acknowledgment, because the research programme presented in this book builds on the domestic ground they prepared.

The first thread concerns the moral experience of teachers. Jiří Mareš introduced to Czech pedagogy the concept of moral distress, originally developed in nursing ethics to describe the suffering of professionals who know what the morally right course of action is but are prevented by institutional constraints from taking it (Jameton, 1984), and argued for its relevance to teachers, mapping the situations in which Czech educators experience such distress (Mareš, 2017). Moral distress is not perceived justice: its focus is the professional's anguish, not the fairness of allocations to pupils. But the two constructs are close kin. Many of the situations that generate moral distress in teaching (being pressed to grade against one's judgment, being unable to give a struggling pupil the support he needs, watching institutional rules produce outcomes one considers wrong) are, in the vocabulary of this book, situations of *thwarted justice*: conflicts between the distributive principles a teacher endorses and the allocations she is able to enact. Mareš's work thus demonstrates two things at once: that the moral and ethical experience of individual Czech teachers is empirically tractable, and that the domestic field possessed, by 2017, a published warrant for studying justice-relevant phenomena at the level of the person rather than the system.

The second thread is the literature on teacher authority and its legitimacy, a long-standing topic of Czech pedagogical discussion (Vališová, 1999) that resurfaced, as noted above, in the ČAPV 2021 volume in the form of a theoretical study of authority and legitimacy (Švaříček & Voňková, 2021). The connection to justice is direct, though rarely drawn explicitly in the Czech debate: the social-psychological research tradition has shown that authorities are granted legitimacy, and voluntarily obeyed, largely to the extent that they are seen to exercise their power through fair procedures (Tyler, 2006). Translated into the classroom, this means that a teacher's authority rests in substantial part on pupils' perception that grades, attention, and sanctions are allocated by fair rules fairly applied. The Czech authority literature thus circles the phenomenon of procedural justice in teacher–pupil interaction without operationalizing it; it names the stakes (legitimacy, discipline, the erosion of authority) while leaving the underlying justice perceptions unmeasured.

3. Justice in Czech Educational Research: The Macro Tradition and the Micro Gap

The third thread is methodological, and for this book it is the most consequential: Hana Voňková's programme of work with anchoring vignettes. The anchoring-vignette method, developed in political science and survey methodology, confronts respondents with short standardized scenarios and uses their ratings of these fixed cases to correct for heterogeneity in how different people use response scales (King et al., 2004). Voňková and colleagues domesticated the method for Czech educational research, applying it among other things to the comparability of pupils' self-assessments (Voňková & Hrabák, 2015). It is telling that one of the editors of the field's justice volume is also its leading vignette methodologist. For the present book, this thread matters in two ways. Practically, it established in the Czech field the legitimacy of vignettes (short, controlled, hypothetical scenarios rated by respondents) as serious measurement instruments. The six-villages task at the heart of Study 1 (Chapter 5) belongs to this broad family: its vignettes are not *anchoring* vignettes, since their purpose is to elicit pre-service teachers' normative judgments of six distributive principles rather than to recalibrate self-report scales, but the underlying logic (standardized scenario, controlled variation, numerical judgment) is shared. Conceptually, the anchoring-vignette thread marks a future direction to which this book returns in Chapter 9: anchoring versions of the justice instrument could one day separate genuine differences in self-perceived justice from differences in scale use, closing the loop between Voňková's methodological programme and the substantive one begun here.

The fourth thread runs through research on assessment. Grades are the most visible distribution in school life, and Czech research has repeatedly approached their fairness: at the system pole, through analyses of the matura examination and admission procedures (Švaříček & Voňková, 2021), and at the classroom pole, through work on formative assessment and on teachers' handling of pupils' errors as an enactment of a fair approach (Švaříček & Voňková, 2021). Assessment is where the abstract principles of distributive justice (equality, needs, merit) and the demands of procedural justice become concrete for every pupil several times a week. Yet in the Czech literature these strands are framed as psychometrics or as didactics, not as justice research: the exam studies ask whether instruments predict, and the formative-assessment studies ask whether practices teach, while the question of

3.4 The Gap and the Claim: Perceived Justice at the Level of Teacher–Pupil Interaction

how the allocation is *perceived* and by what *principles* it is judged remains implicit.

Taken together, these four threads show that the micro level was never empty in Czech research. But they also show why no research line formed. The threads are scattered across pedeutology, general pedagogy, survey methodology, and didactics; they speak different vocabularies (distress, authority, comparability, assessment) with no shared construct of perceived justice; none of them offers a measurement instrument for justice in teacher–pupil interaction; and none defines a population in which the phenomenon could be studied cumulatively across years. They are anchoring points, not a foundation. What is missing is exactly what this book undertakes to supply.

3.4 The Gap and the Claim: Perceived Justice at the Level of Teacher–Pupil Interaction

Consider what the macro tradition's statistics cannot, even in principle, register. Imagine a Czech education system after a successful Strategy 2030+: tracking softened, socioeconomic gradients flattened, regional disparities narrowed. Even in that system, justice would still have to be *done*: every day, in every classroom, by a teacher deciding whom to call on, how to grade an ambiguous performance, how much time to give a struggling pupil at the expense of the rest, whether to bend a rule for a hard case, how to respond to an error. As Chapter 2 argued, the teacher is an allocator: of attention, help, praise, sanctions, and marks. Two pupils identical on every structural indicator the macro tradition measures (the same school type, the same family background, the same region) can experience profoundly different justice in these daily allocations. Aggregate indicators of access and outcome average over precisely the interactions in which pupils actually encounter fairness and unfairness. The macro paradigm does not fail at this task; it simply does not have it. Its blind spot is constitutive, not accidental.

That blind spot would matter little if perceived justice were epiphenomenal. It is not. The organizational-justice tradition (Greenberg, 1987; Colquitt, 2001) and the classroom-justice

3. Justice in Czech Educational Research: The Macro Tradition and the Micro Gap

research built on it have accumulated evidence that perceptions of fairness carry psychological weight of their own: pupils and students who perceive their treatment as unjust respond with diminished motivation and engagement, with resistance and even aggression toward the instructor, while perceived fairness supports trust and the legitimacy of the teacher's authority (Berti et al., 2010; Chory-Assad & Paulsel, 2004; Tyler, 2006). Justice that is not perceived does not do its psychological work; injustice that is perceived does its damage regardless of what the aggregate statistics say. A national research field that measures justice only structurally is therefore measuring a necessary condition of educational justice, not the whole of it.

The claim of this book must be stated with equal care in both directions. This book does *not* argue that the macro tradition is mistaken, exhausted, or in need of replacement; on the contrary, Chapter 9 reads this book's own findings against that tradition's results, and the confrontation is only possible because the tradition exists. Nor could micro-level fairness compensate for structural injustice: the most procedurally impeccable teacher cannot undo, from inside a classroom, an inequitable school structure around it. The claim is one of complementarity. The macro tradition asks whether the *system* distributes educational opportunities justly across social categories; this book asks how the *people who will operate that system* judge the principles by which goods should be distributed, and how their own justice toward pupils can be conceived and measured. These are two layers of one problem. Education policy of the kind envisioned by Strategy 2030+ is not executed by strategies; it is executed by teachers, the street-level bureaucrats of educational equity (Lipsky, 1980), whose situated judgments determine what any policy of justice becomes in practice (MŠMT, 2020). A field that wants justice in education needs to understand both the structure and its enactors.

The book's four research questions map onto this gap systematically, each supplying a layer the macro tradition presupposes but does not examine. RQ1 asks *which* principles of distribution pre-service teachers consider just, and in what hierarchy, using the six-villages vignettes to elicit normative judgments of six distributive principles (Chapter 5). This addresses the missing preference layer: macro research evaluates the system against equity

3.4 The Gap and the Claim: Perceived Justice at the Level of Teacher–Pupil Interaction

criteria fixed by researchers and policymakers, but no Czech study has asked which distributive principles future teachers themselves endorse. RQ2 asks whether self-perceived justice toward pupils can be operationalized as a reliable self-report instrument built on four principles (procedural, equality, needs, and merit), reconstructing the instrument's development across its versions from 2019 to 2023 (Chapter 6). This addresses the missing measurement layer: without an instrument, perceived justice in teacher–pupil interaction cannot enter Czech research as a variable at all. RQ3 asks whether the factor-analytic and network views of the instrument's structure converge, and whether the structure is independent of social desirability (Chapter 7), the construct-validity layer, without which any measurement claim would be premature. And RQ4 asks whether the structure is stable across cohorts and instrument versions from 2020 to 2026 (Chapter 8), the durability layer, testing whether the book describes a stable feature of the measured construct across independent cohorts rather than an artifact of one cohort or one questionnaire version.

Why pre-service teachers? The choice of population is strategic, not opportunistic, on four grounds. First, leverage: teacher education is the one point at which the system can systematically shape its future allocators before practice begins, and each cohort of graduates carries its justice conceptions into classrooms for decades; if there is a lever by which the Czech system can act on the micro layer of justice, it is teacher education. Second, formability: pre-service teachers stand at a biographically open moment, recent enough to remember being on the receiving end of teachers' allocations, yet already rehearsing the allocator's role; their conceptions of justice are still available for reflection and revision in ways that veteran practice tends to foreclose. Third, applicability: instruments developed on this population, both the vignettes and the self-report questionnaire, can serve directly as reflective tools within teacher-education curricula, a practical implication developed in Chapter 9. Fourth, feasibility: pre-service teachers can be reached in successive annual cohorts, enabling the design on which the book's claims to stability rest. The obvious cost (samples drawn from a single faculty of education, with the limits to generalization that entails) is acknowledged here and weighed in Chapters 4 and 9.

The gap, then, is defined, and the response can be announced. What follows in Chapter

3. Justice in Czech Educational Research: The Macro Tradition and the Micro Gap

4 is the design of one research programme comprising two strictly separated streams (a self-report questionnaire of self-perceived justice built on the four principles, and the six-villages vignette task) administered across seven cohorts of Czech pre-service teachers between 2019 and 2026, yielding 1,815 questionnaire responses plus 315 vignette responses (response counts, not necessarily unique persons). The cohorts are repeated cross-sections of independent respondents, not a panel, and the two streams are never pooled; what may be compared across which cohorts is governed by an explicit three-tier comparability framework. With that design, the micro layer of justice in Czech education (perceived, principled, and measured) ceases to be a gap and becomes an object of study.

One feature of that response is methodologically distinctive in the Czech setting. To establish the structure of self-perceived justice, the programme uses network psychometrics, which represents a construct as a system of partial-correlation relations among its items rather than as the reflection of a latent cause. This toolkit remains uncommon in Czech educational research, though several applications now exist: exploratory graph analysis was used to examine dimensionality during validation of a physical-education motivation scale among Czech high-school students (Harbichová et al., 2025); a Ministry of Education pilot analysis of beginning teachers' and graduates' perceived preparedness estimated regularized item networks and used community detection and bootstrap stability checks (MŠMT, 2023); and a teacher-research study used a psychometric network to examine student-perceived inclusive teaching (Bula & Nehyba, 2025). Published network-model applications appeared earlier in Czech psychology (e.g., Klocek et al., 2022). Network psychometrics should not be conflated with the better-established Czech tradition of social network analysis, which maps relations among persons rather than among items, as in the analysis of pupils' classroom-interaction ties by Lintner et al. (2024).

Chapter summary

Czech educational research has built a robust and policy-consequential macro tradition that operationalizes justice as equity of access and educational inequality, from the foundational stratification and tracking studies through the *víceletá gymnázia* debate to Strategy 2030+, which made reducing inequality one of the state's two strategic goals (MŠMT, 2020). The ČAPV 2021 volume documents both this dominance and the field's first openings toward perception and interaction, and scattered domestic threads (moral distress, teacher authority and legitimacy, anchoring-vignette methodology, assessment fairness) already touch the micro level without consolidating it. What is missing is a systematic study of perceived justice in teacher–pupil interaction: which distributive principles future teachers endorse, and how their self-perceived justice can be measured. The book fills this gap as a complement to the macro tradition, and the research programme designed to do so (two streams, seven cohorts, four research questions) is presented in Chapter 4.

4. Research Design and Methodology

Aim of the chapter

This chapter lays out the research programme that carries the book's four research questions: two strictly separated data streams, a self-report questionnaire and a vignette task, collected across independent cohorts of Czech pre-service teachers between 2019 and 2026. It specifies the samples and their limits, the evolution of the instruments, the three-tier comparability framework that governs every cross-cohort claim in Chapter 8, the analytic strategy and software, and the transparency and ethics safeguards on which the empirical chapters (5–8) rest. The chapter's function is regulative: it fixes the rules of comparison *before* any result is shown, so that the reader can check each later claim against a standard announced in advance.

4.1 Overall Design: Two Streams, Seven Cohorts, Four Research Questions

The programme consists of two empirical streams that share a population of origin but not identified persons, and its first design rule follows from that fact: the streams are never pooled. Stream #2, the six-villages vignette task (Chapter 5), was collected in spring 2019 from 315 respondents. Stream #1, the self-perceived justice questionnaire whose construction Chapter 6 traces, was administered in six data-bearing cohort waves between 2020 and 2026 (the 2019 version exists as item wording only), totaling 1,815 questionnaire responses

4. Research Design and Methodology

. No individual-level identifier links a vignette rater to a questionnaire respondent, or a respondent in one wave to a respondent in another; the programme is a series of repeated cross-sections of independent samples, and every analysis in this book treats it as such. Because participation was anonymous, occasional repeated participation by the same student in different years cannot be ruled out, which is one more reason the counts above are reported as *responses*, not unique persons.

Table 4.1 maps the four studies onto the research questions, cohorts, and analyses.

Table 4.1. Master design table: studies × research questions × data × analysis

Study (chapter)	RQ	Data	n	Instrument / scale	Core analysis
1: Six Villages (5)	RQ1	vignettes, 2019	315	6 villages, 1–7 fairness	Friedman + Kendall’s W; post hoc Wilcoxon; CLMM; exploratory clustering
2: Building the Instrument (6)	RQ2	versions 2019–2023	375 / 433 / 263 (+ final waves)	21→23→36→26 items; agreement → fairness	documentary reconstruc- tion; replicated 2021 EFA
3: Structure (7)	RQ3	pooled 2023+2024+2026	744	26 items, fairness 1–7	ordinal CFA/ESEM/bifactor; EBICglasso + EGA; BIDR

Study (chapter)	RQ	Data	n	Instrument / scale	Core analysis
4: Stability (8)	RQ4	all question- naire cohorts	per tier	per tier	invariance ladder; congruence + EGA; ω trajectory

Note. Cohort inventory with collection windows in data/processed/DATA_README.md. Stream #2 (vignettes) is analyzed only in Study 1; streams are never pooled.

4.2 Samples and Data Collection 2019–2026

All cohorts are convenience samples of full-time pre-service teachers at a Czech faculty of education, recruited in connection with teacher-preparation coursework; no probability sampling was used, and the book’s claims are calibrated accordingly: they concern structure and within-sample relations, not population prevalence (the limitation is confronted, not footnoted, in Chapter 9). The vignette task was administered on paper as the opening activity of a self-experiential seminar (Section 5.3.1), with the villages story printed for every participant. The questionnaire waves were administered electronically via online forms; per-cohort sample sizes and collection windows are documented in the data inventory: 2020 ($n = 375$; February–August 2020), 2021 ($n = 433$ raw pool, of which 346 formed the published analytic subset; February 2021–March 2022), 2022 ($n = 263$; February–May 2022), 2023 ($n = 247$; February–April 2023), 2024 ($n = 333$; February–April 2024), and 2026 ($n = 164$; February–April 2026). There was no 2025 collection. The 2026 window is documented as closed from the collection dates in the data; any later collection would constitute a new cohort, not an extension.

4. Research Design and Methodology

Two sampling facts deserve emphasis because later chapters lean on them. First, demographic description is deliberately thin throughout: the vignette stream carries demographics for about two-thirds of the 315 respondents (sex 204, age 206; Section 5.3.2), and the questionnaire’s demographic block (gender, birth year, study subjects, teaching experience) is available in the 2023–2026 waves. Second, one cohort was excluded by design: a small 2020 sample of combined-mode (part-time) students was dropped before analysis as a different population, so that all analyzed cohorts share the full-time pre-service population.

4.3 Instruments and Their Evolution

Stream #2 uses a single fixed instrument: six parallel vignettes, each describing a village that distributes goods by one principle of the wide taxonomy of Chapter 2 (equality, chance, need, merit, equal opportunity, caste), each rated on a seven-point scale from *completely unjust* to *completely just* (materials in Section 5.2 and Appendix B).

Stream #1 is a moving target by design, and the book treats its movement as evidence rather than noise. The questionnaire exists in five versions: 2019 (21 third-person descriptive items, wording only), 2020 (23 third-person *normative* items, “a teacher should...”), 2021 (a 36-item first-person pool reduced to a published 26-item version), 2022 (the same 26 items consolidated), and 2023 (two items reworded, the item order fixed, and, the single most consequential change, the response scale moved from seven-point *agreement* to seven-point *fairness* anchors). Chapter 6 documents every step at item level from the crosswalk (`item_map_full.csv`), in which versions are aligned by wording, never by item number: the item numbering changed between versions (including a 16↔17 swap between the 2021 and final numbering), so positional alignment would silently compare different items. The 2023 version is final: the 2023, 2024, and 2026 cohorts answered an identical instrument. The scale change is announced here because it defines the deepest comparability boundary in the data, between 2022 and 2023, whose consequences the next section regulates.

4.4 The Three-Tier Comparability Framework

The central methodological threat to a multi-cohort programme with an evolving instrument is the illegitimate comparison: treating cohorts as interchangeable when their measurement is not. The framework below assigns every pair of cohorts to the strongest comparison its shared measurement can bear; it was fixed in the analytic protocol before the cross-cohort analyses were run, and Chapter 8 applies it without exception. Figure 4.1 shows the framework at a glance.

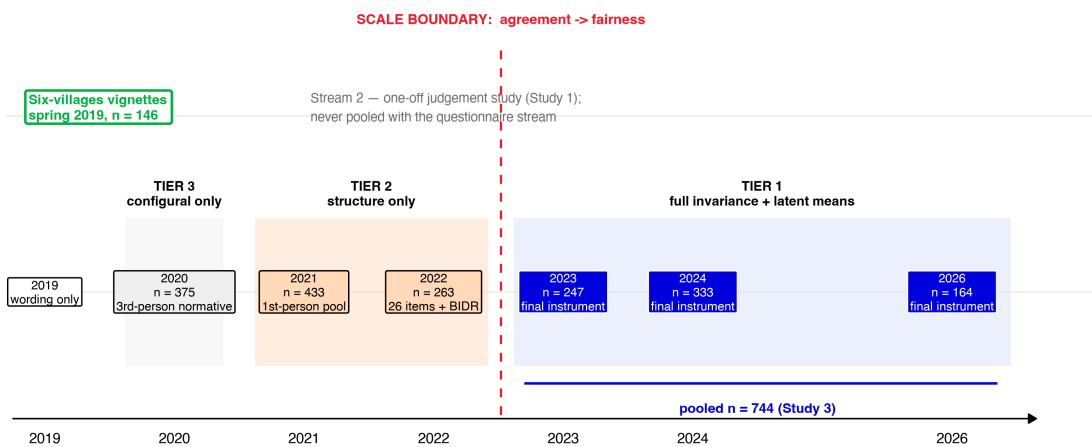


Figure 4.1. The research programme on a timeline: two never-pooled streams, seven cohort-years (2019–2026), and the three comparability tiers, with the agreement→fairness scale boundary between 2022 and 2023 marked. Source: `analyzy/scripts/fig_04_timeline.R`; cohort ns from `DATA_README`.

- Tier 1, identical instrument: {2023, 2024, 2026}. Same 26 items, same first-person wording, same fairness scale. Full measurement-invariance testing is licensed, and, only here, latent means may be compared if scalar invariance holds. Pooling these three cohorts ($n = 744$) for structural analysis (Chapter 7) is likewise licensed, subject to the invariance evidence of Chapters 7.2 and 8.2.
- Tier 2, same items, different scale: {2021, 2022}. Essentially the same 26 items, but on the agreement scale, across the 2022/2023 scale boundary. Only *structure* may be

4. Research Design and Methodology

compared: factor congruence, community recovery, and descriptive network similarity. Means, thresholds, and any inferential network comparison are out of bounds, because the response metric is not shared.

- Tier 3, different construct frame: {2020}. Third-person normative items measure beliefs about what a teacher *should* do, a different frame from first-person self-perception. Only the configural question (does a four-principle organization appear at all?) is admissible.

Three prohibitions follow and bind every page of this book: no pooling of raw scores across the scale boundary; no mean or threshold comparison between Tier 2 and Tier 1; and no growth, trend, or developmental language across cohorts: the samples are independent, non-probability, and measured with an evolving instrument, so cross-cohort differences are never interpretable as change within persons. The book's shorthand, *the structure travels, the scores do not*, is a statement about what measurement permits, and Chapter 8 reports exactly as much comparison as each tier licenses, and no more.

4.5 Analytic Strategy and Software

The division of labor is fixed: Python (3.11) performs data preparation, extraction, anonymization, wording-based item alignment, and the construction of versioned analytic datasets (scripts 01–03), while R (4.5.3; R Core Team, 2026) performs all inferential analysis in reproducible Quarto notebooks, one per empirical chapter, with package versions pinned by renv. The workhorse packages are lavaan (Rosseel, 2012) with semTools for factor models and invariance, psych for exploratory factor analysis, and bootnet, qgraph, EGAnet, NetworkComparisonTest, networktools, and mgm for the network analyses; full session information is preserved with each notebook.

Because the items are seven-point ordered ratings with documented ceiling effects (37% of responses in the top category on average; Chapter 7.2), the default estimation treats them

as ordered categories: polychoric correlations, WLSMV estimation with theta parameterization for factor models, and rank-based or cumulative-link methods for the vignette analyses. Where the ordinal machinery is infeasible (empty response categories within single cohorts make per-group thresholds inestimable in multigroup models), the analysis falls back to robust maximum likelihood (MLR) with the substitution stated and justified at the point of use (Chapters 7.2 and 8.2), never silently.

Three regulative devices discipline the analyses themselves. First, a written analytic protocol (the project's plan document) was fixed before the analyses were run; it pre-specifies the models, criteria, and decision rules, and deviations from it are documented where they occur; the protocol functions as an internal pre-registration, with the honesty that label implies rather than the public registration it does not have. Second, temporal splitting: structural discovery and confirmation never share a sample (exploration on 2023, confirmation on 2024, holdout on 2026; Chapter 7.3). Third, demotion rules fixed in advance: exploratory results (respondent typologies, hierarchical community structures) are reported as exploratory and are demoted to descriptive status when stability checks fail (as happened, and is reported, in Chapter 5.5).

One supplementary analysis stands apart from this pipeline because its units are concepts rather than respondents. The theory-guided check reported in Section 2.7 builds a similarity tree from short working definitions of justice concepts. One representation uses theory-defined features scored by three separately run language-model coders; agreement is evaluated per feature, and a small human-coded pilot provides a plausibility check. A complementary representation uses text embeddings of the same concept names and working definitions but none of the feature scores. Because the two routes share their concept set, definitions, and tree-building procedure, their partial agreement is treated as a sensitivity check on representation choice, not as independent validation. Appendix E documents the bounded concept frame, uneven feature reliability, comparison procedure, and limits. The result is a literature-bounded similarity map, not a reconstruction of the concepts' historical development or an objective taxonomy.

4.6 Transparency and Reproducibility

Every number in the empirical chapters is traceable to a script and a versioned dataset. The mechanism is the number manifest: each analysis notebook ends by writing the statistics it produced to a chapter-specific manifest file (`<chapter>_cisl.csv`), and the prose of the book may cite only numbers present in a manifest or in a frozen output table. Analyses precede prose in a fixed sequence: *analysis freeze* (the notebook runs end-to-end from the versioned data and its manifest is locked) before any results passage is written, and *chapter freeze* before translation. Stochastic steps carry fixed seeds; computationally heavy bootstraps are cached with their seeds. The full pipeline (anonymized datasets, preparation scripts, analysis notebooks, and rendered reports) is openly available in the [companion repository](#) [↗] (Nehyba, 2026b). Links to the specific notebooks, scripts, and data files are embedded directly in figure and table captions throughout the book.

4.7 Ethics and Anonymization

The data are anonymized at source and the anonymization is scripted, versioned, and auditable. Questionnaire exports were processed by a dedicated script that removes direct identifiers (login e-mail addresses and free-text comment fields), truncates timestamps to dates, and assigns synthetic respondent identifiers; the raw exports containing identifiers were deleted from the working environment after processing, and an automated check confirms that no e-mail address survives anywhere in the data tree. Retained variables are limited to quasi-identifiers that do not permit individual identification (gender, birth year, study subjects, prior experience). The vignette data carry no direct identifiers. In the text of this book, no respondent is identifiable; all illustrative names (Chapter 5.6) are fictional and belong to teaching materials, not to data. Participation in all waves was voluntary and anonymous, in the context of regular teacher-education coursework. Before answering, respondents read a standard information-and-consent statement (reproduced in full, with its English translation, in Appendix D) that named the researcher and affiliation, described the

study's purpose and the anonymous academic use of the data, affirmed the right to stop at any time, and gave a contact address; proceeding to the questions constituted informed consent. No separate prior approval from the Masaryk University Research Ethics Committee was required: under the committee's published guidance, anonymous, non-interventional questionnaire research with adults that collects no special-category (sensitive) personal data, involves no minors, and uses no deception falls outside the cases for which ethics review is mandated, and this study meets each of those conditions. Because responses were fully anonymous and could not be linked to individuals or to any course assessment, participation carried no consequence for students' academic standing. The information-and-consent text and the anonymization protocol are collected in Appendix D.

Chapter summary

The book's claims rest on a deliberately partitioned design. Two streams that are never pooled; six questionnaire cohorts and one vignette cohort that are repeated independent cross-sections, not a panel; an instrument whose evolution is documented item-by-item and aligned by wording; a three-tier comparability framework, fixed in advance, that assigns every cross-cohort claim exactly the strength its measurement affords; an ordinal-first analytic strategy with pre-specified models, temporal splits, and demotion rules; and a manifest system that makes every printed number traceable to a script and a dataset. The rules are now on the table before any result, and Chapter 5 opens the empirical sequence where the programme began, with six villages and the question of which distributions future teachers call just.

Evidence

5. Study 1: Which Distributions Do Future Teachers Call Just? The Six Villages

Aim of the chapter

This chapter answers RQ1 by reporting Study 1, a vignette study in which Czech pre-service teachers ($n = 315$) rated the justice of six village-level distribution regimes, each embodying one distributive principle of the wide taxonomy developed in Chapter 2. The chapter establishes an empirical hierarchy of principles (equality clearly at the top, a rejected bottom of position-conditioned regimes, and a closely spaced middle band between them) and uses that hierarchy, together with an analysis of how the six principles translate into classroom practice, to motivate the narrower four-principle taxonomy operationalized in the instrument whose development Chapter 6 traces.

5.1 Focused Background: Vignettes in Justice Research

Asking people directly which principle of justice they endorse is a peculiarly unreliable way of finding out what they consider just. Abstract principles command near-universal assent at the level of slogans (few respondents will vote against fairness, need, or desert when the words appear without content) and the interesting disagreements only surface once a principle is embedded in a concrete allocation with identifiable winners and losers. This is the

5. Study 1: Which Distributions Do Future Teachers Call Just? The Six Villages

core methodological argument for vignettes in justice research: a vignette presents a specific, contextualized state of affairs and asks the respondent to judge it, so that the judgment is anchored in a scenario rather than in the respondent's theory of her own values. The tradition is well established in the psychology of justice (Dalbert & Sallay, 2004), and its underlying logic (that fairness reasoning is context-bound and often runs ahead of, or against, professed principles) echoes the theoretical literature reviewed in Chapter 2, where Walzer (1983) argues that distributive criteria are sphere-specific and Elster (1992) documents how institutions mix allocation principles in ways no single theory predicts. Experimental allocation studies in the tradition of Deutsch (1975) and Schwinger (1980) make the same point from the laboratory side: which principle people apply depends on what is being distributed, to whom, and under what relationship.

In the Czech methodological context, vignettes carry an additional association: the anchoring-vignette method introduced to educational research by Voňková, in which short scenarios serve to calibrate self-assessments across respondents who use response scales differently (King et al., 2004; Voňková & Hrabák, 2015). Study 1 should be explicitly distinguished from that use. The six villages are not anchoring vignettes; no self-assessment is being recalibrated. They are parallel stimulus vignettes: six equally structured descriptions of allocation regimes, rated on a common scale, so that the ratings can be compared within respondents. The design question they answer is comparative and internal (which principles does the same person judge more and less just?) and the analysis below is accordingly built on within-person contrasts rather than between-group comparisons.

One further feature of the design deserves emphasis before the materials are presented. The vignettes deliberately describe villages, not schools. The choice displaces the judgment away from the respondents' emerging professional identity: a student teacher asked to rate the fairness of grading practices already knows what a future teacher is supposed to say, whereas a village distributing goods among its citizens invokes distributive intuitions without cueing professional norms. The cost of this displacement is equally deliberate and is confronted in Section 5.6: what the study measures is the perceived justice of principles in a generic civic context, and any transfer of the resulting hierarchy to classrooms is an inference that requires

argument, not an observation contained in the data.

5.2 Materials: Six Villages, Six Principles

The stimulus material is a short narrative, “The Story of the Villages” (Příběh vesnic), written for the first session of the self-experiential seminar described in Section 5.3 and printed as a handout for every participant. The story frames six villages, each identified by a color, and each allocating goods among its citizens according to a single distributive rule. Together the six rules instantiate the wide taxonomy of Chapter 2: equality, chance, need, merit, equal opportunity, and caste. Table 5.1 summarizes the six regimes; the full vignette texts in Czech with English translations are provided in Appendix B.

Table 5.1. The six villages: colors, principles, and allocation rules (abridged; full texts in Appendix B)

Village	Principle	Allocation rule (abridged)
Blue	Equality	Every citizen receives the same share, regardless of anything.
White	Chance	A lottery: everyone has the same chance; prizes are distributed by luck.
Green	Need	Goods are distributed according to the needs of individual citizens.
Orange	Merit	Those who worked, contributed, or served long receive more.

5. Study 1: Which Distributions Do Future Teachers Call Just? The Six Villages

Village	Principle	Allocation rule (abridged)
Red	Equal opportunity	The same starting line for all; everyone grabs what they can.
Yellow	Caste	Goods are divided among three castes; equality holds only within a caste.

Each village was rated on a seven-point scale from 1 (*completely unjust*) to 7 (*completely just*), yielding six ratings per respondent. Two design decisions built into the materials matter for everything that follows. First, the six vignettes are parallel in form (same narrative frame, same length and register, same response scale) so that differences in ratings are attributable to the allocation rules rather than to surface features of the texts. Second, the vignettes are not normatively innocent paraphrases of textbook principles; they encode the principles as lived regimes, with the frictions a regime brings. The Red village's equal opportunity is not the philosopher's level playing field but a competitive scramble ("everyone grabs what they can") and the Yellow village's caste system pointedly preserves equality *within* castes while denying it between them. These encodings are faithful to how the principles appear in political life, but they have interpretive consequences, particularly for the Red village, that Section 5.6 takes up.

5.3 Sample and Procedure

5.3.1 The pedagogical setting

Study 1 is embedded in a teaching context, and honesty about that embedding is a precondition for interpreting its data. The villages task forms the opening activity of a self-experiential seminar within pre-service teacher preparation at a Czech faculty of education,

a seminar format whose purpose is not to transmit content but to let future teachers examine their own experience, beliefs, and reactions as professional material. In the session's design, participants first read the printed story, rated the six villages individually, and the ratings were collected. The activity then continued (and this matters for what the data can and cannot show) with a 40-minute reflection block ("Reflection on the activity and practice") in which participants searched their own school experience for real manifestations of each of the six types of justice, guided by six parallel prompts: How does justice based on equal division manifest itself at school? How does justice based on specific needs manifest itself? (and likewise for status-based, equal-starting-conditions, random, and merit-based justice). The seminar materials scaffold this reflection with six short fictional classroom exemplars, one per principle, which are reproduced and analyzed in Section 5.6.

The research data of Study 1 are the ratings alone. The reflection block, though pedagogically the heart of the session, was not recorded or collected in any systematic way, and no claim in this book rests on what participants said during it. Section 5.6 discusses one informal observation from these reflections (flagged there, emphatically, as anecdote rather than evidence) and the distinction between the two is maintained throughout.

5.3.2 Sample

The respondents are 315 pre-service teachers who completed the villages task in the spring of 2019. All 315 rated all six villages; there are no missing ratings. Demographic information (sex, age, subjects studied) is available for about two-thirds of the sample (sex for 204 respondents and age for 206) because the demographic section was optional and, in parts of the administration, not distributed at all. The sample description is therefore deliberately thin: the respondents were students of teacher education programs participating in the seminar, predominantly in subject combinations for secondary teaching, and no stronger demographic characterization is defensible from the data. Two further limitations are recorded here rather than discovered later: the sample is a convenience sample of seminar participants, not a probability sample of Czech pre-service teachers; and the vignette respondents

5. Study 1: Which Distributions Do Future Teachers Call Just? The Six Villages

cannot be linked individually to the questionnaire cohorts of Studies 2–4: the two empirical streams of this book share a population of origin but not identified persons, and no analysis in this book pools them.

5.3.3 Analysis plan

The analysis follows the pre-specified plan in the project’s analytic protocol (section E1) and was conducted in R 4.5.3 (R Core Team, 2026). Because the six ratings are ordinal, within-person, and bounded, the primary test of the hierarchy is Friedman’s rank test (Friedman, 1937) with Kendall’s W as the effect size, followed by pairwise Wilcoxon signed-rank tests with Holm’s (1979) correction and matched-pairs rank-biserial correlations as pair-level effect sizes. As a robustness check that respects the ordinal scale while modeling all six ratings jointly, a cumulative link mixed model (Christensen, 2025, R package *ordinal*) with a random intercept per respondent re-estimates the hierarchy on the latent scale. Exploratory analyses of respondent profiles use hierarchical clustering (Ward’s method) on standardized ratings with bootstrap stability assessment (Hennig, 2007, R package *fpc*), triangulated by latent profile analysis (Rosenberg et al., 2018, R package *tidyLPA*) under a pre-registered demotion rule: if the two methods disagree, no typology is claimed. Every number reported below is frozen in the analysis manifest `05_cisla.csv` generated by the notebook `10_vinety.qmd`.

5.4 Results: A Hierarchy of Principles

5.4.1 Descriptive hierarchy

Table 5.2 reports the descriptive statistics; Figure 5.1 shows the full rating distributions. The picture they draw is unusually legible for attitudinal data. Equality stands alone at the top: the Blue village received a mean rating of 5.50 ($SD = 1.38$, $Mdn = 6$), and 79% of respondents placed it on the “just” side of the scale (ratings 5–7), against 10% on the “unjust” side (ratings

5.4 Results: A Hierarchy of Principles

1–3). A closely spaced middle band follows at a clear distance, chance ($M = 4.50$, $SD = 1.70$, $Mdn = 5$), need ($M = 4.22$, $SD = 1.65$, $Mdn = 4$), and merit ($M = 3.93$, $SD = 1.52$, $Mdn = 4$), the three sitting within about half a scale point of one another around the midpoint: chance alone draws majority endorsement (52% just, 27% unjust), need splits roughly two-to-one toward endorsement (47% vs. 33%), and merit divides the sample almost evenly (38% just, 37% unjust). At the bottom, sharply separated, lie the two regimes that condition a person's share on position: equal opportunity in its competitive encoding ($M = 2.36$, $SD = 1.43$, $Mdn = 2$), which 81% of respondents rated unjust, and caste ($M = 2.00$, $SD = 1.15$, $Mdn = 2$), rated unjust by 90%, with only about 3% willing to call the Yellow village just at all.

Table 5.2. Ratings of the six villages: descriptive statistics ($n = 315$; scale 1–7)

Principle (village)	M	SD	Mdn	IQR	% just (5–7)	% unjust (1–3)
Equality (Blue)	5.50	1.38	6	2	79.37	10.48
Chance (White)	4.50	1.70	5	3	52.38	26.67
Need (Green)	4.22	1.65	4	3	46.98	33.33
Merit (Orange)	3.93	1.52	4	2	37.78	37.14
Equal op- portunity (Red)	2.36	1.43	2	2	8.89	81.27
Caste (Yellow)	2.00	1.15	2	2	3.49	90.16

Note. Source: 05_vinety_deskriptiva.csv (notebook 10_vinety.qmd, data vinety_final.csv).

5. Study 1: Which Distributions Do Future Teachers Call Just? The Six Villages

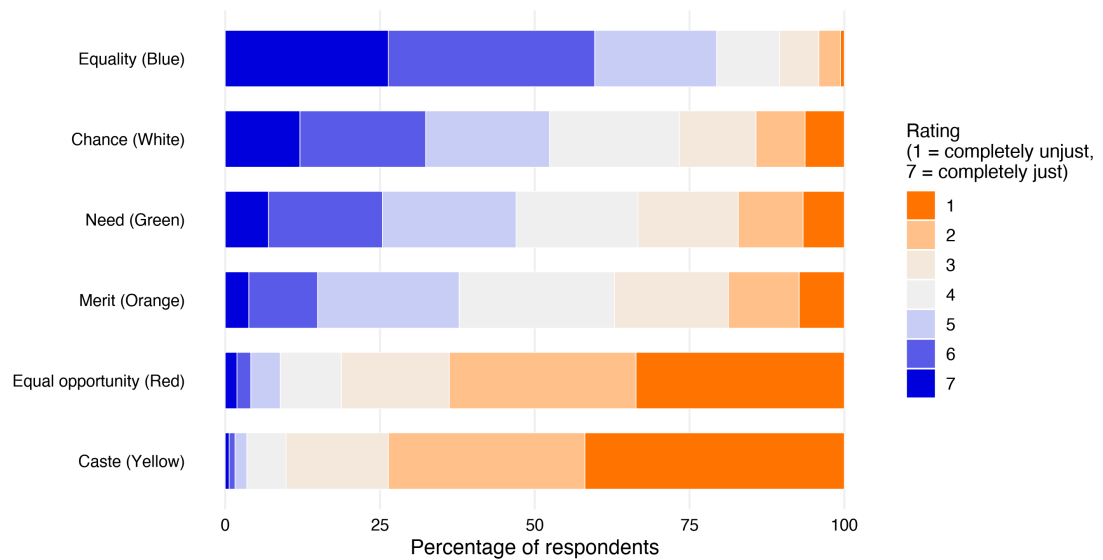


Figure 5.1. Perceived justice of six distributive principles: full rating distributions.

5.4.2 Omnibus and pairwise tests

The hierarchy is not an artifact of averaging. The Friedman test across the six within-person ratings was decisive, $\chi^2(5) = 704.62$, $p < .001$, with Kendall's $W = .45$, a moderate concordance indicating that respondents ranked the principles in substantially similar orders, not merely that means differ. All 15 pairwise contrasts survived Holm correction (Table 5.3): at this sample size even the closely spaced middle band resolves into an ordered sequence. Significance and magnitude, however, tell different stories, and it is the magnitudes that carry the structure. The three contrasts *within* the middle band are small: chance versus need ($r = .11$, adjusted $p = .041$), need versus merit ($r = .13$, adjusted $p = .033$), and chance versus merit ($r = .25$), an order of magnitude weaker than the gaps that separate the middle from equality above ($r = .43$ against chance, rising to $.53$ – $.60$ against need and merit, moderate to large) and from the rejected bottom below (middle-band principles against the bottom two: $r = .63$ – $.78$, all large). The reading the data license is therefore a graded ordering bracketed into three magnitude-tiers: equality clearly superior; a tightly spaced middle in which chance ranks just above need, and need just above merit, each step real but slight; and a

decisively rejected bottom pair. Within that bottom, the caste village was rated significantly lower than the equal-opportunity village, though the difference is small ($r = .18$, adjusted $p = .001$): respondents rejected both, with a marginal extra severity for explicit caste.

Table 5.3. Pairwise Wilcoxon signed-rank tests with Holm correction and matched-pairs rank-biserial effect sizes ($n = 315$)

Contrast	p (Holm)	r (rank-biserial)	Magnitude
Equality vs. chance	< .001	.43	moderate
Equality vs. need	< .001	.53	large
Equality vs. merit	< .001	.60	large
Equality vs. equal opportunity	< .001	.81	large
Equality vs. caste	< .001	.85	large
Chance vs. need	.041	.11	small
Chance vs. merit	< .001	.25	small
Chance vs. equal opportunity	< .001	.73	large
Chance vs. caste	< .001	.78	large
Need vs. merit	.033	.13	small
Need vs. equal opportunity	< .001	.67	large
Need vs. caste	< .001	.76	large
Merit vs. equal opportunity	< .001	.63	large
Merit vs. caste	< .001	.73	large
Equal opportunity vs. caste	.001	.18	small

Note. Source: 05_vinety_posthoc.csv. Exact test statistics and unadjusted p values are reported in the source table.

5.4.3 Robustness on the latent scale

Because rank tests discard metric information and treat each pair in isolation, the hierarchy was re-estimated with a cumulative link mixed model: all six ratings modeled jointly as ordinal responses, a fixed effect for principle with equality as the reference category, and a random intercept per respondent. The model reproduces the descriptive ordering exactly. All five contrasts against equality are negative and significant (all $p < .001$), and their magnitudes recover the three tiers on the latent logit scale: chance -1.20 , need -1.52 , and merit -1.84 form the middle band, while equal opportunity (-3.74) and caste (-4.18) fall more than twice as far below the reference (source: `05_vinety_clmm.csv`). The hierarchy of Table 5.2 is thus not a fragile consequence of one test's assumptions; it survives a change of analytic machinery that shares none of the Friedman test's mechanics.

5.5 Respondent Profiles

The tier structure describes the sample's aggregate judgment. A natural further question is whether it conceals distinct *types* of judges, subgroups whose preference profiles differ in kind rather than degree. The analysis plan treated this as strictly exploratory, under a demotion rule: a typology is entertained only if two methods that share no machinery, hierarchical clustering and latent profile analysis, agree on the number of groups. They do agree, on two groups; but on inspection the honest reading is not two kinds of people so much as a single attitudinal axis (how far a respondent will let *need* justify departing from an equal share) of which the "two groups" are only the high and low ends.

The two-group split is real but weak, and three facts show it is better read as one continuum than two kinds. First, its bootstrap stability is low: a mean Jaccard similarity of .55 (minimum .48), above Hennig's (2007) $\approx .50$ dissolution floor but below the .60 of a trustworthy pattern and far below the .85 of a stable one, with an average silhouette width of only .09, so the groups overlap heavily. Second, the division is almost entirely a single variable: one rule, *rate the Need village 3 or below*, reproduces 86% of the two-group assignment. Third,

the ratings themselves form one broad peak rather than two: the Need scores rise to a single mode near the mid-scale and fall away on both sides (a formal bimodality check agrees: .47 on the Need rating and .36 on the overall between-respondent axis, both below the .56 that would flag two humps; Figure 5.3). The clustering did not uncover two populations; it drew a line through the middle of one continuous distribution, and latent profile analysis returning “two” (BIC lowest at two classes, entropy .85) does not change this: a smooth continuum is routinely fit best by two overlapping components. What is real and carries forward, therefore, is not a typology but the *axis itself*.

The two ends of that axis are easy to read, and both share the sample’s backbone: each rates equality highest (5.4 and 5.7), places chance in the middle (≈ 4.5), and rejects competitive equal opportunity and caste (profile means frozen in `05_vinety_cluster_profiles.csv`; Figure 5.2). They part only on need. The upper end ($n = 218$, 69%) is *need-inclusive*: it endorses the Need village ($M = 4.95$) and merit (4.14) alongside equality, a broadly pluralist acceptance of several distributive grounds. The lower end ($n = 97$, 31%) is *equality-first and need-skeptical*: it rates equality even higher (5.71) but pulls need down to near-rejection (2.59) and merit below the midpoint (3.47), reserving justice for identical treatment and distrusting departures from it. The axis dividing the sample, in other words, is not whether equality is just (all agree it is) but whether *need* licenses just departures from equal shares. That is exactly the fault line the instrument of Chapters 6–8 must later measure, and it is developed in Section 5.6.

5. Study 1: Which Distributions Do Future Teachers Call Just? The Six Villages



Figure 5.2. The two ends of the need-acceptance axis (the $k = 2$ split): a need-inclusive majority and an equality-first, need-skeptical minority. The profiles share the whole hierarchy and diverge almost only on need; latent profile analysis agrees on two groups.

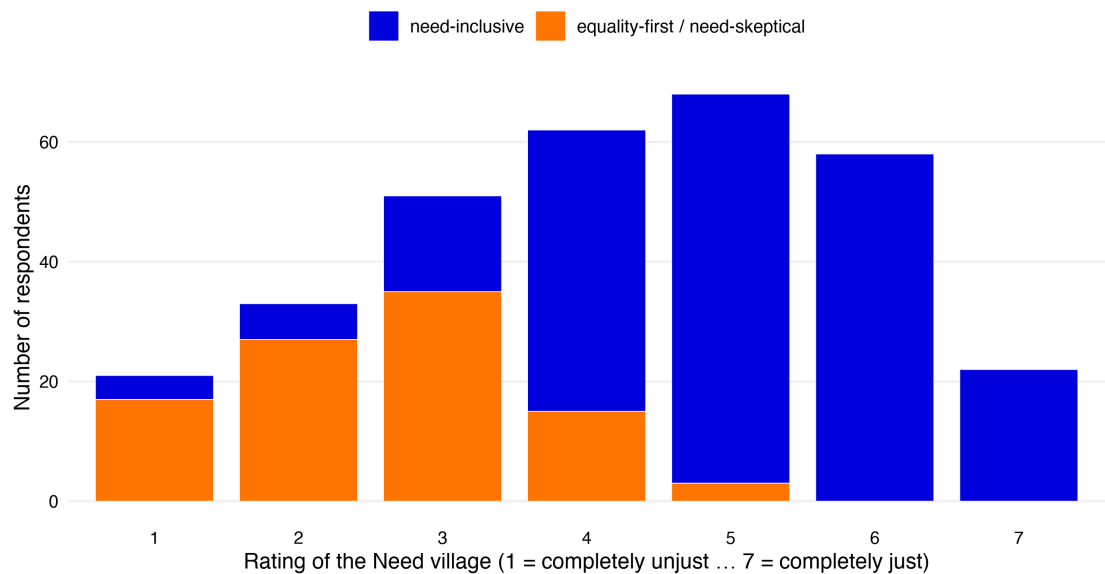


Figure 5.3. The need-acceptance axis is one distribution, not two. Ratings of the Need village form a single broad peak; the two-group split (color) merely divides that peak near its middle

(need-skeptical respondents concentrated in the lower ratings, need-inclusive in the higher) rather than marking two separated clusters.

5.6 Interim Discussion: The Equality Preference, the Equal-Opportunity Paradox, and the Road to Classrooms

5.6.1 The equality preference

The strong preference for flat equality is the study's most consistent finding, and it is worth stating precisely what it is a preference *for*: the Blue village distributes identical shares “regardless of anything,” regardless, therefore, also of need, effort, and contribution. A sample of future teachers placing this regime more than a full scale point above need (a gap of 1.28) and more than a point and a half above merit (1.57) is a substantive result, not a platitude. It resonates with the experimental tradition in which equality operates as the default allocation heuristic, the choice that requires no justification, no information about recipients, and no interpersonal comparison (Deutsch, 1975; Messick, 1993). It also sets up the central tension of this book's later chapters: the questionnaire of Chapters 6–8 asks whether a comparable population of pre-service teachers, judging *their own* teaching behaviors, sustains equality as a separable principle alongside need and merit, or whether, in a setting that daily forces departures from identical treatment, these intuitions are organized differently. Because the vignette respondents and the questionnaire cohorts are different persons (Section 5.3.2), any such comparison is cross-sectional and structural, not a claim about change within individuals.

5.6.2 The equal-opportunity paradox

The second striking result is the collapse of equal opportunity, the principle most celebrated in liberal political rhetoric, to the bottom tier, just above caste (the two differ significantly but only by a small margin, $r = .18$). The interpretation defended here is deflationary and

5. Study 1: Which Distributions Do Future Teachers Call Just? The Six Villages

framing-based. The Red village couples the equal starting line with unregulated competitive appropriation: “everyone grabs what they can.” What respondents rejected, on this reading, is not the ideal of equal chances but the *outcome regime* the vignette attaches to it, a scramble in which formal equality of position licenses unlimited inequality of result. The reading is supported by a contrast internal to the materials: the White village also gives everyone “the same chance,” yet, encoded as a lottery rather than a competition, it sits comfortably in the middle tier, more than two scale points above Red. Same formal property, different social mechanism, opposite judgment; though, because the two villages differ in more than their mechanism and were not designed as a matched framing manipulation, this cross-vignette contrast is suggestive rather than a controlled test. It is, even so, consistent with the sphere-sensitivity of distributive intuitions that the vignette method exists to detect, and it counsels strongly against reading the Red village’s rating as “these pre-service teachers reject equal opportunity,” a formulation the data do not license. The paradox nonetheless earns its place in Chapter 9’s confrontation with the Czech equity discourse, where “equal opportunities” functions as a policy master-term (MŠMT, 2020): if the term’s competitive encoding repels the very future teachers expected to implement it, the policy language and the profession’s intuitions may be speaking past each other.

5.6.3 From villages to classrooms, and the limits of that road

The seminar session from which these data come does not end with the villages; it ends by walking the six principles into school. In the 40-minute reflection block, participants search their own school biographies for manifestations of each type of justice, and the seminar materials seed the search with six fictional classroom exemplars (Table 5.4; all names are fictional and belong to the teaching materials, not to the data).

Table 5.4. Classroom transpositions of the six principles used as reflection scaffolds in the seminar (fictional exemplars; translated from the seminar materials)

5.6 Interim Discussion: The Equality Preference, the Equal-Opportunity Paradox, and the Road to Classrooms

Principle	Fictional classroom exemplar
Equality	Teacher Petra takes care to give all pupils the same amount of time for a written test and to grade everyone “by the same yardstick.”
Need	Teacher David gives extra time on tests to pupils with a documented specific learning disorder.
Merit	Teacher Ondřej gives an end-of-year reward to the pupils with the best grade average on their report card.
Chance	Teacher Martina does not want her sympathies to influence whom she calls on, so she calls on pupils according to a random system.
Equal opportunity	Teacher Lucie tells her pupils at the start of the school year that each of them has the same chance of earning a good final grade.
Caste (status)	Teacher Adam, like all teachers, does not have to change into slippers in the school building, while pupils must wear slippers the whole time they are at school.

Three observations about this transposition matter for the book’s argument, and their epistemic status must be stated with care, because the first of them rests on no collected data at all.

First, an informal observation from years of running the seminar: participants typically *succeed* at the reflection task; for each of the six principles, including caste, groups generally

5. Study 1: Which Distributions Do Future Teachers Call Just? The Six Villages

locate recognizable school manifestations. This is reported here as an instructor's anecdote, nothing more. The reflections were never collected, coded, or counted; the observation carries no evidential weight in this book, and no conclusion below depends on it. It is mentioned because it motivates a genuine research gap: a systematic study (collecting student reflections as data, or mining public accounts of school practice) testing whether all six principles are recoverable in the lived institution of schooling would convert the anecdote into evidence. This book does not perform that study.

Second, and independently of any anecdote: *occurrence is not justification*. Even where a principle demonstrably operates in school, its presence says nothing about its legitimacy: the descriptive question of what schools do and the normative question of what would be just are different questions, and sliding from the first to the second is a naturalistic fallacy this book refuses. The caste exemplar makes the point sharply. Status asymmetries of exactly Adam's kind, rules that bind pupils but not teachers, are mundane, ubiquitous school reality; and caste is simultaneously the principle these respondents rated most unjust ($M = 2.00$, with 90% rejection). If the anecdotal observation above is even directionally right, the institution routinely enacts the one principle its future professionals most decisively reject. That tension is a finding about neither schools nor the data alone; it is a hypothesis-generating friction between them, and it belongs to the questions this book opens rather than settles.

Third, the transposition subtly *transforms* the principles, and the transformations are analytically instructive. Chance, in the village a distributive lottery over unequal prizes, becomes in Martina's classroom a device of *procedural impartiality*, randomness deployed to protect allocation from favoritism. Equal opportunity, in the village a competitive scramble, becomes in Lucie's classroom a verbal assurance about grading prospects, closer to a procedural commitment to consistency and transparency than to any distributive mechanism. The classroom, in other words, absorbs several distributive principles into the *procedures* of teaching. This observation does real work in the next section: it is one of the reasons the instrument of Chapter 6 includes a procedural-justice principle that the villages taxonomy lacks.

5.7 From Six Principles to Four: Rationale for the Instrument's Narrower Taxonomy

Study 1 rates six principles; the instrument whose development Chapter 6 traces measures four: procedural justice, equality, needs, and merit. The reduction is not a loss of nerve but a change of object, and the vignette results, read together with Section 5.6.3, supply its rationale.

Equality, need, and merit transfer directly. They occupy the top and middle of the empirical hierarchy, they are the three classic principles of the distributive-justice literature (Deutsch, 1975), and, decisively for a self-report instrument, each corresponds to concrete, first-person teacher behaviors of the kind Table 5.4 illustrates: giving everyone the same time, giving some pupils more time for documented reasons, rewarding performance. They can be asked about as things a teacher *does*.

Chance does not transfer as a distributive principle. Its middle-tier rating notwithstanding, a teacher does not distribute grades or attention by lottery as a matter of allocative policy; where randomness legitimately enters teaching, it enters, as in Martina's exemplar, as a procedural safeguard of impartiality. The principle's classroom content is thus absorbed into procedural justice rather than retained as a separate distributive subscale.

Equal opportunity is excluded for a methodological reason the vignette data make vivid: its rating is plausibly shaped by how its mechanism is framed (Section 5.6.2, offered as interpretation rather than controlled test). Two different villages that share the formal feature of an equal ex-ante chance, the Chance lottery ($M = 4.50$) and the competitive Red scramble ($M = 2.36$), were rated more than two scale points apart, and a principle whose endorsement may hinge this strongly on the portrayal of its mechanism is a poor candidate for decontextualized self-report items. Its benign classroom transposition, Lucie's assurance, again dissolves into procedural commitments to transparency and consistency, and is captured there.

Caste is excluded as a self-report subscale for reasons that need hardly be labored: items of the form "I allocate benefits to pupils according to their status group" are unanswerable

5. Study 1: Which Distributions Do Future Teachers Call Just? The Six Villages

in the first person; floor-level endorsement and maximal social-desirability pressure would render the subscale psychometrically dead on arrival. The principle's near-unanimous rejection in Study 1 (90% unjust) documents the normative consensus; its analytic interest survives not as a subscale but as the critical lens of Section 5.6.3, the institution's routine status asymmetries against the profession's rejection of status as a principle.

Procedural justice, finally, is the taxonomy's addition rather than a survivor of the reduction. It enters from the organizational- and classroom-justice literature reviewed in Chapter 2 (Leventhal, 1980; Colquitt, 2001; Chory-Assad & Paulsel, 2004), and Study 1's transposition analysis independently argues for it: when distributive principles are walked into classrooms, a substantial part of their content reappears as properties of procedure; impartiality, transparency, consistency, voice. An instrument that measured only distributive principles would miss the very dimension into which the classroom translates them. The result is the four-principle taxonomy (procedural, equality, needs, merit) whose 26-item operationalization, wording history, and scale decisions occupy Chapter 6.

5.8 Chapter Summary

Study 1 asked which distributions future teachers call just (RQ1) and answered with a graded, three-magnitude-tier hierarchy of unusual clarity. Rating six village-level allocation regimes ($n = 315$, spring 2019), the sampled Czech pre-service teachers placed flat equality alone at the top ($M = 5.50$; 79% rating it just); chance, need, and merit in a closely spaced middle band ($M = 4.50, 4.22, 3.93$; every pairwise contrast now reaches Holm-corrected significance, but the differences within the band are small, $r = .11-.25$, against large gaps to the tiers above and below); and position-conditioned regimes at the rejected bottom: competitive equal opportunity ($M = 2.36$; 81% unjust) and caste ($M = 2.00$; 90% unjust). The hierarchy is decisive by omnibus test ($\chi^2(5) = 704.62$, $p < .001$, Kendall's $W = .45$) and survives re-estimation with an ordinal mixed model. Beneath the aggregate ordering the sample spreads along a single need-acceptance axis rather than

splitting into distinct types: a two-group description (a need-inclusive majority, 69%, and an equality-first, need-skeptical minority, 31%) only approximates it; the split is weakly separated (silhouette .09), a single need-rating threshold recovers 86% of it, and the ratings form one broad peak, not two; so what carries forward is the axis, not a typology of kinds: the sample divides not on whether equality is just but on whether need justifies departures from it. Two interpretive results carry forward: the equal-opportunity paradox is read as a framing effect, two vignettes sharing an equal-chance feature (the Chance lottery and the competitive Red scramble) judged more than two scale points apart, rather than a rejection of the ideal; and the transposition of the six principles into classroom practice, illustrated by the seminar's reflection scaffolds, shows several distributive principles dissolving into properties of procedure. All three results (the hierarchy, the need fault line, and the procedural transposition) motivate the four-principle instrument (procedural, equality, needs, merit) whose construction is the business of Chapter 6.

6. Study 2: Building an Instrument for Self-Perceived Justice of Teaching

Aim of the chapter

This chapter answers RQ2 by reconstructing, version by version, how self-perceived justice toward pupils was operationalized between 2019 and 2023, from third-person descriptive items to the final 26-item first-person instrument organized around the four principles of procedural justice, equality, needs, and merit.

6.1 Focused Background: How Justice Has Been Measured

Anyone who sets out to measure justice in the classroom inherits a measurement tradition that was built for other settings and, more importantly, for other informants. The dominant line is organizational. After Greenberg (1987) consolidated the field of organizational justice, Colquitt (2001) gave it its canonical instrument: a construct-validated measure in which employees report how fairly they are treated: whether the procedures that produce decisions about them allow voice and are applied consistently, whether their outcomes reflect their contributions, and whether the people enacting those procedures treat them with dignity and candor. The measure has been enormously productive, and Chapter 2 adopted several of its conceptual distinctions, including the justice rules formulated by Leventhal (1980). But

6. Study 2: Building an Instrument for Self-Perceived Justice of Teaching

its perspective is fixed: the respondent is the *recipient* of just or unjust treatment, judging an authority from below.

Classroom justice research imported this logic more or less intact. In the line opened by Chory-Assad and Paulsel (2004), students rate the fairness of their instructor's grades, procedures, and interpersonal conduct, and these ratings predict consequential outcomes such as student aggression and resistance. Berti et al. (2010) broadened the informant base by studying both students' and teachers' representations of classroom justice, but even where teachers speak, they speak as observers characterizing what happens in classrooms, not as allocators auditing their own conduct on a standardized instrument. The just-world tradition around Dalbert and Sallay (2004) examined justice in school and family contexts (including the justice of grading, often through vignettes describing concrete evaluative episodes), yet here again the psychometric target is the justice that adolescents *experience* or the beliefs they hold about a just world, not the justice a teacher attributes to their own actions.

Across all three lines, then, the rater's role is constant: one judges justice received or observed. What no existing instrument captures is the allocator's own view of their conduct: a teacher, or a pre-service teacher anticipating the teacher role, judging how justly they themselves distribute what a classroom obliges them to distribute. Chapter 2 argued that the teacher is precisely an allocator: of time in oral examinations, of attention across thirty pupils, of grades, of praise, of exceptions. If perceived justice at the micro level of teacher-pupil interaction is worth studying (the claim developed against the Czech macro tradition in Chapter 3), then the allocator's self-perception is not a curiosity but a missing half of the phenomenon. For teacher education in particular it is the actionable half: pre-service teachers form their allocation habits before they have classrooms in which to enact them, and an instrument that makes those habits inspectable has an obvious reflective use (a point taken up in Chapter 9.5).

The instrument this chapter reconstructs was therefore designed to meet four requirements that no available measure satisfied jointly. First, it had to be a first-person self-report: the respondent judges their own (anticipated) conduct, not a generalized teacher's. Second, it

6.1 Focused Background: How Justice Has Been Measured

had to be situated in concrete, recurring allocation situations of school life (classroom rules, oral examination, the grading of written work, homework, the recognition of pupil performance) rather than in abstract endorsements of fairness, which invite unanimity and carry little information. Third, its content had to follow the four-principle framework of this book (Chapter 2.5): procedural justice plus the three distributive principles of equality, needs, and merit (Deutsch, 1975), the narrow taxonomy that Chapter 5.7 justified as the defensible core of the wider six-principle set used in the vignettes. Fourth, it had to be answerable by pre-service teachers with limited classroom experience, which means its items describe conduct that can be judged prospectively. One further difficulty was visible from the outset: an instrument that asks people to appraise their own virtue courts socially desirable responding. How the development process gradually took that threat seriously (first with an ad hoc screen, later with a standardized inventory) is part of the story below, and Chapter 7.6 gives it a full empirical treatment.

No instrument meeting these requirements existed; one had to be built. It was not built in a single pass. Between 2019 and 2023 the questionnaire went through five versions, and the remainder of this chapter reconstructs that development openly, including the version whose raw data did not survive, the version that turned out to measure a different construct, and the late decision to change the response scale, whose cost for comparability is accounted for rather than hidden. Figure 6.1 charts the five versions and the change that defines each transition.

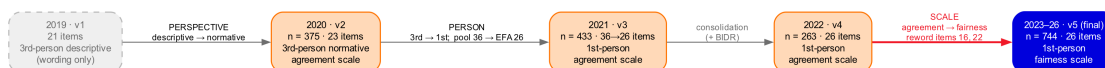


Figure 6.1. Evolution of the self-perceived-justice instrument across five versions. Source: [analyzy/figures/06_evolve_nastroje.dot](#); ns and item counts from DATA_README.

6.2 Version 1 (2019): Third-Person Descriptive Items, a Documentary Reconstruction

The first version of the questionnaire, administered in 2019, consisted of 21 items written in the third person and in a descriptive register: declarative statements about what “the teacher” does, such as “Učitel všem žákům věnuje stejnou míru pozornosti” (“The teacher devotes the same degree of attention to all pupils”) or “Učitel si na začátku roku zřetelně nastavuje pravidla fungování třídy” (“The teacher clearly sets up the rules of how the class works at the start of the year”). What survives of this version is its item dictionary (the full list of wordings) but not the raw responses. The pilot predates the systematic data archiving that this research programme later adopted, and the response files can no longer be located. This is stated plainly: no analysis of Version 1 is reported anywhere in this book, no number is attributed to it, and everything claimed about it concerns wording, not data. Its role in the argument is documentary: it is the root of the item lineage.

That lineage is real and traceable. The cross-version crosswalk constructed for this book (`item_map_full.csv`; Appendix A) matches the 2019 wordings to the final instrument by textual similarity, and by that matching, 12 of the 26 final items have a plausible 2019 ancestor, several of them nearly verbatim once the grammatical person is changed. The six formerly uncertain pairings across the 2019 and 2020 versions were manually confirmed by the author before the Tier-3 analysis, and every cross-version analysis uses only confirmed links. The continuity matters because it shows that the situational content of the instrument (rules announced at the start of the year, evenly distributed examination time, criteria communicated in advance) was present from the first draft. What changed across five versions was almost never *what* conduct the items describe; it was *who* is described, in what modality, and how the respondent is asked to judge it.

The lesson of Version 1 was the perspective problem. A respondent rating “The teacher devotes the same degree of attention to all pupils” must first decide who “the teacher” is. A typical teacher they have observed? An ideal teacher? Themselves, projected into the

role? The descriptive third-person frame leaves the target of the judgment indeterminate, suspended between a description of how teachers are and an expectation of how teachers ought to be, and none of the available readings is the construct of interest, namely the respondent's perception of their own justice. Rating "a teacher" is not rating oneself. The first revision resolved this ambiguity, but in the direction of the norm rather than the self, and thereby produced a well-defined instrument for the wrong construct, the subject of the next section.

6.3 Version 2 (2020): Third-Person Normative Items

The 2020 version made the modality explicit. All 23 items were rewritten as normative statements with the stem "Učitel by měl..." ("The teacher should..."), and several items were negated, "Učitel by NEMĚL..." ("The teacher should NOT..."), functioning as reverse-coded items. Respondents rated their agreement on a seven-point scale anchored from strong disagreement to strong agreement. The version was administered to $n = 375$ pre-service teachers between February and August 2020, making it the first wave of the questionnaire stream with archived data.

Version 2 is a coherent instrument, but for a different construct. A statement of the form "the teacher should announce grading criteria in advance" asks the respondent to endorse or reject a professional norm. What it yields is a measure of normative beliefs about just teaching: a picture of what this population thinks a teacher owes pupils. That is not a defective version of self-perceived justice; it is a distinct construct frame, and the distinction is easy to state. A respondent can wholeheartedly agree that a teacher should use the same assessment criteria for every pupil while doubting that they themselves manage to do so, and conversely, a respondent may describe their own conduct as just in ways their endorsed norms would not license. Norm endorsement and self-perception can diverge in both directions, and an instrument pitched at the norm cannot detect the divergence. A further, conceptual limitation follows from the content: norms of fairness are close to what survey methodology calls

6. Study 2: Building an Instrument for Self-Perceived Justice of Teaching

consensus statements, and agreement with them can be expected to concentrate near the top of the scale, limiting the discriminating power of the frame, an expectation this book examines empirically where the 2020 data are analyzed (Chapter 8.4).

Within the three-tier comparability framework that governs all cross-cohort claims in this book (Chapter 4.4; applied in full in Chapter 8.1), the 2020 cohort therefore occupies the third tier: because it operationalizes a different construct frame, it supports only configural and qualitative comparison with the later self-report cohorts, no score comparison, no invariance testing, no trend language. But tier 3 is not a euphemism for “useless.” The normative cohort earns its keep twice over: methodologically, as the contrast that makes visible what the later versions changed, and substantively, as a norm-level benchmark against which the self-report cohorts can be read interpretively (Chapter 9.1), what future teachers believe a teacher should do, set beside what later cohorts say they themselves do.

The lesson of Version 2 completed the lesson of Version 1. Resolving the descriptive–normative ambiguity toward the norm produced clean items and a well-defined construct, but the construct was still located outside the respondent. To measure self-perceived justice, the instrument had to stop asking about “the teacher” altogether and ask about *me*. That turn (grammatically small, conceptually decisive) defines Version 3.

6.4 Version 3 (2021): The 36-Item First-Person Pool and the Published 26-Item EFA

The 2021 version rewrote the instrument in the first person. Every item now describes the respondent’s own (anticipated) conduct in a concrete classroom situation, “Na začátku roku je pro mě důležité zřetelně stanovit pravidla fungování ve třídě” (“At the start of the year, it is important to me to clearly establish the rules of how the class works”), and the item pool was expanded to 36 items to give the subsequent psychometric reduction room to work. The pool deliberately oversampled each of the four principles across the recurring situational contexts identified in Section 6.1 (classroom rules and conduct, oral examination, grading of written

work, homework, and the recognition of pupil performance), following the standard logic of scale construction in which a broad, theory-driven pool precedes empirical trimming (Clark & Watson, 1995). Responses were again collected on the seven-point agreement scale. Two further features of the wave deserve note. First, data collection extended over an unusually long window: the raw file comprises $n = 433$ respondents collected between February 2021 and March 2022. Second, the questionnaire carried, for the first time, an explicit check on self-presentation, a screen of six social-desirability (“lie”) items. The screen was rudimentary, and Version 4 would replace it with a standardized inventory, but its presence marks the point at which the programme began treating the flattery risk of first-person virtue reports as a measurement problem rather than a footnote.

The psychometric reduction of the pool was published openly in an exploratory factor analysis available in a public repository (Nehyba, 2021a) and reported to the Czech educational research community in a contribution to the justice-themed ČAPV 2021 conference volume (Nehyba, 2021b, pp. 120–123; see Chapter 3.2). On the cleaned analytic sample of $n = 346$, the repository analysis (Nehyba, 2021a) proceeded on polychoric correlations in the R package psych (Revelle, 2021). Sampling adequacy was acceptable, $KMO = .80$ (Kaiser, 1974), and Bartlett’s test rejected sphericity on the polychoric matrix, $\chi^2(325) = 2991.1$, $p < .001$; parallel analysis (Horn, 1965) indicated four factors, which together accounted for approximately 40% of the common variance. (These sampling-adequacy diagnostics, KMO, Bartlett, parallel analysis, live in the archived repository notebook rather than in the printed conference summary, which reports the four-factor solution, the variance recovered, and the subscale reliabilities below.) The four factors were interpretable as the four intended principles (procedural justice, equality, needs, and merit), and on the basis of this solution the pool was trimmed from 36 to the 26 items of the published version. One reverse-coded item survived into the published set: the equality item “S některými žáky zacházím jinak než s ostatními” (“I treat some pupils differently than others”), whose later fate Section 6.6 describes.

The published subscale reliabilities were adequate for research use: standardized Cronbach’s alphas ranged from .70 to .80. These are respectable values for short principle subscales, but they should not be over-read as the last word on measurement quality in

6. Study 2: Building an Instrument for Self-Perceived Justice of Teaching

either direction. Each subscale is short (six to eight items) and deliberately heterogeneous: the equality subscale, for instance, spans equal time in oral examination, equal attention to pupils' problems, and identical criteria for assessing conduct, which are distinct behavioral facets of one principle, not paraphrases of a single sentiment. Coefficient alpha rewards item redundancy over exactly this kind of content breadth (Cronbach, 1951; Sijtsma, 2009), so a pool built to cover facets rather than repeat them earns its alpha the harder way. The procedural conclusions this book draws are therefore independent of the alpha value itself: reliability is re-examined at the latent level with coefficients appropriate to ordinal items (categorical omega, Chapter 7), and simple sum scores are not recommended for individual diagnostics. What the 2021 analysis established was more modest and more important: that a four-factor structure corresponding to the four principles is recoverable from first-person items at all.

Because the published analysis predates this book's reproducible pipeline, it was replicated end-to-end on the archived data (`justice_2021_clean.csv`, $n = 346$) in the notebook `20_psychometrie.qmd`, with every statistic regenerated and frozen to the chapter's number manifest (`06_cisla.csv`; R package `psych`, version 2.6.5). The replication substantiates the published analysis on every anchor it can reach. Sampling adequacy was reproduced exactly ($KMO = .80$); Bartlett's polychoric test statistic matched the archived notebook to the first decimal ($\chi^2 = 2991.1$ vs. 2991.08); and the four-factor solution on polychoric correlations accounted for 39.2% of the variance, the "approximately 40%" of the published report. Two aspects of the replication warrant transparent comment rather than silent agreement. First, parallel analysis in the replication suggested five factors rather than four, a reminder that resampling-based retention criteria are unstable with ordinal, ceiling-prone items; the four-factor solution itself, which is what the published reduction relied on, was reproduced without difficulty, and the book's own dimensionality evidence (Chapter 7.3) rests on the convergence of theory and exploratory graph analysis rather than on parallel analysis alone. Second, when raw-score alphas are computed for the four *a priori* principle subscales in the final numbering, they range from .69 to .76, consistent with the published standardized range of .70 to .80; the small difference reflects raw versus

standardized coefficients and the published solution's own item-to-scale assignment. The two ranges are not in conflict; they answer slightly different bookkeeping questions about the same data, and both are preserved: the published standardized range as a claim of the source (Nehyba, 2021b), the replicated raw-score range in the manifest. The substantive conclusion is unchanged: a four-principle structure is recoverable from the 2021 items, and the instrument's reliability question is best settled not by any alpha but by the latent-level, ordinal-appropriate coefficients of Chapter 7.

6.5 Version 4 (2022): Consolidation on the Agreement Scale

The 2022 version changed nothing on purpose. The same 26 items, in the same order as the published 2021 version, were administered on the same seven-point agreement scale to a new cohort of $n = 263$ pre-service teachers between February and May 2022 . The point of this deliberate stasis was consolidation: after the reduction from 36 to 26 items, the item set needed to demonstrate that it behaves consistently in a fresh cohort before any further revision could be justified. In the architecture of the book, the 2022 wave thereby completes the pair of cohorts, {2021, 2022}, that share (nearly) identical items on the agreement scale, the second tier of the comparability framework, on which structural replication is examined in Chapter 8.3.

The one genuine addition concerned social desirability. The ad hoc six-item screen of 2021 was replaced by a 20-item version of the Balanced Inventory of Desirable Responding (BIDR; Paulhus, 1991), targeting self-deceptive enhancement . The substitution upgraded the flattery check from an improvised set of lie items to a standardized instrument with a research tradition behind it, and it set the pattern for all subsequent waves, which carry a 19-item BIDR variant . The payoff of this investment is collected in Chapter 7.6, where the independence of the justice structure from socially desirable responding is tested directly.

For the item lineage, the 2022 wave has one bookkeeping consequence worth recording: the two items that Version 5 would reword (the reverse-coded equality item and the

6. Study 2: Building an Instrument for Self-Perceived Justice of Teaching

examination-turns item, items 16 and 22 in the final numbering) still appear here in their old wordings. The archived 2022 dataset preserves those old wordings unrecoded, and every cross-version analysis in Chapter 8 that touches these items goes through the documented crosswalk rather than through item numbers.

6.6 Version 5 (2023): Rewording and the Scale Decision

The final revision, introduced with the 2023 cohort, consisted of two rewordings, a renumbering, and one decision that outranks everything else in this chapter: the change of the response scale.

The first rewording eliminated the instrument's last reverse-coded item. The equality item "S některými žáky zacházím jinak než s ostatními" ("I treat some pupils differently than others") became the positively worded "Se všemi žáky se snažím zacházet stejně" ("I try to treat all pupils the same"). The methodological case against reversed items is well documented; they are misread at higher rates, attract careless-response errors, and tend to generate method artifacts rather than substantive information (Weijters & Baumgartner, 2012; van Sonderen et al., 2013); and a single reversed item among 26 buys none of the intended protection against acquiescence while incurring all of these costs. But the decisive argument was local: under the new response scale described below, respondents judge the justice of the conduct an item describes, and asking them to place "I treat some pupils differently than others" on a scale from completely unjust to completely just stacks a negation on top of an evaluation, a double operation that invites confusion precisely where the item should discriminate. The second rewording concerned the examination-turns item: "Při zkoušení vyvolávám všechny žáky rovnoměrně" ("During examination, I call on all pupils evenly") became "Při zkoušení se snažím všechny žáky vyvolávat se stejnou pravidelností" ("During examination, I try to call on all pupils with the same regularity"). The revision replaced an assertion of accomplished evenness with an expressed effort and made the behavioral referent more specific, a formulation better matched to a population that largely does not

yet run its own classrooms. Alongside these rewordings, two items exchanged positions: the reworded equality item and the adjacent needs item (extra time for a pupil who was ill) swapped places 16 and 17 relative to the 2021/2022 ordering. The swap is trivial in itself but consequential for bookkeeping, which is why all cross-version item identification in this book runs through the crosswalk in `item_map_full.csv` (Appendix A), never through bare item numbers.

The scale decision requires more than bookkeeping. Through 2022, respondents had rated every item on an agreement scale (“Silně nesouhlasím ... Silně souhlasím”, strongly disagree to strongly agree). From 2023 onward, the anchors changed to justice: “Zcela nespravedlivé ... Zcela spravedlivé” (completely unjust to completely just), with the seven-point format retained. The rationale is alignment between the judgment and the construct. Under agreement anchors, a respondent rating “I devote the same amount of time to every pupil during oral examination” is asked, in effect, whether this description of their conduct is accurate; the instrument then measures self-described behavior, and its justice content rides entirely on the item wordings. Under justice anchors, the respondent is asked to pass a justice judgment on their own conduct as the item describes it: the evaluative act itself becomes the datum. For a construct defined as *self-perceived justice*, this is not cosmetic: it moves the construct out of the item stems and into the response process. It also opens a possibility the agreement format forecloses: a respondent can now judge a practice they in fact engage in as less than fully just, or a practice they struggle with as just in principle, so that the instrument registers the perceived justice of conduct rather than its reported frequency.

The cost is comparability, and it is stated here as plainly as the benefit. Anchors are part of the measurement model: responses collected under agreement anchors and under justice anchors do not lie on a common continuum, and no recoding puts them on one. The boundary between the 2022 and 2023 cohorts is therefore a hard comparability boundary. Raw scores, means, and thresholds are never compared across it anywhere in this book; across the boundary, only structure (factor configurations, network topologies) is compared, under the rules of the three-tier framework introduced in Chapter 4.4 and applied in full in Chapter 8.1. This is also why the book systematically avoids growth or trend language across cohorts:

6. Study 2: Building an Instrument for Self-Perceived Justice of Teaching

the cohorts are independent samples, and from 2023 the instrument itself judges differently. What the scale decision purchased with this cost is a final instrument whose response process matches its construct, and stability thereafter: the 2023 version was administered without any further change to the 2023, 2024, and 2026 cohorts, creating the identical-instrument tier on which the validation study of Chapter 7 and the invariance analyses of Chapter 8.2 rest.

6.7 The Final Instrument

The final instrument, in use unchanged since 2023, consists of 26 first-person items covering the four principles: eight items for procedural justice and six each for equality, needs, and merit. The items describe concrete, recurring allocation situations of classroom life (setting and co-creating rules, oral examination, the grading of written work, homework, and the recognition of pupil effort and performance), and the respondent rates each described behavior on a seven-point scale from 1 = completely unjust (*zcela nespravedlivé*) to 7 = completely just (*zcela spravedlivé*). The instrument contains no reverse-coded items: after the 2023 rewording of item 16, all items are positively keyed, so that the judgment task is uniform across the instrument, one evaluative operation, applied to one described behavior at a time. The trade-off is acknowledged: without balanced keying, response styles such as acquiescence cannot be separated from content by design, which is one reason the validation study measures socially desirable responding directly (BIDR) and models the structure at the latent level (Chapter 7.6). Table 6.1 presents the items by principle in English working translation; the authoritative Czech wordings, the full English versions, and the complete cross-version crosswalk are given in Appendix A.

Table 6.1. The final 26-item instrument by principle (English draft wordings)

Item	Wording (EN draft)
Procedural justice (8 items)	

Item	Wording (EN draft)
1	At the start of the year, it is important to me to clearly establish the rules of how the class works.
2	I clearly explain to pupils what I emphasize during oral examination.
6	I inform my pupils in advance how I handle any conflicts in the class.
7	I tell pupils they can discuss any discrepancies in the grading of a test with me at any time.
8	I announce to pupils in advance the criteria for grading their written work.
23	I explain to pupils how to give peer feedback, e.g., when assessing each other's work.
24	I give pupils room to co-create the rules of behavior in the class.
26	It is important to me to have clear criteria by which I assess pupils.
Equality (6 items)	
3	I devote the same amount of time to every pupil during oral examination.
10	I devote the same degree of attention to every pupil's problems.
14	I give the same degree of attention to all pupils.

6. Study 2: Building an Instrument for Self-Perceived Justice of Teaching

Item	Wording (EN draft)
15	When assessing every pupil's behavior, I always use the same criteria.
16	I try to treat all pupils the same.
22	During examination, I try to call on all pupils with the same regularity.
Needs (6 items)	
4	If a pupil is going through a difficult life situation (e.g., parents' divorce), I take it into account when grading.
5	I give more care to the pupil who, in my view, needs it most.
9	When grading, I take into account whether the subject is difficult for a particular pupil.
11	To a pupil with a long-term problem understanding the material, I assign different homework than to others.
17	To a pupil who was ill, I give extra time so they can catch up on the material.
18	I would initially adjust the assessment of a pupil who transferred from another school.
Merit (6 items)	
12	I reward pupils who finish their classwork ahead of the deadline (e.g., they may work on their own things).
13	If a pupil completes an extra task during the lesson, I give due recognition to their effort.

Item	Wording (EN draft)
19	I acknowledge a pupil's exceptional effort in the lesson appropriately.
20	I praise a pupil who successfully represents the school in a knowledge competition.
21	I give recognition to a pupil who showed effort beyond the assigned task.
25	I appreciate a pupil who can carry out various class activities.

The composition of the subscales reflects the theory of Chapter 2 rather than statistical convenience. The procedural items operationalize the classic procedural rules (transparency of criteria, advance notice, voice, and the possibility of correction) translated into teachers' everyday acts. The three distributive subscales deliberately confront the respondent with principles that can conflict: the equality items assert sameness of treatment across pupils, the needs items assert justified departures from sameness, and the merit items assert differentiation by effort and performance. A respondent who rates all 26 behaviors as completely just is thereby endorsing tensions the instrument is designed to expose; how the four principles actually organize themselves in respondents' judgments (as four correlated factors, a hierarchy, or a network of local dependencies) is precisely the structural question the instrument hands to Chapter 7.

6.8 Chapter Summary: Open Questions for Validation

Five versions in five cohort-years took the instrument from a plausible first draft to a defensible operationalization, and the development can be compressed into three lessons: perspective, person, and scale. The lesson of perspective (Versions 1 and 2): items about "the

6. Study 2: Building an Instrument for Self-Perceived Justice of Teaching

teacher” measure a generalized other; the descriptive frame of 2019 left the judgment target indeterminate, and the normative frame of 2020 resolved the indeterminacy into a well-defined but different construct, normative beliefs about just teaching, preserved in this book as the tier-3 cohort. The lesson of person (Versions 3 and 4): only first-person items reach self-perception, and the 2021 pool with its published EFA established that a four-factor structure matching the four principles is recoverable from such items, with adequate standardized subscale alphas of .70 to .80 earned by short, deliberately heterogeneous behavioral facets rather than by item redundancy, a range this book nonetheless revisits with latent-level reliability estimation. The lesson of scale (Version 5): even first-person items measure self-described behavior if the anchors ask for agreement; the 2023 move to justice anchors placed the justice judgment itself into the response process, at a hard, openly accounted comparability cost between the 2022 and 2023 cohorts.

RQ2 asked whether self-perceived justice toward pupils can be operationalized as a reliable four-principle self-report instrument. What this chapter establishes is the operationalization: a 26-item, four-principle, first-person instrument with a uniform justice-judgment response format, stable across the 2023, 2024, and 2026 cohorts, with its full item lineage documented back to 2019. What it deliberately does not establish is the psychometric verdict, and three open questions carry forward. First, structure: does the four-factor organization survive confirmatory testing on independent cohorts, and does a network lens, which imposes no latent variables at all, recover the same organization (Chapter 7.3–7.5)? Second, reliability: do latent-level coefficients appropriate to ordinal items confirm, at the individual-diagnostic level, subscales whose published sum-score alphas were adequate but built from short, heterogeneous facets (Chapter 7.3, with the cross-cohort reliability record in Chapter 8.5)? Third, contamination: is the structure independent of socially desirable responding, the threat the development history took seriously from the six lie items of 2021 to the BIDR of 2022 onward (Chapter 7.6)? The instrument built here is, in short, a set of well-motivated claims awaiting verification, and Chapter 7 is where they are put on trial.

7. Study 3: The Structure of Perceived Justice: Factor and Network Views

Aim of the chapter

This chapter answers RQ3 by testing whether the factor-analytic and the psychometric-network view of the final 26-item self-perceived justice instrument converge on the same four-principle structure (procedural, equality, needs, merit) in the pooled sample of the three fully comparable cohorts, and whether that structure is independent of social desirability.

7.1 Focused Background: Two Ways to Think About a Construct

What kind of thing is “self-perceived justice of teaching”? The question sounds philosophical, but it has two concrete statistical translations, and the choice between them changes what counts as evidence of structure. The classical translation is the reflective latent variable: each of the four principles is a common cause of its items, so that a student teacher’s standing on, say, the equality principle explains why her answers to the six equality items rise and fall together. On this view the items are interchangeable symptoms of an underlying attribute, their correlations exist *because of* the attribute, and factor analysis is the appropriate measurement model. The alternative translation, developed over the past fifteen years in network psychometrics, drops the common cause: a construct is a system of

7. Study 3: The Structure of Perceived Justice: Factor and Network Views

mutually reinforcing components, and items correlate not because something behind them moves them jointly but because the behaviors and judgments they describe influence one another directly (Borsboom & Cramer, 2013). For illustration, not as anything this study tests, a teacher who lets pupils voice objections during oral examination might, by that very practice, come to apply criteria more transparently, and that transparency might in turn make equal treatment easier to enact. On this view the structure of the construct is the pattern of direct dependencies, and the appropriate model is a network.

The workhorse of the network translation is the Gaussian graphical model (GGM): items are nodes, and edges are partial correlations: the association that remains between two items after conditioning on all the others. Because dozens of weak, spurious edges would make such a graph uninterpretable, the model is estimated with regularization (EBICglasso), which shrinks small edges to exactly zero and retains a sparse, interpretable web (Epskamp, Borsboom, & Fried, 2018). Communities of densely interconnected nodes then play the structural role that factors play in the classical translation, and exploratory graph analysis (EGA) detects them algorithmically, without being told how many to find or which items belong together (Golino & Epskamp, 2017). For an educational-research audience the essential contrast can be put in one sentence: factor analysis asks *what common causes could have generated these correlations*, network analysis asks *which items are directly connected once everything else is held constant*, and the two questions have no obligation to return the same answer.

That last point is what this chapter exploits. Where the two lenses are treated as rivals, one must be declared the true model, a metaphysical commitment the present data cannot adjudicate. This book treats them instead as complementary lenses on the same correlation structure, which turns their agreement into an empirical result. If a method that presupposes four latent causes and a method that presupposes none converge on the same four-fold partition of the 26 items, that convergence is structural-validity evidence of a strong kind, because it cannot be an artifact of either method's assumptions. If they diverge, the divergence localizes exactly which items behave "factor-like" and which behave "network-like," and the instrument's users learn where its theoretical map is reliable. Sections 7.3 and 7.4

apply the two lenses in turn; Section 7.5 quantifies their agreement.

7.2 Sample and Data

The analytic sample pools the three cohorts that sit in the identical-instrument tier of the comparability framework (Chapter 4): the 2023 wave ($n = 247$), the 2024 wave ($n = 333$), and the 2026 wave ($n = 164$), for a total of $n = 744$ pre-service teachers. All three cohorts answered the same 26 items, in the same first-person wording, on the same seven-point justice scale (1 = *completely unjust* ... 7 = *completely just*); there are no missing values on any item. One pair of respondents shares an identical 26-item response pattern; rather than deciding whether the duplication is substantive coincidence or an artifact, the pattern was left in the main analyses and every key model was re-estimated without its second occurrence ($n = 743$) as a sensitivity check, with, as Section 7.3 reports, no visible consequence.

Pooling three cross-sectional cohorts into one analytic sample presupposes that the instrument measures the same thing in each; that presupposition is itself an empirical claim, and this book treats it with a two-stage discipline. Within this chapter, a minimal invariance gate (configural versus loading-constrained multigroup models) screens whether pooled estimation is defensible at all: constraining loadings to equality across the three waves cost $\Delta CFI = .004$, comfortably under the conventional .01 bound (Cheung & Rensvold, 2002), so the pooled analyses below are provisionally supported. Two caveats qualify that support. Because several items have empty response categories within single waves, a direct consequence of the ceiling effects documented next, the gate was estimated with a robust continuous approximation rather than the full ordinal machinery; and it constrains loadings only, leaving threshold invariance, the ordinal requirement most exposed to the ceiling problem, untested here. Chapter 8 therefore applies the fullest estimable ladder: configural through strict invariance under the MLR continuous approximation, followed by latent-mean comparisons once scalar invariance holds. A full ordinal threshold ladder proved inestimable because of the empty categories. Pooling is consequently supported by the loading gate here

7. Study 3: *The Structure of Perceived Justice: Factor and Network Views*

and the later strict MLR result, while untested threshold invariance remains an explicit limitation rather than a pending promise.

The items themselves behave as the instrument's history predicts. Distributions are strongly ceiling-shifted: on average 37% of respondents choose the top category of an item, and the most extreme item draws 75% of responses to the scale maximum. Skew of this magnitude is not a nuisance to be normalized away; it is the expected signature of virtue self-reports and one more reason the analyses below treat the items as ordered categories throughout (polychoric correlations, WLSMV estimation with theta parameterization for factor models, and rank-appropriate methods in the network sections) rather than as continuous scores.

7.3 The Factor View

7.3.1 Exploration: the 2023 wave

The factor sequence follows the pre-registered temporal split: explore on 2023 ($n = 247$), confirm on 2024 ($n = 333$), validate on the untouched 2026 holdout ($n = 164$). Exploration began with adequacy checks on the polychoric matrix ($KMO = .775$, Bartlett's test $p < .001$) and with the dimensionality question put to two methods with different assumptions. Exploratory graph analysis recovered exactly four communities, matching the theoretical taxonomy. Parallel analysis suggested five factors; the same procedure had wavered between four and six across data sets and reruns elsewhere in this programme, and its instability with ordinal, ceiling-prone items is a documented weakness, so retention rested on the convergence of theory and EGA, with parallel analysis recorded rather than obeyed. A four-factor EFA (minres, oblimin, polychoric) accounted for 44.2% of the variance, with 23 of the 26 items loading saliently ($\geq .40$) on some factor and the pattern interpretable in terms of the four principles; the three items falling short of the salience threshold were retained: the instrument enters confirmation intact, as pre-specified, with no silent trimming. The full 26×4 loading matrix is given in Appendix C (Table C.1).

7.3.2 Confirmation: competing models on the 2024 wave

Confirmation asked five models to account for the 2024 wave (ordinal confirmatory factor analysis [CFA], WLSMV estimation, theta parameterization; Table 7.1). The one-factor model fails decisively ($CFI = .519$, $RMSEA = .139$): whatever self-perceived justice is, it is not one undifferentiated disposition, and the scaled likelihood-ratio test against the four-factor model is unambiguous ($p < .001$). The focal four-factor correlated model (the instrument's theoretical blueprint translated literally, each item loading on exactly one principle) reaches only mediocre global fit ($CFI = .812$, $RMSEA = .088$, $SRMR = .086$), and the higher-order variant does no better ($CFI = .797$). The bifactor model improves the fit ($CFI = .848$, $RMSEA = .082$), but its tandem indices dismantle the interpretation a naive reader might draw from that improvement: the general factor claims less than half the common variance ($ECV = .476$) and reaches $\omega_H = .632$ with $PUC = .775$, far below the benchmarks at which a total score could be treated as essentially unidimensional (Rodriguez, Reise, & Haviland, 2016). The general factor of self-perceived justice, in other words, exists but is weak; the group factors carry the reliable specificity, and a single summed “justice score” is not licensed, a conclusion this book states here, before any reviewer does.

Table 7.1. Competing measurement models, 2024 confirmation wave ($n = 333$; ordinal WLSMV)

Model	χ^2 (scaled)	df	CFI	TLI	RMSEA	
					[90% CI]	SRMR
One factor	2164.31	299	.519	.477	.139 [.132, .146]	.136
Four factors, correlated (focal)	912.57	293	.812	.792	.088 [.080, .095]	.086
Higher-order	1060.50	295	.797	.776	.091 [.083, .098]	.095

7. Study 3: The Structure of Perceived Justice: Factor and Network Views

Model	χ^2 (scaled)	df	CFI	TLI	RMSEA	
					[90% CI]	SRMR
Bifactor	840.81	273	.848	.820	.082 [.074, .090]	.080
ESEM (four factors, geomin)	417.49	227	.916	.879	.067 [.057, .076]	.040

Note. Source: 07_cfa_modelfit.csv (notebook 20_psychometrie.qmd; manifest 07_cisla.csv). Robust (scaled) indices. Bifactor verdict rests on tandem indices (ECV = .476, ω_H = .632, PUC = .775), not on fit alone. Information criteria (AIC/BIC) are not reported: under the WLSMV estimator appropriate to ordinal items there is no maximum-likelihood value to compute them from, so model comparison follows the ordinal-CFA standard of the scaled χ^2 , CFI/TLI, RMSEA, and SRMR shown here.

The best-fitting model is exploratory structural equation modeling (ESEM; Asparouhov & Muthén, 2009), the four-factor model re-estimated with all cross-loadings freed under geomin rotation, at CFI = .916, RMSEA = .067, and SRMR = .040. That ESEM fits better is in one sense guaranteed rather than discovered: freeing every cross-loading spends 66 additional parameters (df = 227 vs. 293), so an ESEM solution cannot fit worse than the independent-clusters CFA nested within it. The preference for it therefore rests not on the fit gain itself but on two substantive checks. First, the reason the independent-clusters CFA fails is diagnostic: it fails not because the four principles are wrong but because it forbids every secondary loading, and items deliberately written as heterogeneous behavioral facets (Chapter 6.4) carry real, small secondary relationships to neighboring principles: announcing assessment criteria in advance is procedurally transparent *and* equality-tinged. ESEM prices those relationships in and the four-factor organization snaps into good fit. Second, the pre-registered inflation check confirms the fit gain was not bought by letting the fac-

tors blur together: ESEM's factor correlations are in fact *lower* than the focal model's, not higher (mean absolute $r = .283$ vs. $.433$; maximum $.540$ vs. $.603$), which is the signature of genuine cross-loadings rather than over-correlated factors. The focal model's own correlations already tell the story of distinct-but-related principles: the strongest association is procedural–equality ($r = .603$), the weakest equality–needs ($r = .112$), theoretically apt, since equality asserts sameness of treatment while needs asserts justified departure from it, and none approaches the level at which two factors would collapse into one.

7.3.3 Reliability at the latent level

The reliability verdict that the instrument's alpha history ($\alpha \approx .69$ – $.80$ across cohorts and scale assignments; Chapter 6) could not by itself deliver arrives here. Categorical omega on the latent-response scale of the 2024 four-factor model yields procedural $\omega = .877$, equality $\omega = .853$, needs $\omega = .720$, and merit $\omega = .860$; the pooled sample reproduces the same band ($.732$ – $.871$), and the 2026 holdout independently replicates it ($.735$ – $.897$). Needs is consistently the weakest subscale (short, heterogeneous, at $\omega = .720$ only just past the conventional threshold, and the one to watch in the network sections), but every subscale clears conventional adequacy at the latent level. These coefficients estimate reliability at the latent level, not the reliability of any sum score a user might compute, consistent with the verdict above against a single composite. With that qualification, the published sum-score alphas (adequate in the $.70$ – $.80$ band, as Chapter 6 reports) understate, rather than cap, the instrument's latent-level reliability, which these ordinal-appropriate coefficients estimate directly.

7.3.4 Holdout: the 2026 wave, reported without cosmetics

The untouched 2026 wave ($n = 164$) received the focal and winning models exactly as fixed on 2024, with no modifications permitted. The result is a partial replication, and it is reported as one. Global fit degrades for both models ($CFI = .751$, $RMSEA = .111$ for the four-factor CFA; $CFI = .792$, $RMSEA = .116$ for ESEM) while everything about the *content* of the

7. Study 3: *The Structure of Perceived Justice: Factor and Network Views*

solution replicates: the loading pattern is congruent with 2024 in configuration (Tucker $\varphi = .991$) though its magnitudes correlate only moderately ($r = .743$), φ indexing agreement in shape, the correlation the stricter agreement in size, and the subscale omegas come out at 2024 levels or above (.735–.897). Two readings of this dissociation are possible. The benign reading notes that WLSMV fit statistics with 26 ordinal indicators are unstable at $n = 164$ and that congruent loadings with replicated reliabilities are exactly what parameter-level replication looks like. The skeptical reading takes the fit indices at face value and concludes that the model's misfit is amplified, not attenuated, in the smallest cohort. The pre-registered evaluation standard for the holdout (fit replication *and* parameter congruence, with explicit caution against strict cutoffs at this sample size) was written before these numbers existed, and by that standard the holdout supports parameter-level replication while leaving global fit an open question that the full invariance analysis of Chapter 8, with all three cohorts modeled jointly, is positioned to settle.

7.3.5 Pooled estimates and sensitivity

With the invariance gate passed (Section 7.2), the four-factor model was re-estimated on the pooled $n = 744$ for the parameter values that later chapters cite: CFI = .819, RMSEA = .081, SRMR = .075, with pooled subscale omegas of .867 (procedural), .836 (equality), .732 (needs), and .871 (merit). The duplicate-pattern sensitivity re-run ($n = 743$) moved the pooled fit by one point in the third decimal (CFI .818 vs. .819): the duplicated pair is analytically irrelevant. One bookkeeping deviation from the analysis plan is recorded here rather than hidden: the plan anticipated “two duplicate rows” ($n = 742$); the data contain one duplicated pair, so the sensitivity sample is $n = 743$.

The factor view's summary judgment: four correlated principles, distinct but not orthogonal, best represented with cross-loadings admitted (ESEM), reliable at the latent level, with no license for a single total score. What the factor view cannot say is whether this four-fold organization is an artifact of asking latent-variable questions. Section 7.4 asks a method that has never heard of latent variables.

7.4 The Network View

7.4.1 Estimation and stability

The network lens was applied to the pooled sample with the pre-registered pipeline (notebook 30_network.qmd; all numbers frozen in 07_network_cisla.csv). A redundancy screen (unique variable analysis) preceded estimation: three item pairs exceeded the moderate-redundancy threshold ($wTO > .25$), two equality pairs (items 15–16, $wTO = .288$; items 10–14, $wTO = .262$) and one merit pair (items 12–13, $wTO = .283$). All three sit *within* their a priori principle blocks (redundancy of facet wording inside a principle, not blurring between principles) and, per the pre-registered rule, they are reported here and retained: no item was removed (all nine pairs above $wTO = .20$ retained in the frozen screen output are tabulated in Appendix C, Table C.2).

The regularized partial-correlation network (EBICglasso, $\gamma = .5$) retains 143 of 325 possible edges (density .44). Two robustness observations frame everything that follows. First, the solution is robust to the regularization choice in the sense that matters: lowering γ to .25 barely moves the retained-edge count (145), and although the unregularized-prior extreme $\gamma = 0$ admits substantially more edges (200), the edge weights correlate at $r = 1.00$ and .975 with the focal solution: it is the weights, not the raw counts, that carry every interpretation below. Second, the estimation emitted the standard dense-network caution (with a low selected penalty, the smallest retained edges may include false positives), so no interpretation below leans on any individual weak edge. Stability of what is interpreted was checked the mandatory way (Epskamp et al., 2018): case-drop bootstrap correlation-stability coefficients reached $CS = .672$ for strength and $CS = .75$ for expected influence, both clearing not just the minimal .25 but the preferred .50 threshold; edge weights carried acceptable nonparametric bootstrap intervals (1,000 resamples each).

7.4.2 Centrality and predictability

Following the pre-registered restriction, only strength and expected influence are interpreted; betweenness and closeness are not computed, in line with the critique of borrowing flow-based centralities for psychological networks (Bringmann et al., 2019). The strongest node in the network is item 8, announcing grading criteria for written work in advance, and the weakest is item 12, the reward-for-early-finishers merit item, whose idiosyncratic behavioral content connects it only loosely to the rest of the system. Node predictability (Haslbeck & Waldorp, 2020; mean $R^2 = .306$, maximum .485) indicates that on average roughly 31% of an item's variance is accounted for by its direct neighbors, a moderate level typical of attitude and self-report networks: the items form a genuinely interconnected system, but no item is reducible to its neighborhood.

7.4.3 Communities: the network's answer to the dimensionality question

Exploratory graph analysis, given no information about the four principles, returned exactly four communities, and so did every robustness probe of the *count*: the Louvain algorithm agreed with Walktrap, and across 1,000 resampled networks the median number of dimensions was four, with 99.6% of bootstraps returning exactly four . The dimensionality verdict of the network lens is thus unusually emphatic. The *composition* of the communities is almost as clean, with one localized and instructive exception discussed in Section 7.5: 25 of 26 items land in the community corresponding to their a priori principle (Figure 7.1 shows the same network under both colorings). The same softness surfaces at item level (these are two views of one phenomenon, not independent corroborations): mean item stability across bootstraps is .958, with only two items below the .70 benchmark, item 22 (.632) and item 26 (.316); and structural consistency is essentially perfect for the procedural, needs, and merit communities (1.00, .99, .99) but low (.32) for the equality community, whose boundary with the procedural block is exactly where those two items wobble. The *count* of communities is rock-solid; the *composition* is clean for three communities and soft along one seam, and the seam is interpretable.

A hierarchical EGA, run as an exploratory coda (the implementation is flagged experimental by its authors), suggested eight lower-order facets nesting under three higher-order groupings, consistent with the reading that the four principles are themselves composed of situational facets (Chapter 6.1), and is reported for completeness rather than leaned on.

7.4.4 The same network in all three cohorts

Finally, the network comparison test (NCT; van Borkulo et al., 2023) was applied pairwise to the three waves, with identical estimation settings throughout, the within-tier check that the pooled network does not average over heterogeneous cohort networks. Nothing approaches significance: network structure invariance $p = .434$ (2023 vs. 2024), $.859$ (2023 vs. 2026), and $.405$ (2024 vs. 2026); global strength $p = .312$, $.396$, and $.255$; and after Holm correction not a single individual edge differs in any pairwise comparison. Non-significance is absence of detectable difference, not a demonstration of equivalence (the permutation test can only fail to find what may be too small to see), but within the identical-instrument tier it is exactly the pattern pooling presupposes: the network lens detects no cohort differences at any of the three levels it can test. Figure 7.2 makes the point visible: the three cohort networks, drawn on a *shared* layout so that node positions are comparable across panels, are near-indistinguishable: the same communities in the same places, the same strong intra-community edges. Chapter 8 develops the corresponding invariance evidence on the factor side.

7. Study 3: The Structure of Perceived Justice: Factor and Network Views

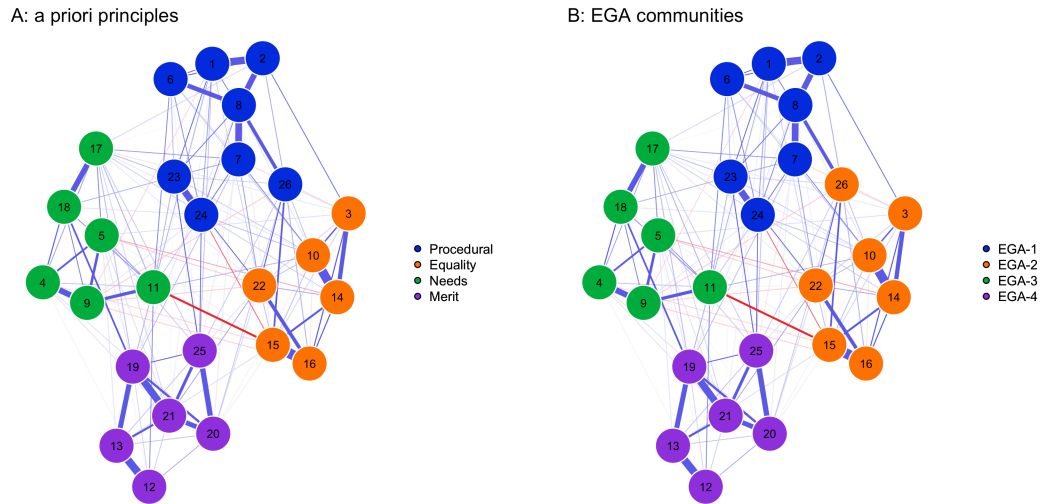


Figure 7.1. The pooled justice network ($n = 744$) under a priori (A) and data-driven EGA (B) coloring, one shared layout. Node colors: procedural (blue), equality (orange), needs (green), merit (purple). Source: `analyzy/scripts/fig_07_networks.R`.

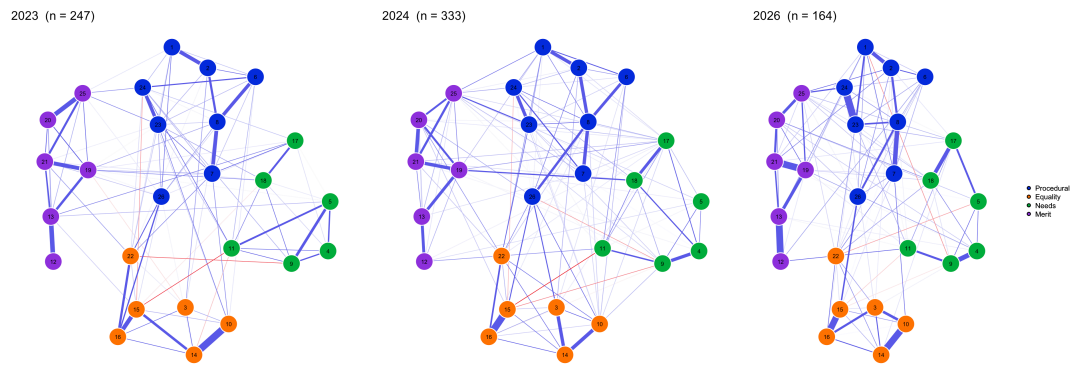


Figure 7.2. The justice network estimated separately in each identical-instrument cohort (2023, 2024, 2026), drawn on a single shared layout so positions are comparable across panels. The visual near-identity is the qualitative counterpart of the null network-comparison tests above. Source: `analyzy/scripts/fig_07_networks.R`.

7.5 Convergence and Divergence of the Two Views

The chapter's central question was whether a method that presupposes latent causes and a method that presupposes none return the same organization. They nearly exactly do. Agreement between the EGA communities and the a priori four-principle assignment is $ARI = .885$ ($NMI = .918$); between EGA and the exploration-wave EFA assignment $ARI = .790$; and between the EFA and the a priori taxonomy $ARI = .887$, high on every pairing, by any conventional reading of adjusted chance-corrected agreement (Table 7.2 gives the full cross-tabulation). The factor lens and the network lens, applied to the same 744 respondents, recover the same four-principle architecture that the instrument was built to have.

Table 7.2. Cross-tabulation of a priori principles $\times$ EGA communities (pooled $n = 744$)

A priori principle	EGA-1	EGA-2	EGA-3	EGA-4
Procedural (8 items)	7	1	0	0
Equality (6 items)	0	6	0	0
Needs (6 items)	0	0	6	0
Merit (6 items)	0	0	0	6

Note. Source: 07_network_crosstab_apriori_ega.csv. The single off-diagonal item is item 26.

The one disagreement is as informative as the agreement. The migrating item is item 26 (*“It is important to me to have clear criteria by which I assess pupils”*), a procedural item by design that the network assigns to the equality community, and the least stable node in the bootstrap (.316). Its companion soft spot, item 22 (*“During examination, I try to call on all pupils with the same regularity”*), is an equality item by design, the second-least stable node (.632), and the network's strongest *bridge* between communities (Jones et al., 2021; bridge strength .534, the maximum in the network). Both items live on the same conceptual seam: clear, evenly applied criteria are simultaneously a property of fair procedure and the mechanism

7. Study 3: The Structure of Perceived Justice: Factor and Network Views

by which equal treatment is enacted. Chapter 5 met this seam from the other direction, when the classroom transpositions of distributive principles kept dissolving into properties of procedure (Section 5.6.3), and Section 7.3.2 met it again in the item-level cross-loadings that ESEM needed to price in. Three complementary indications of the same boundary constitute, in this book's reading, less a flaw of the instrument than a finding about the construct: in teachers' self-perception, procedural justice and equality are distinct organizing principles that *share a hinge*: the transparent, uniformly applied criterion. The two readings are, however, not mutually exclusive: the same numbers that support the hinge interpretation also flag item 26, with its low bootstrap stability, as a natural candidate for rewording should the instrument be revised.

7.6 Is the Structure Contaminated by Social Desirability?

First-person virtue reports invite flattering self-presentation, and the instrument's development history treated that risk as a measurement problem from 2021 onward (Chapter 6). The final validation asks the discriminant-validity question directly: do the justice factors reflect socially desirable responding? The 2023–2026 waves carried a 19-item self-deceptive-enhancement scale drawn from the Czech adaptation of the Balanced Inventory of Desirable Responding (BIDR; Paulhus, 1991; Czech version BIDR-CZ, Preiss & Mačudová, 2013). The administered items correspond to BIDR items 1–17, 19, and 20 of the source version; item 18 was omitted deliberately, on ethical rather than psychometric grounds: the item asks respondents to evaluate their abilities as lovers, and in a teacher-education setting, where the questionnaire is administered by instructors to their own students, an intimate question of this kind is disproportionate to the study's purpose and inappropriate to the asymmetric instructor–student relationship; under the data-minimization principle of Chapter 4.7, information touching on intimate life is data the study had no scientific reason to collect. Dropping this single item leaves the self-deceptive-enhancement construct covered by the remaining 19 items and is immaterial for a scale used here solely as a discriminant-validity check. The keying (nine reversed items describing admitted frailties, the canonical even-

numbered items plus item 20) was reconstructed from item wordings and confirmed against the source version of the Czech instrument. Scored continuously after reversal ($M = 3.92$, $SD = 0.64$, $\alpha = .711$, $n = 744$), the resulting score (called *self-deception* below, the customary shorthand for self-deceptive enhancement) is essentially unrelated to all four justice factors: $r = .012$ (procedural), $.161$ (equality), $-.070$ (needs), and $.040$ (merit), with Spearman coefficients telling the same story. Under Paulhus's stricter dichotomized scoring the largest coefficient rises to $.235$, still a small effect. At the item level no justice item correlates with self-deception beyond $|r| = .168$, and, pre-empting the worry that the network's core might be socially saturated, an item's centrality is unrelated to its social-desirability loading (rank correlation between strength and $|r|$ with BIDR) $= .095$).

The only coefficient that even invites comment is equality's $r = .161$: under 3% shared variance, small by every convention, but directionally sensible; of the four principles, flat equal treatment is the most publicly creditable thing a future teacher can claim, and a trace of enhancement in its self-report is exactly where one would predict it. The conclusion for the validity argument is nonetheless plain: the four-principle structure recovered twice in this chapter is not an artifact of impression management. What the instrument measures may be self-*perception* rather than observed practice (Chapter 9 returns to that boundary), but it is not, on this evidence, self-*flattery*.

7.7 Chapter Summary

RQ3 asked whether the factor and network views of the final instrument converge on the four-principle structure, and whether that structure survives a social-desirability audit. The answer to both is yes, with precisely mapped qualifications. The factor lens rejected a single justice dimension outright, found the rigid independent-clusters translation of the four principles too strict for deliberately heterogeneous behavioral items, and settled on four distinct-but-related factors with cross-loadings admitted (ESEM; CFI = .916), reliable at the latent level ($\omega = .72-.88$ in the confirmation wave, with the pooled and holdout samples in the

7. Study 3: The Structure of Perceived Justice: Factor and Network Views

same band) but licensing no summed total score (bifactor ECV = .476, ω_H = .632). Without using a priori principle labels in community detection, the network lens put the number of dimensions at four in 99.6% of 1,000 resamples, with stable centralities ($CS \geq .67$) and edge weights carrying acceptable bootstrap intervals; it detected no cohort differences across the three waves (all NCT comparisons null); and its community composition agrees with the a priori taxonomy at ARI = .885, cleanly for three communities, softly along the equality–procedural seam (structural consistency .32). The two lenses disagree about a single item (the clear-criteria item 26), and their disagreement, together with the bridging behavior of item 22 and the echoes in Chapters 5 and 6, localizes a substantive hinge between procedural justice and equality rather than a measurement defect. Social desirability explains under 3% of any factor’s variance in the primary continuous scoring (at most ~5.5% under the stricter dichotomized scoring) and is unrelated to the network’s core. What remains open is the outer boundary of generalization: the three pooled cohorts share one instrument and one scale, but the research programme spans seven cohorts and five instrument versions, and whether the structure validated here also travels across those boundaries is the question Chapter 8 was built to answer.

8. Study 4: Stability Across Cohorts and Instrument Versions

Aim of the chapter

This chapter answers RQ4 by testing whether the four-principle structure of perceived justice established in Chapter 7 is stable across the cohorts of 2020–2026 and the instrument versions documented in Chapter 6. Every comparison is disciplined by the three-tier comparability framework from Chapter 4: each set of cohorts is compared only at the level of rigor their shared measurement warrants, and no growth or trend language is used: the cohorts are independent samples of different people, and this chapter asks whether a *structure* recurs, not whether individuals *changed*.

8.1 What Can and Cannot Be Compared: The Three-Tier Framework Applied

Before a single coefficient is shown, this section states what the multi-cohort programme permits and forbids (Study 4 tests the six questionnaire cohorts of 2020–2026; the seven-cohort-year count of Chapter 1 includes the 2019 vignette cohort, which belongs to Study 1) because the central threat to a multi-cohort argument is not a weak result but an illegitimate comparison. The cohorts differ along three measurement dimensions at once (item set, item

8. Study 4: Stability Across Cohorts and Instrument Versions

wording, and response scale) and the three-tier framework assigns each pair of cohorts to the strongest comparison its shared measurement can bear, and no stronger.

Tier 1: identical instrument {2023, 2024, 2026}. These three cohorts answered the same 26 items, in the same first-person wording, on the same seven-point justice scale. Only here is the full apparatus of measurement invariance licensed, and only here can latent means be compared: this is the one place in the book where a difference in *level* between cohorts is interpretable rather than an artifact of changed measurement.

Tier 2: same items, different scale {2021, 2022}. These cohorts used essentially the same 26 items but on the earlier *agreement* scale, before the 2023 move to *fairness* anchors (Chapter 6.6). The scale boundary falls between 2022 and 2023. Across it, only *structure* can be compared (the pattern of loadings and the community organization), never means or thresholds, whose metric is not shared. Comparisons here take the form of factor congruence, community agreement, and descriptive (never inferential) network similarity.

Tier 3: different frame {2020}. The 2020 cohort answered third-person *normative* items (“A teacher should...”), a different construct frame (Chapter 6.3). Only a configural question is admissible: does a four-principle organization emerge at all? No parameter equivalence is tested.

Three things are therefore explicitly out of bounds, and the chapter never does them: pooling raw scores across the scale boundary; comparing latent means or thresholds between 2022 and 2023 (the metric is not shared); and any growth, trend, or developmental language across cohorts (they are non-probability samples of different students under a changing instrument). The rhetorical shorthand for the whole chapter (*the structure travels, the scores do not*) means precisely that construct structure can be compared everywhere the items align, whereas score levels can be compared only inside Tier 1, by measurement necessity rather than by empirical discovery.

8.2 The Identical-Instrument Tier: Invariance and Latent Means 2023–2024–2026

A note on estimator. The pre-registered plan called for ordinal three-group invariance (WLSMV; Wu & Estabrook, 2016). That route proved infeasible for a data reason worth stating plainly: 18 of the 26 items have at least one empty response category within a wave, always at the bottom of the scale (categories 1–3), a direct consequence of the ceiling effect documented in Chapter 7, and most severe in the small 2026 cohort. Empty categories make per-group thresholds jointly inestimable under WLSMV. The invariance ladder is therefore estimated with robust maximum likelihood (MLR), treating the items as continuous (the standard, estimable route to a latent-mean comparison for ceiling-prone Likert data). Separately estimated per-wave ordinal CFAs provide a descriptive complement, not a test of multigroup configural or threshold invariance; on the network side, the null network comparison tests reported in Chapter 7.4.4 provide another complementary view. The convergence of these routes, not any one of them, carries the claim.

Invariance holds to the strictest level tested (Table 8.1). Constraining the model step by step across the three cohorts costs almost nothing in fit: the change in robust CFI is .004 from configural to metric (equal loadings), a further .002 to scalar (equal intercepts), and .007 to strict (equal residual variances), every step comfortably under the conventional .01 criterion (Chen, 2007; Cheung & Rensvold, 2002). Under this continuous approximation, measurement invariance is tenable up to and including residual (strict) invariance across the 2023, 2024, and 2026 cohorts, a rung beyond the scalar level the pre-registered plan set as the endpoint for latent-mean comparison, added here as a conservative check (Meredith, 1993). Two qualifications travel with the claim, and neither is cosmetic: it holds in the MLR (continuous) metric, and ordinal *threshold* invariance is not claimed, because the thresholds could not be tested across groups. Separately estimated ordinal CFAs provide only a descriptive complement: their similarly imperfect fit (robust CFI = .772/.812/.751, robust RMSEA = .089/.088/.111 for 2023/2024/2026) shows no single wave as an obvious outlier, but these models do not test multigroup configural or threshold invariance.

8. Study 4: Stability Across Cohorts and Instrument Versions

Table 8.1. Three-group measurement invariance, Tier 1 (MLR; 2023/2024/2026, $n = 744$)

Model	CFI	RMSEA	SRMR	Δ CFI
Configural	.840	.062	.074	
Metric (loadings)	.836	.061	.080	.004
Scalar (+ intercepts)	.834	.060	.080	.002
Strict (+ residuals)	.827	.060	.084	.007

Note. Source: 08_tier1_invariance.csv (40_crosswave.qmd). Robust indices; Δ CFI relative to the previous, less constrained model.

Latent means are, with one small exception, equivalent. Because scalar invariance holds, latent means can be compared, the one legitimate level comparison in the book. Of the eight cohort-versus-2023 contrasts (four factors $\times$ two later cohorts), seven are non-significant and negligible in size. The single exception is the procedural factor in 2026, standardized latent-mean difference $d = 0.33$ ($p = .003$), a small-to-moderate elevation, the largest absolute difference in the set, and the one result to read with caution: the MLR model returned a non-positive-definite covariance warning whose smallest eigenvalue (-6.7×10^{-16}) is at machine precision, indicating numerical rounding rather than genuine rank deficiency, so the standard error is treated as stable while the lone significant contrast is not over-interpreted. In substantive terms the identical-instrument cohorts are close to interchangeable in level as well as in structure, with only a hint that the 2026 students rated their own procedural fairness slightly higher. This within-Tier-1 stability of level is a genuine bonus: the framework promised that scores *could* be compared here, and when compared they prove stable, reinforcing the network-side finding (Chapter 7.4.4) that the three cohorts' networks are statistically indistinguishable.

8.3 The Same-Items Tier: Structural Replication 2021 and 2022

Across the scale boundary, the question narrows to structure. The 2022 cohort ($n = 263$) already carries the final 26-item numbering; its one reverse-worded item (item 16, in its pre-2023 form) was recoded to the positive direction for structural comparison, and analyses were rerun without the two old-wording items (16 and 22) as a sensitivity check. The 2021 cohort was realigned to the final numbering item-by-item by wording (the drift documented in Chapter 6 [notably the 16↔17 swap] means position cannot be trusted), using the cross-walk in `item_map_full.csv`; the realigned analytic sample is the full 2021 pool restricted to the 26 published items ($n = 433$, larger than the $n = 346$ analytic subset of the published EFA, since for a structural comparison it is the wording-faithful realignment, not the exclusion rule, that matters; the exclusion that produced the published 346, most plausibly the 2021 lie-item screen, is not recorded in the working data, so a matched $n = 346$ re-run is not reproducible here, and the structural comparison accordingly rests on the full realigned pool).

Factor congruence is good for 2022 and mixed for 2021 (Table 8.2). Tucker's congruence coefficients between each cohort's four-factor solution and the final 2023 solution average .89 for 2022 (minimum .86: all four factors at least "fair" by the .85 benchmark; Lorenzo-Seva & ten Berge, 2006) and .86 for 2021, but with one 2021 factor congruent at only .65. The picture is therefore an honest "mostly, with a soft spot," not a clean replication: three of the four principles recover their configuration across the scale boundary in both cohorts, while one factor in the earliest first-person cohort does not align well under rotation. The 2022 sensitivity analysis without the two old-wording items leaves the conclusion intact (minimum congruence .87).

Community structure agrees for 2021 and partly collapses for 2022. Without using the a priori principle labels in community detection, exploratory graph analysis recovered exactly four communities in 2021 with a community assignment identical to the a priori principles (adjusted Rand index = 1.00, a partition-level agreement coarser than factor congruence, which is why it can be perfect while one factor's loadings are only weakly congru-

8. Study 4: Stability Across Cohorts and Instrument Versions

ent), whereas 2022 returned three communities ($ARI = .51$), one principle merging with a neighbor. That the two structural lenses disagree about *which* cohort replicates more cleanly (congruence favors 2022, community recovery favors 2021) is itself informative: it says the four-principle organization is present but not equally crisp on the agreement scale, and it counsels against any strong claim that the pre-2023 instrument measured exactly what the final one does.

Table 8.2. Structural replication across the scale boundary (reference: 2023 four-factor solution)

Cohort	n	Congruence		ARI	Edge similarity vs. 2023 (Spearman)
		(mean / min)	EGA dims		
2021	433	.86 / .65	4	1.00	.49
2022	263	.89 / .86	3	.51	.58

Note. Source: 08_tier2_congruence.csv, 08_cisla.csv. Edge similarity is descriptive only; no significance test is computed across the scale boundary. 2022 item 16 reverse-coded; sensitivity without items 16/22 leaves congruence $\geq .87$.

The descriptive network comparison is reported in the same spirit: the cohort partial-correlation networks correlate with the 2023 network at Spearman $\rho = .49$ (2021) and $.58$ (2022) across the edge set, a moderate structural resemblance that is offered as description, not as an inferential claim, because the underlying scales differ and a network comparison test would be illegitimate here.

8.4 The Different-Frame Tier: The 2020 Normative Cohort

The 2020 cohort ($n = 375$) answered third-person normative items (“A teacher should...”), a different construct frame (Chapter 6.3), and therefore admits only the configural ques-

tion fixed in Section 8.1: does a principle-based organization emerge at all? The analysis rests on the item crosswalk, and the crosswalk had to be completed by hand before anything could be computed. Normative and self-descriptive phrasings do not always align one-to-one, and six pairings were flagged for manual review rather than trusted to an automated match (`item_map_full.csv`); all six were subsequently confirmed by the author as the same item across frames, including the two where the wording had shifted in content rather than merely in grammar: the equality item whose object moved from assessing *performance* to assessing *conduct* (a deliberate re-operationalization toward the more concrete behavior), and the needs item that changed from *postponing a test* for a pupil who was ill to *granting extra time to catch up* (a more realistic formulation of the same accommodation). The confirmations are recorded in the crosswalk. Twelve of the 26 final items have a 2020 counterpart (four procedural, three equality, three needs, and two merit), and the configural check runs on these twelve aligned items, with the item-count imbalance stated up front.

Both configural lenses recover a recognizable, but not a complete, four-principle organization. An exploratory four-factor solution on polychoric correlations, matched against the 2023 reference solution restricted to the same twelve items, yields Tucker congruence of .94 for procedural, .93 for equality, and .90 for needs, fair factor similarity by conventional benchmarks (Lorenzo-Seva & ten Berge, 2006), and .82 for merit. The weak spot is prefigured by the design: merit enters the comparison with only two aligned items, and a two-item factor can barely be identified, let alone matched precisely; the .82 is read here as an item-count artifact at least as much as a substantive difference. Exploratory graph analysis, asked for its own answer, finds three communities rather than four: the procedural items and the merit items each form their own community, while the equality and needs items fuse into one (ARI against the a priori four-principle assignment = .66).

The configural verdict is therefore differentiated, in both senses. Under a normative teacher-should frame, on the twelve pairable items, procedural justice and merit are already visibly separate organizations of the item space; the child-responsive pair, equality and needs, is present but not yet differentiated into two communities. That fusion is informative rather than embarrassing: it echoes the soft spots of the neighboring tier (the 2022 community col-

8. Study 4: Stability Across Cohorts and Instrument Versions

lapse of Section 8.3), and it concerns exactly the two principles whose items share a surface vocabulary of attention and accommodation. Nothing stronger is claimed, exactly as Section 8.1 promised: no loading, threshold, or mean of the 2020 cohort is compared to anything, and the 2020 cohort's role in the book remains that of a documented earlier construct frame (Chapter 6.3). For RQ4 the tier-3 answer is: yes, a principle-based organization emerges even under the different frame, with the four-way differentiation partial rather than complete.

8.5 Reliability Across Cohorts

Categorical ω for the four subscales, computed per cohort on the aligned items, traces a coherent trajectory (Table 8.3; no significance tests, by design). Procedural and merit reliabilities are stable and high throughout ($\omega \approx .82-.90$). Equality is the most scale-sensitive: it sits at $\omega = .66$ (2021) and $.62$ (2022) on the agreement scale but rises to $.80-.85$ on the fairness scale (2023–2026), consistent with the reading that asking for fairness judgments, rather than agreement, sharpened exactly the principle most exposed to social desirability (Chapter 7.6). Needs is the weakest subscale on the fairness scale ($\omega \approx .72-.75$ in 2023–2026), the short, heterogeneous scale flagged repeatedly in Chapters 6 and 7; on the earlier agreement scale it is equality, not needs, that sits lowest. Read against this trajectory, the published 2021 subscale alphas of $.70-.80$ (Chapter 6.4) sit comfortably within this reliability record; the lowest coefficients anywhere in the programme are the agreement-scale equality omegas ($\omega \approx .62-.66$), which the ordinal-appropriate, fairness-scaled final instrument brings up to conventional adequacy.

Table 8.3. Categorical ω by cohort and subscale (aligned items)

Cohort	Scale	Procedural	Equality	Needs	Merit
2021	agreement	.82	.66	.74	.85
2022	agreement	.82	.62	.69	.89
2023	fairness	.84	.80	.75	.87

8.6 Chapter Summary: A Structure That Travels, Scores That Do Not

Cohort	Scale	Procedural	Equality	Needs	Merit
2024	fairness	.88	.85	.72	.86
2026	fairness	.89	.84	.74	.90

Note. Source: 08_omega_trajectory.csv. Latent-response-scale categorical ω ; descriptive trajectory only.

8.6 Chapter Summary: A Structure That Travels, Scores That Do Not

RQ4 asked whether the four-principle structure holds across cohorts and instrument versions, and the answer is tiered because the evidence must be. In the identical-instrument tier, the instrument is measurement-invariant to the strictest level tested: loadings, intercepts, and even residuals equal across 2023, 2024, and 2026 (all $\Delta\text{CFI} \leq .007$), under a continuous (MLR) approximation and without a claim about ordinal thresholds; and, where the framework permits a level comparison, the cohorts' latent means are equivalent but for a single small elevation of procedural justice in 2026; the network lens concurs (Chapter 7.4.4). In the same-items tier, structure replicates across the agreement→fairness scale boundary well enough to be recognizably the same four principles, but with honest soft spots: strong factor congruence for 2022 alongside a collapsed community count, and perfect community recovery for 2021 alongside one poorly congruent factor. In the different-frame tier, the 2020 configural check on the twelve author-confirmed item pairings finds a partial four-principle organization: procedural and merit stand apart (congruence .94 and .82, the latter on two items), while equality and needs fuse into a single community (three EGA communities, ARI .66), a recognizable organization under a different construct frame, claimed at configural strength only. The reliability trajectory shows the published 2021 alphas sitting comfortably within a record whose weakest coefficients (the agreement-scale equality omegas) the

8. *Study 4: Stability Across Cohorts and Instrument Versions*

final, fairness-scaled instrument lifts to conventional adequacy. The chapter's shorthand can now be stated precisely: the four-principle *structure* recurs everywhere the items align (the construct travels) while *scores* are comparable only inside the identical-instrument tier, not because levels are unstable but because measurement forbids the comparison elsewhere. This tiered stability, together with the hierarchy of Study 1, the instrument of Study 2, and the converging factor-and-network structure of Study 3, is the integrated result that Chapter 9 now interprets.

Integration

9. Discussion

Aim of the chapter

This chapter integrates the findings of the four studies into a single argument (from the hierarchy of distributive principles pre-service teachers endorse to the stable measured structure of self-perceived justice) and confronts that argument with the Czech equity tradition, with theory, with the programme's limitations, and with its implications for teacher education.

9.1 The Arc of Findings: From a Hierarchy of Principles to a Stable Measured Structure

The four studies were designed as one arc, and read together they support one integrated claim: the justice thinking of the sampled Czech pre-service teachers is coherent, equality-led, four-principled in structure, measurable at the latent level, and structurally recurrent across cohorts, within precisely stated limits of measurement and generalization.

Study 1 (RQ1) established the evaluative landscape. Asked to judge six village-level distribution regimes, respondents ($n = 315$) produced a graded hierarchy of unusual clarity: flat equality alone at the top ($M = 5.50$), a closely spaced middle band of chance, need, and merit ($4.50 / 4.22 / 3.93$), and a decisively rejected bottom of competitive equal opportunity (2.36) and caste (2.00). The hierarchy survived every robustness check; and beneath it the sample spread along a single axis rather than into distinct types: how far a respondent lets

9. Discussion

need justify departing from an equal share, with a need-inclusive majority at one end and an equality-first, need-skeptical minority at the other (the two-group split is weak and largely reducible to the need rating alone, so this is one continuum, not two kinds). That fault line is the first appearance of a theme the whole book turns on: equality is uncontested, but the status of need is where justice reasoning divides, and where, in Studies 3–4, the needs subscale proves the weakest-measured principle.

Study 2 (RQ2) turned from what students judge to how their self-perception can be measured, and its five-version history distilled into three lessons: perspective (third-person items measure a generalized other), person (only first-person items reach self-perception), and scale (agreement anchors measure self-description; fairness anchors put the justice judgment itself into the response). The result was a 26-item, four-principle instrument whose development is documented item-by-item, including the honest price of its design: deliberately heterogeneous short subscales with historically modest alphas.

Study 3 (RQ3) put the instrument's structure before two complementary methods with different modeling assumptions. The factor lens rejected a single justice dimension, found the rigid one-item-one-factor translation too strict, and settled on four distinct-but-related factors with cross-loadings admitted (ESEM, CFI = .916), reliable at the latent level ($\omega = .72$ –.90 across samples) but licensing no summed total score. Without using the a priori principle labels in community detection, the network lens found four communities in 99.6% of a thousand resamples and agreed closely with the a priori taxonomy at ARI = .885. The two methods converged on the same broad architecture with one item-level disagreement, and the structure proved essentially unrelated to socially desirable responding (no factor beyond $|r| = .161$ with self-deception in the primary scoring). That disagreement (the clear-criteria item 26, with the criterion-regularity item 22 as the network's strongest bridge) localizes a substantive hinge between procedural justice and equality rather than a measurement accident: the same seam also surfaced in the classroom transpositions of Study 1 and the cross-loadings of the factor view, a reading that item 26's low bootstrap stability does not by itself exclude (Chapter 7.5); the rewording test proposed in Section 9.6 is the check.

Study 4 (RQ4) asked whether any of this travels, and answered by tier. Where the instrument is identical (2023, 2024, 2026), measurement invariance holds to the strictest level tested under a continuous approximation (all $\Delta\text{CFI} \leq .007$), no differences between the cohorts' networks were detected (all comparisons null), and, in the book's only licensed level comparison, latent means are equivalent in seven of eight contrasts, with a single small elevation of procedural justice in 2026 ($d = 0.33$). Where the items are shared but the scale differs (2021, 2022), structure replicates recognizably but imperfectly: strong congruence with a collapsed community count in 2022, perfect community recovery with one weakly congruent factor in 2021. Where the construct frame differs (2020), the configural check, run on twelve author-confirmed item pairings, finds a partial four-principle organization: procedural and merit already stand apart, while equality and needs fuse into a single community (three EGA communities, ARI .66; factor congruence .82–.94, the low end on a two-item merit factor). The shorthand earned by these results is exact: *the structure travels, the scores do not*, and the scores' immobility is a fact about measurement, not about instability.

9.2 A Micro Lens on a Macro Debate: Confronting the Czech Equity Tradition

Czech educational research on justice has been, for two decades, predominantly *macro*: a tradition centered on equity of access and the system's reproduction of inequality (early tracking, school selectivity, and the socioeconomic gradient of educational chances; Matějů & Straková, 2006; Straková, 2007), culminating in a national strategy whose master-term is the reduction of inequalities and the promise of equal opportunities (MŠMT, 2020). This book neither contradicts that tradition nor competes with it. It changes the grain: from the system that allocates chances to the teacher who allocates daily (time, attention, grades, exceptions) in the sense in which street-level actors *are* the policy their institutions promise (Lipsky, 1980). Three of its findings, with the readings this book attaches to them, speak directly back to the macro debate.

9. Discussion

First, the population expected to implement the equity agenda arrives equality-led: flat equal treatment is the one principle the sampled future teachers rate decisively just at the village level (Study 1); in self-perception, equality is a distinct, reliably measured organizing principle (Studies 3–4). A profession-in-formation that treats *sameness* as the default of fairness is fertile ground for universalist policy, and potentially resistant ground for policies that require *justified departures* from sameness.

Second, exactly there lies the friction the macro lens cannot see: need, the distributive engine of inclusion, is the ambivalent principle. In the vignettes it sits in the contested middle tier, dividing the sample roughly into endorsers and skeptics; in the instrument it is also the weakest subscale on the final scale ($\omega \approx .72-.75$), though that weakness partly reflects the subscale's deliberately short, broad design (Chapter 6.4), so the measurement evidence is a supporting voice here, not proof of ambivalence on its own. Inclusion policy asks teachers to depart from equality daily, on grounds of need; on this book's reading, such departures are, for future teachers, neither self-evidently just nor crisply organized in self-perception. That is not an argument against inclusion; it is a location of the pedagogical work inclusion requires.

Third, the equal-opportunity paradox is a finding about language as much as about principle. The policy master-term, encoded in the vignettes as a competitive scramble from a common starting line, was rated barely above caste (2.36 vs. 2.00); the same formal property of equal chances, encoded as a lottery, sat more than two points higher. Read with the framing caution the book attaches to it (the contrast is suggestive, not a controlled manipulation; Section 5.6.2), the practical implication is nonetheless pointed: if "equal opportunities" evokes, for future teachers, a competition that abandons its losers, then policy rhetoric and professional intuition are speaking past each other, and the macro agenda's key word may need translating, not just implementing, at the classroom level.

9.3 Theoretical Contribution

The book makes three contributions it is entitled to claim. The first extends organizational justice into the teacher's allocative role. The classroom justice literature had established pupils as fairness perceivers (Chory-Assad & Paulsel, 2004; Resh & Sabbagh, 2016); this programme operationalizes the other side of the dyad: the teacher as an everyday allocator whose procedural rules (Leventhal, 1980; Colquitt, 2001) and distributive principles (Deutsch, 1975) can be measured as *self-perceived practice*, in the first person, with a justice-anchored response scale. The instrument's documented lineage (perspective, person, scale) is offered as a reusable lesson in how such measures need to be built.

The second is the hinge finding. Procedural justice and equality, kept apart by the literatures that produced them, meet in the classroom at a specific behavioral joint: the transparent, uniformly applied criterion. Three methodologically independent encounters (the dissolution of distributive principles into procedure in the vignette transpositions, the secondary loadings the factor view had to admit, and the migration and bridging of the criterion items in the network) converge on the same seam. The claim is deliberately modest and falsifiable: in teachers' self-perception, fair procedure and equal treatment share their central token, and instruments that force the two apart will misfit precisely there.

The third is methodological: a demonstration of structure-first validation under an evolving instrument. Treating the factor and network views as complementary lenses whose convergence is evidence; pre-specifying demotion rules for exploratory structure; and disciplining a seven-cohort programme by a three-tier comparability framework that licenses exactly as much comparison as measurement affords (*structure travels, scores do not*) together constitute an exportable template for the common, awkward situation in which an instrument matured while its research programme was already running. Within the Czech field, moreover, network psychometrics remains uncommon in educational research, while published network-model applications appeared earlier in Czech psychology (e.g., Klocek et al., 2022). The few educational applications include exploratory graph analysis of a physical-education motivation scale (Harbichová et al., 2025), a Ministry of Education pilot analysis of begin-

9. Discussion

ning teachers' perceived preparedness using regularized item networks, community detection, and bootstrap stability checks (MŠMT, 2023), and the psychometric-network study of student-perceived inclusive teaching by Bula and Nehyba (2025). The approach should not be conflated with the more familiar Czech tradition of social network analysis, which maps ties among persons rather than partial-correlation structure among items (Lintner et al., 2024).

9.4 Limitations

The programme's limits are stated as plainly as its findings. Sampling: all cohorts are convenience samples from a single faculty of education; claims concern structure and within-sample relations, and no population prevalence is estimated. Design: the cohorts are repeated independent cross-sections, not a panel; nothing in this book measures change within persons, and cross-cohort equivalence of latent means is not evidence about individual development. Measurement history: the response scale changed mid-programme (agreement → fairness), creating a boundary across which scores are not comparable at all (handled, but not removed, by the three-tier framework); early-version subscale reliabilities were adequate but modest for the shortest scales: published 2021 standardized subscale alphas of .70–.80, replicated as raw-score a priori-subscale alphas of .69–.76 on the same data (Chapter 6.4), remedied at the latent level but leaving sum scores unrecommended throughout. Data structure: the two streams cannot be linked at the individual level, so the relation between judging distributions (Study 1) and self-perceiving one's practice (Studies 2–4) remains a between-designs inference; vignette demographics exist for about two-thirds of the 315 respondents (sex 204, age 206). Method: ceiling effects forced a continuous (MLR) approximation for multigroup invariance, with ordinal thresholds untested across groups; the 2026 holdout is small, and its global fit indices degraded even as parameters replicated; the 2020 configural comparison rests on only twelve pairable items (merit on just two) and is claimed at configural strength alone. Construct: everything measured is self-perception; the BIDR analysis rules out gross self-flattery, not the gap between perceived and enacted

fairness: pupils and observers may see a different classroom.

9.5 Implications for Teacher Education

Four implications follow, offered as evidence-informed proposals rather than tested interventions. First, make distribution dilemmas explicit in the curriculum. The six-villages activity is a ready-made, low-cost instrument for surfacing the intuitions this book has mapped; its own seminar embedding (Section 5.3.1) shows how the move from villages to classrooms can be scaffolded, with the is/ought guard that finding a principle at school never suffices to justify it. Second, train assessment fairness at the hinge. If transparent, uniformly applied criteria are the shared token of procedural justice and equal treatment, then criterion transparency is the single highest-leverage practice teacher education can rehearse: it serves both perceptions at once, and the data locate it (items 8, 22, 26) at the center of the measured construct. Third, use the instrument formatively, at subscale level. The psychometric verdict has a practical corollary: profile feedback on four principles is defensible; a single summed “justice score” is not: a teacher educator who ranks students by total would be doing what the bifactor indices forbid. Fourth, do explicit legitimacy work for need-based differentiation. The ambivalence around need is exactly where inclusion practice will strain against default intuitions of sameness; naming the tension (when is unequal treatment just?) belongs in the preparation of every teacher who will be asked to differentiate.

9.6 Future Research

The agenda that follows is concrete. A panel design, following the same pre-service teachers from entry through induction, is the natural next step the present design cannot substitute for, and would convert structural stability into a claim about development. The pupils’ perspective completes the dyad: whether pupils’ perceived justice of a teacher aligns with that teacher’s self-perception is an open, consequential question. Anchoring vignettes (King et

9. Discussion

al., 2004; Voňková & Hrabák, 2015) would close the methodological loop the book opened: using short scenarios to calibrate the self-report scale across respondents, and across cohorts, attacking the very comparability boundary this programme had to manage by framework. Behavioral and observational measures would address the perception–practice gap that self-report cannot. Within the instrument, the obvious experiments are a rewording test of item 26 and a strengthened needs subscale; across contexts, replication beyond one faculty and one country. At the conceptual level, Section 2.7 offers a literature-bounded similarity map of justice concepts based on short working definitions (Appendix E); its natural extension is a larger, time-calibrated corpus study that could examine not only present-day resemblance but how the vocabulary of justice has diversified. And one debt the programme owed itself is now paid within these pages: the 2020 configural comparison, completed once the item pairings were confirmed (Chapter 8.4), leaving as the standing extension the author’s continuing interest in restorative justice, present in the taxonomy but deliberately unmeasured here.

Chapter summary

Four studies, one arc: the sampled Czech pre-service teachers judge distributions with an equality-led, three-tier hierarchy; their self-perceived justice is organized into four measurable principles; that organization is recovered closely, with one item-level disagreement, by two complementary methods with different assumptions, is not an artifact of self-flattery, and recurs, as structure, across cohorts and instrument versions exactly as far as measurement licenses comparison. Confronted with the Czech macro tradition, the micro lens neither contradicts nor repeats it: it locates the classroom-level frictions (ambivalent need, a mistranslated equal-opportunity rhetoric, and the procedural–equality hinge) that system-level equity research cannot see and teacher education can act on. The concluding chapter distils this into the book’s final claims.

10. Conclusion

Aim of the chapter

This chapter closes the monograph: it distils the answers to RQ1–RQ4 into the book’s final claims and states, in a form the reader can carry away, what the six-villages research programme adds to Czech educational research.

This book began with six villages and a choice, and it can end by reporting what the choosers chose, and what the choosing turned out to be made of.

The first claim is descriptive. When three hundred and fifteen Czech pre-service teachers judged six ways of dividing a common good, they did not produce scatter; they produced a hierarchy of three magnitude-tiers. Flat equality stood alone at the top. Chance, need, and merit occupied a closely spaced middle, every pairwise difference statistically real once the sample grew, yet small in magnitude (rank-biserial $r = .11-.25$): a shallow gradient, not a plateau, of principles one can argue about, and the sample did. And two regimes were rejected outright: allocation by caste, and equal opportunity in its competitive reading, the same starting line followed by a scramble. That last result is the book’s most quotable finding and its most carefully bounded one: it is best read not as a rejection of the ideal of equal chances (the lottery village, formally just as chance-equal, sat two points higher) but as a rejection of what the phrase can code for: a race that abandons its losers. The sampled future teachers, these data say, arrive equality-led: sameness of treatment is their default of fairness, departures from it require justification, and positions fixed by caste or by unregulated competition strike the great majority of them as unjust. Beneath that shared default ran a

10. Conclusion

single fault line (most would let need, and less often desert, justify departures from strict equal shares, while a stricter-egalitarian minority would not), the division the instrument chapters return to.

The second claim is psychometric. Self-perceived justice of teaching can be measured, but only if three lessons are respected, each learned the expensive way across five instrument versions: items must speak in the first person, or they measure a generalized other; the response scale must ask about *fairness*, not agreement, or it measures self-description; and short, deliberately broad behavioral subscales will look unreliable to coefficient alpha while being adequately precise at the latent level. The final 26-item instrument measures four principles (procedural justice, equality, needs, merit) with latent-level reliabilities of .72 to .90, and it earns a verdict rarely stated so bluntly: there is no license for a single summed “justice score.” The general factor is too weak to carry one; the four principles carry the information, and any practical use of the instrument must respect that.

The third claim is structural, and it is the book’s strongest. The four-principle organization is not an artifact of the method that looks for it. A factor model that presupposes latent causes and a network model that presupposes none, applied to the same 744 respondents, recovered the same architecture (four communities, agreeing with the theoretical taxonomy at $ARI = .885$, with the number of dimensions reproduced in 99.6% of a thousand resamples), and the structure proved essentially untouched by socially desirable responding. Where the two lenses disagreed, they disagreed instructively: at a single seam, the transparent and uniformly applied criterion, where procedural justice and equal treatment share their central token. Three methodologically independent analyses stumbled over that same hinge, and the book’s reading is that this is not a flaw in the instrument but a fact about the construct: in teachers’ self-perception, fair procedure and equal treatment are distinct principles hinged together at the criterion.

The fourth claim is the disciplined one. Across the cohort-years 2019–2026, with an instrument that changed wording, numbering, and finally its response scale, exactly one form of stability could be tested honestly, and the book built a framework to test no more than that:

full measurement invariance (under a continuous approximation), with latent means equivalent in seven of eight contrasts, where the instrument was identical; recognizable structural replication, with honest soft spots, across the scale boundary; and, where the construct frame differed, a configural-only verdict that is itself partial, a recognizable principle organization in the 2020 normative cohort, with procedural and merit distinct and equality and needs not yet differentiated. *The structure travels, the scores do not*, and the scores' immobility is a statement about measurement, not about people. Whether Czech pre-service teachers of 2020 and 2026 differ in how just they perceive themselves to be is a question these data, by their own rules, decline to answer; that the *organization* of their self-perceived justice is the same four-principle structure wherever it can be checked is the answer they give with unusual consistency.

What does the programme add to Czech educational research? A complement, not a correction. The Czech justice tradition has looked at the system (access, tracking, the inheritance of disadvantage), and this book adds the missing grain: the teacher as an everyday allocator, and the future teacher as a bearer of distributive intuitions that policy will eventually pass through. At that grain, the findings locate three things the macro lens cannot see: a profession-in-formation whose fairness default is sameness; an ambivalence about need (the very principle inclusion asks teachers to enact daily) visible in judgment and echoed, more softly, in measurement; and a policy master-term, equal opportunities, whose competitive reading repelled the future teachers sampled here. None of these is an indictment; each is a precise place where teacher education can work, and the book leaves behind the tools for that work: a six-villages activity that surfaces the intuitions in one seminar session, and a validated instrument that maps them at subscale resolution.

Everything here is offered with its limits attached: convenience samples from one faculty, self-perception rather than observed practice, independent cross-sections rather than a panel, and a comparability framework that forbids exactly the trend claims a casual reader might want. Those limits are not disclaimers; they are the book's method. A research programme that grew alongside its own instrument owes its readers a strict account of what may be compared with what, and the account proved to be the contribution: claims sized

10. Conclusion

precisely to their evidence.

The Blue Village divides equally, and the students chose it, by a margin no other village approached, with nearly four in five calling it just (Chapter 5). But the more important discovery lies one step deeper: beneath the choosing there is a stable, four-cornered structure (procedure, equality, need, merit) that organizes how the sampled future teachers perceive the justice of their own teaching, that two methods sharing no assumptions agree on, and that recurred wherever, across seven cohort-years, measurement allowed it to be checked. Teaching will test that structure daily, one small allocation at a time. This book has measured what future teachers bring to it.

Chapter summary

The book ends with one distilled claim (the sampled Czech pre-service teachers hold a coherent, equality-led, and reliably measurable four-principle conception of self-perceived justice, stable in structure wherever measurement permits comparison) and hands the reader to the references and appendices (final instrument with version crosswalk, vignette texts, supplementary tables, ethics protocol).

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List of Abbreviations

Abbreviation	Meaning
ARI	adjusted Rand index (chance-corrected agreement between two partitions; 0 = chance, 1 = identity)
BIC	Bayesian information criterion
BIDR	Balanced Inventory of Desirable Responding
CFA	confirmatory factor analysis
CFI	comparative fit index
CLMM	cumulative link mixed model for ordinal data
EBICglasso	graphical lasso network estimation with the extended Bayesian information criterion
ECV	explained common variance
EFA	exploratory factor analysis
EGA	exploratory graph analysis
ESEM	exploratory structural equation modelling
GGM	Gaussian graphical model (partial-correlation network)
IQR	interquartile range

List of Abbreviations

Abbreviation	Meaning
KMO	Kaiser–Meyer–Olkin measure of sampling adequacy
LPA	latent profile analysis
M, Mdn, SD	mean, median, standard deviation
MLR	robust maximum-likelihood estimator
NCT	network comparison test
RMSEA	root mean square error of approximation
RQ1–RQ4	Research Questions 1–4
SRMR	standardized root mean square residual
TLI	Tucker–Lewis index
WLSMV	weighted least squares mean- and variance-adjusted estimator
wTO	weighted topological overlap
ω , ω_H	categorical omega, hierarchical omega

Appendix A: The Perceived-Justice Instrument (Final 26-Item Version)

This appendix reproduces the final self-report instrument used in the 2023, 2024, and 2026 cohorts (Studies 2–4). The instrument was administered in Czech; the Czech wordings are authoritative, and the English column is a working translation provided for readers of this book, not an officially validated English-language instrument. The full item-by-item lineage across all versions (2019 → 2020 → 2021 pool → 2022 → final) is documented in `data/processed/item_map.csv` and narrated in Chapter 6.

A.1 Instructions and Response Scale

Response scale (7 points). Each item describes a concrete teaching behavior; the respondent rates how *just* that conduct is:

1 = *zcela nespravedlivé* (completely unjust) · 2 · 3 · 4 = *ani spravedlivé, ani nespravedlivé* (neither just nor unjust) · 5 · 6 · 7 = *zcela spravedlivé* (completely just)

The final instrument contains no reverse-coded items: after the 2023 rewording of item 16, all items are positively keyed, so the judgment task is uniform (one evaluative operation applied to one described behavior at a time). See Chapter 6.6 for the rewording and Chapter 6.5 for the scale change (agreement → fairness anchors) that defines the tier boundary at 2022/2023.

A.2 Items by Principle

The instrument comprises 26 items across the four principles: procedural justice (8 items), equality (6), needs (6), and merit (6). Items are listed below in administration numbering; the “Situation” column names the recurring classroom context each item samples.

A.2.1 Procedural justice (8 items)

#	Czech (authoritative)	English (working translation)	Situation
1	Na začátku roku je pro mě důležité zřetelně stanovit pravidla fungování ve třídě.	At the start of the year, it is important to me to clearly establish the rules of how the class works.	conduct
2	Žákům jasně vysvětlím, na co kladu důraz při ústním zkoušení.	I clearly explain to pupils what I emphasize during oral examination.	examination
6	Své žáky předem seznámím s tím, jakým způsobem řeším případné konflikty ve třídě.	I inform my pupils in advance how I handle any conflicts in the class.	conduct

#	Czech (authoritative)	English (working translation)	Situation
7	Žákům sdělím, že mohou kdykoliv konzultovat nesrovnalosti ve vyhodnocení písemky.	I tell pupils they can discuss any discrepancies in the grading of a test with me at any time.	teaching
8	Žákům předem oznamuji, jaká jsou kritéria pro hodnocení jejich písemných prací.	I announce to pupils in advance the criteria for grading their written work.	grading
23	Vysvětlím žákům, jak si mají dávat vrstevnickou zpětnou vazbu, například při vzájemném hodnocení prací.	I explain to pupils how to give peer feedback, e.g., when assessing each other's work.	grading
24	Žákům dávám prostor pro spoluvytvoření si pravidel chování ve třídě.	I give pupils room to co-create the rules of behavior in the class.	conduct

Appendix A: The Perceived-Justice Instrument (Final 26-Item Version)

#	Czech (authoritative)	English (working translation)	Situation
26	Je pro mě důležité, abych měl pro žáky jasná kritéria, podle kterých je hodnotím.	It is important to me to have clear criteria by which I assess pupils.	grading

A.2.2 Equality (6 items)

#	Czech (authoritative)	English (working translation)	Situation
3	Věnuji každému žákovi stejné množství času při ústní zkoušce.	I devote the same amount of time to every pupil during oral examination.	examination
10	Věnuji problémům každého žáka stejnou míru pozornosti.	I devote the same degree of attention to every pupil's problems.	conduct
14	Všem žákům věnuji stejnou míru pozornosti.	I give the same degree of attention to all pupils.	teaching, conduct
15	Při hodnocení chování každého žáka používám vždy stejná kritéria.	When assessing every pupil's behavior, I always use the same criteria.	conduct

#	Czech (authoritative)	English (working translation)	Situation
16	Se všemi žáky se snažím zacházet stejně.	I try to treat all pupils the same.	teaching
22	Při zkoušení se snažím všechny žáky vyvolávat se stejnou pravidelností.	During examination, I try to call on all pupils with the same regularity.	examination

A.2.3 Needs (6 items)

#	Czech (authoritative)	English (working translation)	Situation
4	Pokud žák prochází obtížnou životní situací, jako je například rozvod rodičů a podobně, při hodnocení jeho situaci zohledňuji.	If a pupil is going through a difficult life situation (e.g., parents' divorce), I take it into account when grading.	grading
5	Věnuji více péče žákovi, který to podle mě nejvíce potřebuje.	I give more care to the pupil who, in my view, needs it most.	teaching, conduct

Appendix A: The Perceived-Justice Instrument (Final 26-Item Version)

#	Czech (authoritative)	English (working translation)	Situation
9	Při hodnocení zohledňuji, zda je pro konkrétního žáka předmět obtížný.	When grading, I take into account whether the subject is difficult for a particular pupil.	grading
11	Žákovi, který má dlouhodobý problém s pochopením učiva, zadávám jiné domácí úkoly než ostatním žákům.	To a pupil with a long-term problem understanding the material, I assign different homework than to others.	homework
17	Žákovi, který byl nemocný, dám čas navíc, aby se mohl látku doučit.	To a pupil who was ill, I give extra time so they can catch up on the material.	teaching, conduct
18	Zpočátku bych přizpůsobil hodnocení žáka, který by přišel do třídy z jiné školy.	I would initially adjust the assessment of a pupil who transferred from another school.	grading

A.2.4 Merit (6 items)

#	Czech (authoritative)	English (working translation)	Situation
12	Žáky, kteří stihnou práci v hodině před termínem, oceňuji (například si žák může pracovat v hodině na svých věcech).	I reward pupils who finish their classwork ahead of the deadline (e.g., they may work on their own things).	pupil performance
13	Pokud žák během výuky splní úkol navíc, tak jeho snahu patřičně ocením.	If a pupil completes an extra task during the lesson, I duly appreciate their effort.	pupil performance
19	Výjimečnou snahu žáka v hodině ocením odpovídajícím způsobem.	I acknowledge a pupil's exceptional effort in the lesson appropriately.	pupil performance
20	Oceňuji žáka, který úspěšně reprezentuje školu na vědomostní soutěži.	I praise a pupil who successfully represents the school in a knowledge competition.	pupil performance

Appendix A: The Perceived-Justice Instrument (Final 26-Item Version)

#	Czech (authoritative)	English (working translation)	Situation
21	Dávám uznání žákovi, který projevil snahu nad rámec zadaného úkolů.	I give recognition to a pupil who showed effort beyond the assigned task.	pupil performance
25	Oceňuji žáka, který dokáže zrealizovat nejrůznější třídní aktivitu.	I appreciate a pupil who can carry out various class activities.	pupil performance

A.3 Version Crosswalk

The final 26 items descend from a 36-item pool (2021) and, before it, from earlier third-person and normative wordings (2019, 2020). The complete crosswalk (every version's wording, the reformulations, the 36→26 reduction, and the agreement→fairness scale boundary) is held in machine-readable form in `data/processed/item_map.csv` and is summarized in Chapter 6 (Table 6.1 and Sections 6.1–6.6). Because the scale wording changed between the 2022 and 2023 cohorts, sum scores are not comparable across that boundary; the three-tier comparability framework (Chapter 4.4, applied in Chapter 8) governs every cross-cohort claim.

Appendix B: The Six Villages Vignettes

This appendix reproduces the vignette materials of Study 1 (Chapter 5): six short stories, each describing how one village distributed a windfall of royal gold, and each embodying a distinct principle of distributive justice. The vignettes were administered on paper in Czech in spring 2019 ($n = 315$). The Czech text is authoritative (as administered); the English is a faithful translation for readers of this book. The six principles span a broader taxonomy than the four-principle questionnaire (Studies 2–4): equality, need, merit/desert, chance (lottery), equal opportunity, and caste/social groups (`codebook_vinety.md`).

B.1 Instruction and Response Scale

Respondents first read the instruction and rated each village on a seven-point justice scale, then produced a full ranking.

Zhodnotte do jaké míry se každá vesnice zachovala spravedlivě: 7 = Naprosto spravedlivě, 6 = Spravedlivě, 5 = Spíše spravedlivě, 4 = Ani spravedlivě ani NESpravedlivě, 3 = Spíše NESpravedlivě, 2 = NESpravedlivě, 1 = Naprosto NESpravedlivě.

[EN] Rate the extent to which each village behaved justly: 7 = Completely just, 6 = Just, 5 = Rather just, 4 = Neither just nor unjust, 3 = Rather unjust, 2 = Unjust, 1 = Completely unjust.

B.2 The Six Vignettes

B.2.1 Blue settlement: *equality* (v_rovnost)

[CZ, as administered] Nejblíže královskému hradu byla Modrá osada. Tam starosta svolal všechny občany a pečlivě začal rozpočítávat peníze. Každý občan, ať mladý nebo starý, pracovitý nebo líný, starosta či služka, dostal stejné množství zlatáků do hrsti, až se rozdaly všechny královské peníze. Někteří byli spokojeni, jiní tak trochu a další si mysleli, jak jim bylo ukřivděno. Starosta i delegace však byli spokojeni s dobře odvedenou prací.

[EN] Nearest the royal castle was the Blue settlement. There the mayor summoned all the citizens and carefully began to divide up the money. Every citizen (young or old, hardworking or lazy, mayor or maid) received the same handful of gold coins, until all the royal money had been distributed. Some were satisfied, some only a little, and others felt they had been wronged. The mayor and the delegation, however, were satisfied with a job well done.

B.2.2 Green village: *need* (v_potreby)

[CZ, as administered] Když delegace dorazila do Zelené vesnice, starosta nakázal všem občanům vyplnit dotazníky jak velkou částku peněz právě potřebují a nač by je užili. Někteří psali, že potřebují rekonstrukci chýše, jiní čekali dítě a další chtěli na zaslouženou dovolenou. Byli mezi zelenými i tací, kteří psali, že dukátek navíc na úspory by se hodil, ale jinak zrovna nic nepotřebují. Vše si pak starosta zkontroloval. A moudře dal těm, co potřebovali, ale ti kteří si řekli o málo, s málem také odešli. Něktěm snad i bylo líto vidět sousedy odlétat na drahou dovolenou z peněz, které delegáti do vesnice přivezli. A tak tedy stejně jako v modré osadě ta velká spravedlnost přinesla štěstí i mrzení.

[EN] When the delegation arrived at the Green village, the mayor ordered all citizens to fill in questionnaires stating how much money they needed right now and what they would use it for. Some wrote that they needed to rebuild their cottage, others were expecting a child, and still others wanted a well-deserved holiday. Among the Greens there were also some who wrote that an extra ducat for savings would come in handy, but otherwise they needed nothing in particular. The mayor then checked everything. And wisely he gave to those who were in need, but those who had asked for little also left with little. Some perhaps even felt sorry to see neighbors flying off on an expensive holiday paid for from the money the delegates had brought to the village. And so, just as in the Blue settlement, that great justice brought both happiness and chagrin.

B.2.3 Orange town: *merit / desert* (v_zasluchy)

[CZ, as administered] V městečku Oranžová byli hrdi na svou historii a dbali na tradici a zásluhy a podle nich starosta také přiděloval peníze. Více tak dostali tací, kteří tvrdě pracovali, dobrovolničili o sobotách na obecních pozemcích, žili ve vesnici déle než 30 let. Starosta částečně zohlednil i ty jejichž otcové a dědové bojovali před 100 lety v bitvě u Bílé a přinesli obci velké vítězství a oběti. Taková starostovi i ve jménu svých dědů děkovali za prozřetelnou spravedlnost, avšak zbylo i dost těch, kteří měli jaksi pocit, že se na jejich zásluhy v pravou chvíli nevzpomnělo.

[EN] In the town of Orange they were proud of their history and cared about tradition and merit, and it was by these that the mayor allotted the money. More was given to those who had worked hard, volunteered on Saturdays on the communal land, and lived in the village for more than 30 years. The mayor also partly took into account those whose fathers and grandfathers had fought 100 years ago in the battle of White and brought the community a great victory and

made great sacrifices. Such people thanked the mayor, in the name of their forefathers too, for his providential justice; yet there remained plenty of those who somehow felt that their merits had not been remembered at the right moment.

B.2.4 White settlement: *chance / lottery (v_nahoda)*

[CZ, as administered] V Bílé osadě měli pocit, že ta dvě kolečka zlaťáků stejně nejsou dost pro všechny, aby způsobily radost, a tak udělali loterii. Jméno každého občana bylo v osudí právě jednou, díky čemuž měli všichni spravedlivou šanci na peníze dosáhnout. 1. cena byla polovina všech zlaťáků, druhá cena pak byla polovina té zbylé poloviny a tak dále. Celkem mohli občané bílé osady vyhrát deset cen. Většina odešla z loterie malinko zklamaná, ale hrál je pocit, že alespoň deset jejich sousedů si sáhlo na opravdové štěstí.

[EN] In the White settlement they felt that those two wheels of gold coins were not enough for everyone to bring joy anyway, and so they held a lottery. Each citizen's name was in the urn exactly once, so that all had a fair chance of getting the money. The first prize was half of all the coins, the second prize half of the remaining half, and so on. In total the citizens of the White settlement could win ten prizes. Most left the lottery a little disappointed, but were warmed by the feeling that at least ten of their neighbors had touched real fortune.

B.2.5 Red village: *equal opportunity (v_rovne_prilezitosti)*

[CZ, as administered] V Červené vsi vysypali peníze uprostřed náměstí a udělali kolem nich kruh tak, aby na jeho obvodu mohli stát všichni občané. Každý dopředu věděl, že budou mít stejnou šanci a mohli si sebou přinést jakýkoliv jeden nástroj - někdo přišel s lopatou, s kbelíkem či cedníkem, jiný za sebe někoho poslal. Když odbyla čtvrtá hodina odpolední, všichni se ve stejnou chvíli mohli rozběhnout doprostřed náměstí a kdo si co nabral, to mu patřilo. Starosta byl

se sebou spokojený, protože dal každému stejnou šanci a mnozí mu za to byli vděční, jen na ty, co si to nedokázali zařídit nezbylo mnoho.

[EN] In the Red village they poured the money out in the middle of the square and made a circle around it so that all citizens could stand on its perimeter. Everyone knew in advance that they would have an equal chance and could bring with them any one tool: someone came with a shovel, a bucket, or a sieve; another sent someone in their place. When the fourth hour of the afternoon struck, everyone could rush into the middle of the square at the same moment, and whatever anyone scooped up was theirs. The mayor was pleased with himself, because he had given everyone an equal chance and many were grateful to him for it; only for those who could not manage it was there little left.

B.2.6 Yellow town: *caste / social groups* (v_kasty)

[CZ, as administered] Poslední Žluté město bylo odjakživa svázáno pevným sociálním a kulturním systémem, tak když dorazila delegace s kolečky zlata, starosta hovořil o spravedlivém rozdělení pro všechny tři skupiny. Každá skupina dostane třetinu peněz a v rámci skupiny se pak peníze rozdělí spravedlivě rovným dílem, tak bude spravedlnosti napříč skupinami i v nich zachována. Na vrcholu pyramidy byla městská rada, tu tvořilo 8 nejmoudřejších a starosta. Druhou skupinou byli nepostradatelní, sem patřili vědci, učitelé, vojevůdci, diplomaté, prostě městská inteligence. Bylo jich 50 a dostali druhou třetinu. Třetí třetina připadla sedlákům a obchodníkům, kteří tvořili třetí a poslední skupinu, ve Žlutém městě jich žilo několik tisíc.

[EN] The last town, the Yellow town, had always been bound by a firm social and cultural system, so when the delegation arrived with wheels of gold, the mayor spoke of a just distribution for all three groups. Each group would receive a third of the money, and within a group the money would then be divided justly

Appendix B: The Six Villages Vignettes

in equal shares, so that justice would be preserved both across the groups and within them. At the top of the pyramid was the town council, made up of the 8 wisest and the mayor. The second group were the indispensables: scientists, teachers, commanders, diplomats, in short the town's intelligentsia. There were 50 of them and they received the second third. The third third fell to the farmers and merchants, who formed the third and last group; several thousand of them lived in the Yellow town.

B.3 Ranking Task

After rating all six villages, respondents completed a forced ranking:

[CZ, as administered] Zapište si pořadí vesnic od nejspravedlivější po nejméně spravedlivou vesnici.

[EN] Write down the order of the villages from the most just to the least just village.

Appendix C: Supplementary Tables

This appendix retains only numerical detail that is not already printed in the main chapters: the complete exploratory factor-loading matrix and all nine item pairs retained in the frozen redundancy-screen output. The shorter network-sensitivity, network-comparison, ordinal-fit, and invariance results remain in Chapters 7 and 8, where they are interpreted. No model was re-estimated for this appendix; displayed trailing zeros are formatting only.

C.1 Full Exploratory Factor Loadings (2023)

Table C.1 gives every loading from the four-factor exploratory factor analysis of the 2023 exploration wave ($n = 247$; minimum residual extraction, oblimin rotation, polychoric correlations). The raw factor codes are retained in parentheses, each labeled by its item content: MR3 = merit, MR2 = procedural justice, MR4 = equality, and MR1 = needs. Signs and all secondary loadings are shown; no loading is suppressed. As reported in Chapter 7.3.1, 23 of 26 items have an absolute loading of at least .40 on some factor. Items 17, 22, and 26 are the three below that descriptive salience threshold; all were retained.

Table C.1. Complete four-factor EFA loading matrix, 2023 ($n = 247$)

Item	Merit (MR3)	Procedural (MR2)	Equality (MR4)	Needs (MR1)
item_01	0.069	0.537	0.006	-0.134
item_02	-0.087	0.642	-0.023	-0.008
item_03	-0.135	0.123	0.488	0.073

Appendix C: Supplementary Tables

Item	Merit (MR3)	Procedural (MR2)	Equality (MR4)	Needs (MR1)
item_04	0.013	0.017	0.011	0.567
item_05	0.140	-0.089	0.055	0.645
item_06	-0.028	0.665	0.018	0.053
item_07	0.117	0.571	0.068	-0.065
item_08	0.074	0.630	0.080	0.012
item_09	-0.039	-0.011	0.012	0.713
item_10	-0.090	0.238	0.592	0.145
item_11	-0.026	0.266	-0.193	0.561
item_12	0.514	-0.195	0.119	0.096
item_13	0.727	-0.114	0.113	0.096
item_14	-0.069	0.061	0.731	-0.045
item_15	0.041	-0.137	0.724	-0.084
item_16	0.069	-0.024	0.694	0.009
item_17	0.196	0.143	0.152	0.280
item_18	0.218	0.101	-0.040	0.402
item_19	0.758	0.146	-0.066	0.007
item_20	0.735	-0.042	-0.090	-0.030
item_21	0.799	0.027	0.024	0.026
item_22	0.208	0.328	0.322	-0.258
item_23	0.106	0.654	0.101	0.062
item_24	0.030	0.643	-0.105	0.159
item_25	0.694	0.153	-0.061	-0.060
item_26	0.167	0.330	0.328	-0.043

Note. Source: 07_efa2023_loadings.csv; the counts match 07_cisla.csv (23 salient items and 3 below |.40|). Item wordings and a priori principle assignments appear in Appendix A.

C.2 Pair-Level Redundancy Diagnostics

Weighted topological overlap (wTO) measures how similarly two items connect to all other items; higher values flag possible redundancy. Table C.2 reproduces the nine pairs above .20 stored by the frozen routine, not all 325 possible pairs or the network's partial-correlation edges.

Table C.2. Nine frozen wTO pairs above .20

Item i	Item j	wTO
item_15	item_16	.288
item_12	item_13	.283
item_10	item_14	.262
item_19	item_21	.235
item_04	item_09	.218
item_01	item_02	.217
item_07	item_08	.214
item_20	item_21	.213
item_23	item_24	.210

Note. Source: 07_network_uva.csv; the three pairs above the pre-specified .25 threshold also appear in 07_network_cisla.csv. All remain within their a priori blocks; no item was removed (Chapter 7.4.1).

Appendix D: Ethics, Informed Consent, and Anonymization

This appendix documents the ethical basis of the research: the information-and-consent statement presented to every respondent, the reasoning under which the study required no prior ethics-committee review, and the anonymization protocol applied to the data.

D.1 Information and Consent Statement

All questionnaire waves were administered online (Google Forms). The questionnaire opened with the following information-and-consent statement, which respondents read before reaching any item. The statement is reproduced here verbatim in Czech, the language of administration, followed by an English translation. Proceeding past this screen to the questionnaire items constituted informed consent.

D.1.1 As administered (Czech)

Informace o výzkumu a souhlas s účastí

Dobrý den,

tento anonymní dotazník se zaměřuje na vnímání spravedlnosti u studentů učitelství. Výzkum realizuje Jan Nehyba z Katedry pedagogiky Pedagogické fakulty Masarykovy univerzity.

Appendix D: Ethics, Informed Consent, and Anonymization

Účast je dobrovolná a vyplňování můžete kdykoli ukončit. Získaná data budou anonymně zpracována a mohou být v anonymní podobě použita k akademickým a výzkumným účelům.

V případě dotazů se můžete obrátit na: Jan Nehyba, jan.nehyba@gmail.com.

Pokračováním k otázkám potvrzujete, že jste si uvedené informace přečetl/a a dobrovolně souhlasíte s účastí ve výzkumu.

D.1.2 English translation

Information about the research and consent to participate

Dear participant,

This anonymous questionnaire concerns the perception of justice among pre-service teachers. The research is carried out by Jan Nehyba of the Department of Education, Faculty of Education, Masaryk University.

Participation is voluntary and you may stop completing the questionnaire at any time. The data collected will be processed anonymously and may be used, in anonymous form, for academic and research purposes.

If you have any questions, you may contact: Jan Nehyba, jan.nehyba@gmail.com.

By proceeding to the questions, you confirm that you have read the above information and voluntarily consent to participate in the research.

The consent statement thus discloses (a) the anonymous nature of the data, (b) the researcher and institutional affiliation, (c) the study's purpose, (d) the voluntary nature of participation and the right to withdraw at any point, (e) the academic and research use of the anonymized data, and (f) a contact for questions.

D.2 Ethics Review

The study required no prior approval from the Masaryk University Research Ethics Committee (*Etická komise pro výzkum MU*). Under the committee's published guidance, anonymous survey research with adults generally does not require ethics approval; prior review is recommended only for defined cases: research involving children, vulnerable groups, or participants with limited capacity to consent; psychology projects that employ deception (a cover story); and the collection of special-category (sensitive) personal data under the GDPR, such as data on racial or ethnic origin, political opinions, religious or philosophical beliefs, trade-union membership, genetic or biometric data, or data concerning health, sex life, or sexual orientation.

The present study falls into none of these categories. It is an anonymous, non-interventional online questionnaire administered to adult university students; it involves no minors, no deception or cover story, and no intervention, and it collects no special-category personal data. The only demographic variables retained (gender, birth year, study subjects, prior experience) are ordinary, non-sensitive quasi-identifiers that do not permit individual identification.

Two safeguards address the one nuance the committee flags: that participants who could be seen as being *in a dependent position* relative to the researcher warrant particular care. First, participation was explicitly voluntary, with the right to withdraw at any time stated up front. Second, and decisively, responses were fully anonymous: they could not be linked to any individual student, nor to any course assessment or grade, so participation had no bearing whatsoever on a student's academic standing. The design therefore removes the dependency concern at its root rather than merely disclosing it.

Basis: Masaryk University Research Ethics Committee, "Ethics review" guidance (<https://www.muni.cz/en/about-us/organizational-structure/boards-and-committees/research-ethics-committee/ethics-review> [↗]).

D.3 Anonymization Protocol

The anonymization is scripted, versioned, and auditable; the operative script is `analyzy/scripts/03_*` and the full protocol is documented in `data/processed/DATA_README.md`.

In summary:

- Direct identifiers removed at source. The questionnaire exports carried a login e-mail address (Užívateľské meno) and a free-text comment field (Jakákoľvek poznámka) that could contain personal text. Both columns are dropped during processing and never enter the analytic data tree.
- Timestamps truncated. Submission timestamps are truncated to dates.
- Synthetic identifiers assigned. Each respondent receives a synthetic identifier (w<year>_<seq>, e.g., w2024_017) and a wave label; no natural identifier is retained.
- Automated verification. An automated check confirms that no e-mail address (no @) survives anywhere in `data/processed/` or `data/raw/`.
- Raw identifier-bearing exports deleted. After processing, the raw exports containing identifiers were deleted from the working environment (documented in `DATA_README.md`); the one combined-studies file that contained full participant names (2020komb, n = 46) was excluded from the programme entirely and deleted.
- Retained variables. Only non-sensitive quasi-identifiers remain (gender, birth year, study subjects, prior experience). The vignette stream (Study 1) carries no direct identifiers.

In the text of this book no respondent is identifiable; all illustrative pupil and teacher names (e.g., Chapter 5.6) are fictional and belong to teaching materials, not to data.

Appendix E: A Conceptual-Similarity Map of Justice Concepts: Method

This appendix documents the supplementary exploratory similarity map of justice concepts reported in Chapter 2.7. It compares two representations of the same 24 concepts: one based on theory-defined features scored by model coders, the other on the wording of short working definitions. A *dendrogram* is the branching diagram produced by either route: concepts joined farther to the right are more similar, while the vertical order of labels has no substantive meaning. The exercise maps resemblance within this literature-bounded set; it does not reconstruct the historical development of the concepts, test Figure 2.1, or establish a uniquely correct taxonomy.

E.1 Design and concept set

No concept families are defined in advance or used. The design is nevertheless not theory-free: the author selected the reference works, synthesized the working definitions, and specified the feature set. The two representations also share the same concept names and definitions and use the same tree-building procedure. Their partial agreement therefore serves as a sensitivity check on representation choice, not as an independent replication.

The concepts were assembled from a bounded set of sources already used in Chapter 2: the *Stanford Encyclopedia of Philosophy* entry on justice (Miller, 2026), the social-justice

handbook of Sabbagh and Schmitt (2016), and organizational- and classroom-justice reviews (Colquitt, 2001; Colquitt & Rodell, 2015; Chory-Assad & Paulsel, 2004). Without a preliminary division into families, the resulting set contains 24 concepts: equality, need, merit/desert (equity), the difference principle, sufficiency, fair equality of opportunity, luck egalitarianism, the capabilities approach, procedural, interpersonal, informational, retributive, restorative, corrective, compensatory, social, commutative, entitlement, utilitarian, and formal justice, justice as fairness, complex equality, recognition justice, and relational equality.

The 24 working definitions and their source hints are archived with the coding materials. Because no complete source-by-source inclusion and synonym log was created, this should be read as an author-assembled, literature-bounded frame rather than an exhaustive census of justice theory.

E.2 Theory-defined feature representation

The first representation describes each concept on 16 distinctions drawn from justice theory. Table E.1 groups them for readability; the coding manual defines every permitted state, including not-applicable and mixed states where needed.

Table E.1. The 16 conceptual features, grouped into four readable domains

Feature domain	Distinctions coded	Main theoretical anchors
Object and comparison	focus; comparative status; temporal orientation; end-state, patterned, or historical structure	Colquitt (2001); Feinberg (1974); Nozick (1974)

Feature domain	Distinctions coded	Main theoretical anchors
Basis of allocation	currency; sensitivity to need; sensitivity to desert/contribution; equal shares as default	Sen (1980); Deutsch (1975); Leventhal (1980)
Response to a wrong	prior wrong; punishment; repair/reconciliation	rectificatory and retributive traditions; Karp and Breslin (2001)
Procedure, relationship, and context	voice; interpersonal treatment; transparency/information; micro/macro level; domain- general/sphere-specific scope	Thibaut and Walker (1975); Bies and Moag (1986); Colquitt (2001); Walzer (1983)

Three separately run language-model coders from different providers (Anthropic, OpenAI, and Zhipu) assigned one permitted state to every concept–feature cell. The coding prompts supplied the concept names, working definitions, feature names, and the permitted response options; the fuller coding manual documents the intended meanings but was not itself included in them. The consensus state was the majority code; in the two cells where all three codes differed, the pre-specified tie-break used the first coder’s state.

Agreement was evaluated feature by feature because the 16 features have different state spaces. Krippendorff’s α ranged from .26 to 1.00; nine of the 16 features met the .667 reporting threshold and seven did not. A purposive five-concept check by two human coders produced exact-match rates close to those between each human and the model consensus. This small check supports the plausibility of the coding, not its correctness for the full matrix. The low-agreement features may reflect conceptual ambiguity, underspecified guidance, or coder error; the sensitivity analyses below therefore examine their influence explicitly.

The consensus feature matrix was converted to categorical Gower distances and grouped with Ward's hierarchical procedure. Figure 2.2 displays the resulting tree.

E.3 What the feature tree shows

The number of coloured branches in Figure 2.2 was selected with a simple average-silhouette diagnostic over cuts $k = 2-8$. Four groups produced the highest value (.395), narrowly ahead of three groups (.381); the .014 difference is small. The $k = 4$ cut is therefore only an exploratory reading aid. With just 24 concepts it is not evidence for four natural types. The full diagnostic is stored in [E5_cluster_diagnostics.csv](#)[↗].

The four principles measured in the book are marked by a diamond and bold type, not by a second family-colour scheme. Procedural justice lies beside informational and interpersonal justice; equality and need are immediate neighbours; and merit lies in the same broad branch near fair equality of opportunity and luck egalitarianism. These locations describe this input- and method-conditioned map; they do not test Figure 2.1.

E.4 Wording-based representation

The second representation uses *text embeddings*: numeric representations designed so that similarly worded passages lie closer together. The embedded inputs combined each concept name with its working definition and used none of the 16 feature scores. Cosine distance summarized wording similarity, after which the same Ward procedure produced Figure E.1.

The two representations agree partly on some neighbourhoods but differ substantially in detailed branching (Figure E.2); the full convergence diagnostics are archived with the analysis outputs. This is useful precisely because it limits the claim: neither map validates the other, and both remain conditional on their shared concept set and working definitions.

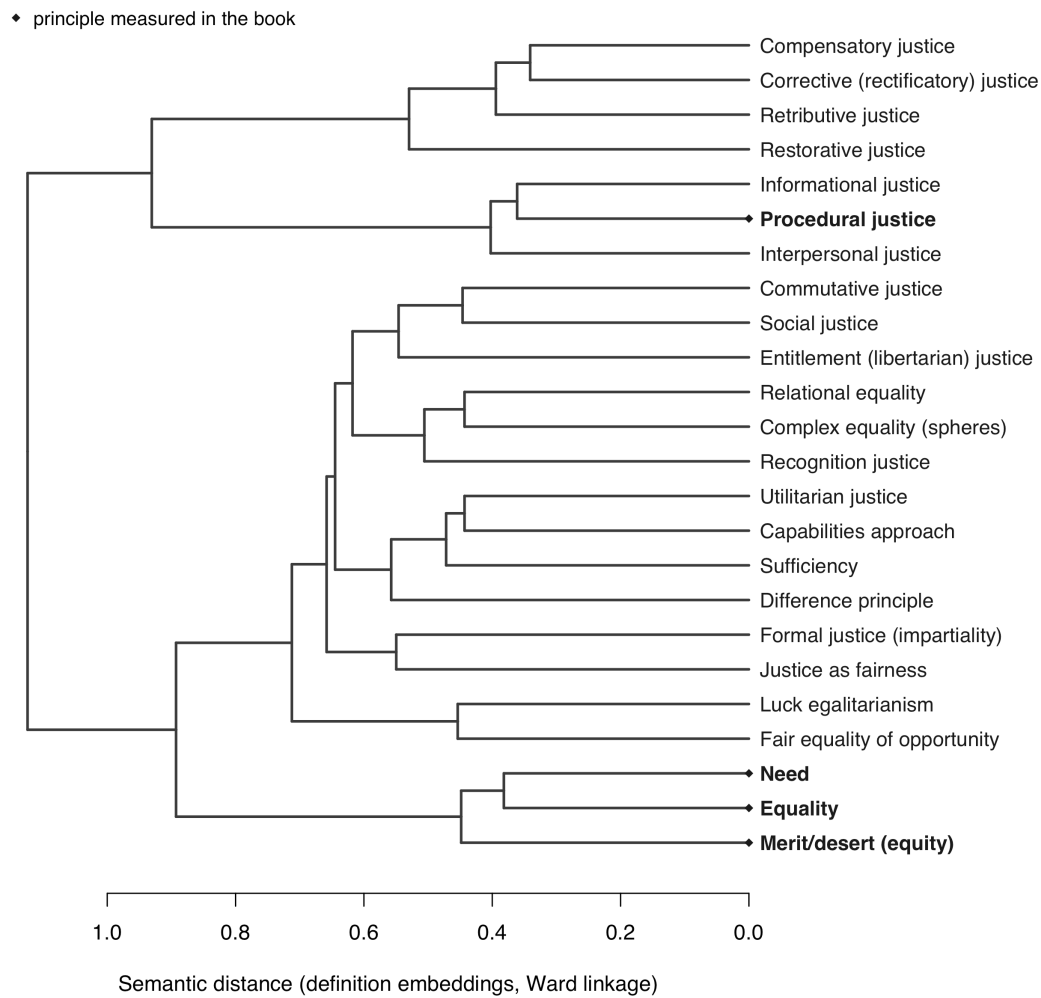


Figure E.1. Wording-based dendrogram built from the names and working definitions of the 24 concepts, using none of the 16 feature scores and no pre-assigned families. Diamonds and bold labels mark the four principles measured in the book. Concepts joined farther to the right are more similar. Source: [fig_E5_trees.R script](#).

Appendix E: A Conceptual-Similarity Map of Justice Concepts: Method

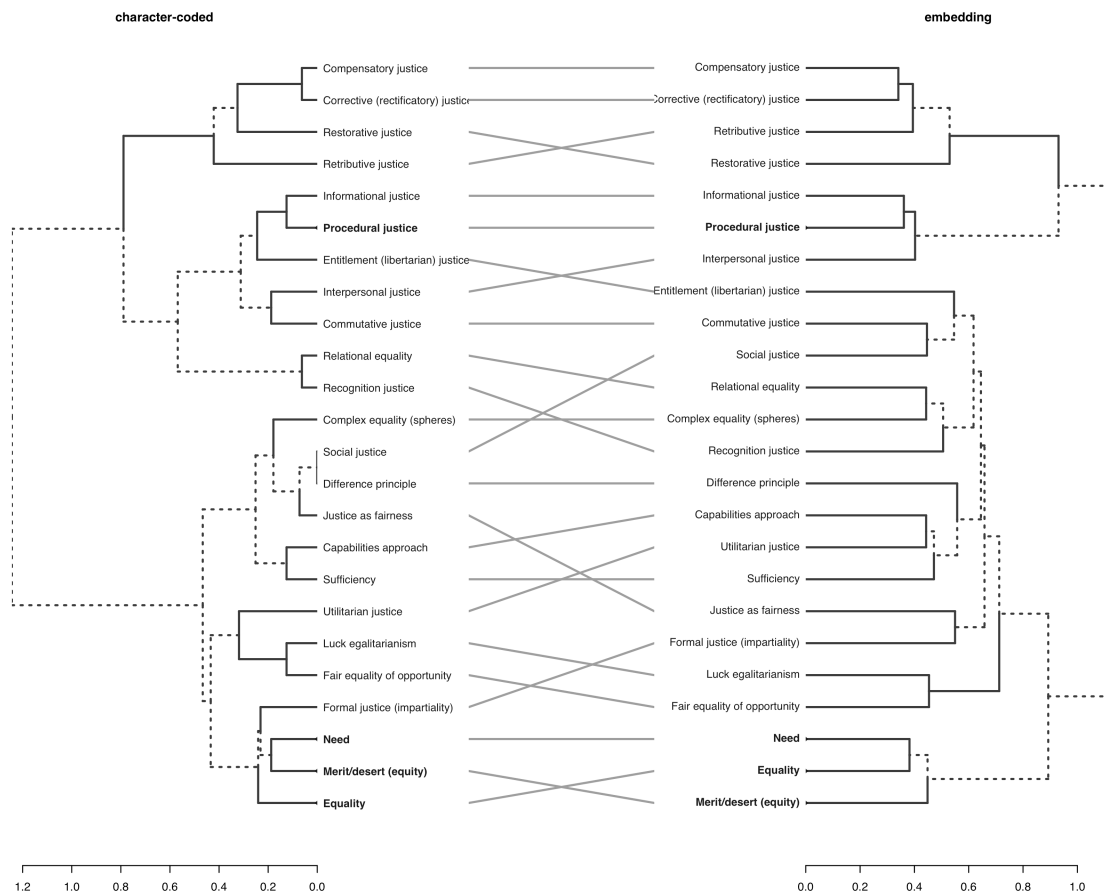


Figure E.2. Tanglegram of the feature-score (left) and wording-based (right) dendrograms. Each neutral line links the same concept; after the branches are rotated to align the trees, remaining crossings summarize differences in leaf order. They are a visual summary, not concept-level tests of agreement. Source: [fig_E5_trees.R script](#).

E.5 Reproducibility and limits

Archived materials include the concept table and definitions, feature manual, coder and consensus matrices, human pilot, wording distances, and numerical diagnostics. From the stored matrices onward, the analysis and figures are scripted; exact regeneration of provider outputs is not claimed.

Results remain sensitive to source selection, author-written definitions, uneven feature reliability, and the small pilot. They offer a transparent exploratory similarity map, not a validated taxonomy or a test of the didactic diagram in Figure 2.1.

